

Thread - 2026-02-23

<https://x.com/RiikonenMarko/status/2025907323097465003>

1/32 - 2026-02-23 12:15 UTC

22/23 Feb 2026, Rovaniemi. After two nights of no conditions we are back in business.

In a dwindling display in Koskenkylä I noticed flip-cna that seemed to be too close to the light. I corrected mentally the identification to "dual-path" 46° contact arc (lower arrow) and then I also noticed the actual flip-cna above it (upper arrow). This was my first visual of the 46° contact arc so it was real pleasure.

3 x 30s max stack. I'll see if a better stack can be worked from more photos. Visually both arcs were clear even if the display was weak.

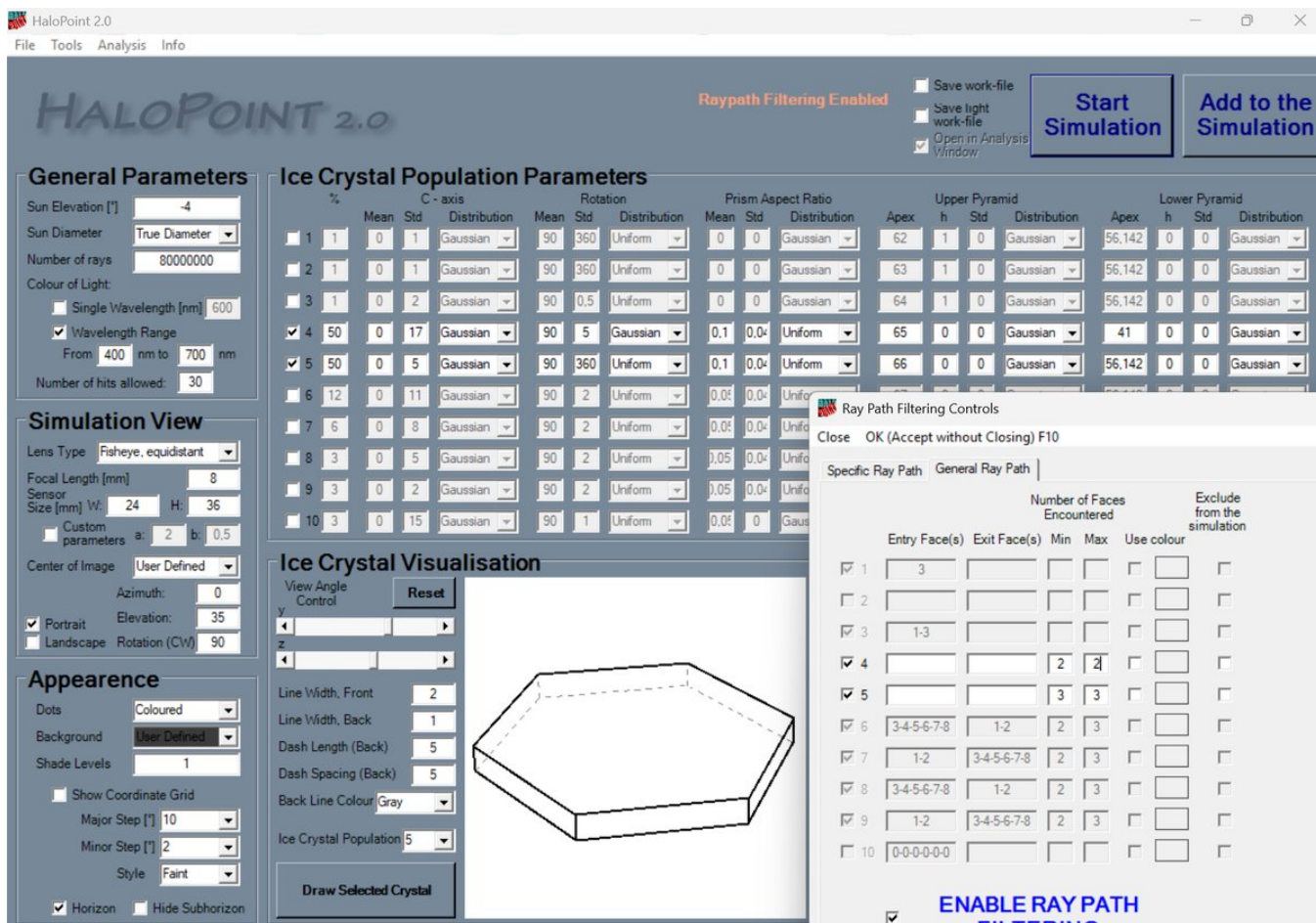
Upon arrival at the location car thermometer -22 C, leaving -25 C. Ice fog was really whimsical this night, not staying long in one place. I was thinking of giving up around 1 am and then realized it's a wrap in any case because once again the snow gun had stopped running.

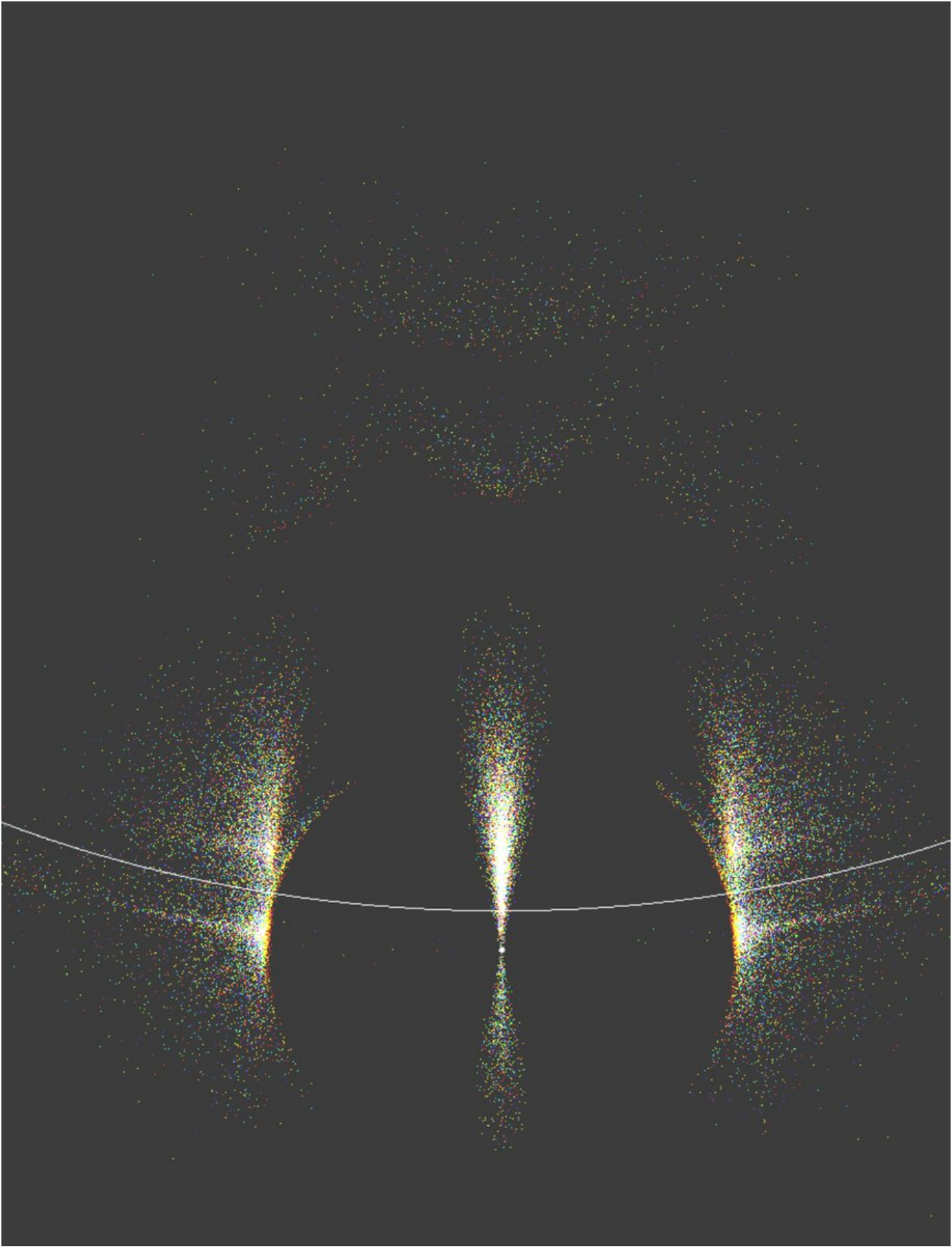


2/32 - 2026-02-25 15:31 UTC

22/23 Feb 2026, Rovaniemi. Looks like single images work best to show the flip-cna and dual-path 46° contact arc. Yeah, seeing and understanding this live was one of the most exquisite things. A quick sketch of simulation provided.

As to my earlier similar case in December 2016 <https://t.co/3jPvAh974u> , I am not sure if the 46° contact arc is tapering down to the lamp. Probably just an illusion created by the pillar that extends and opens ups upwards at the same time as flip-cna turns into 46° contact arc.

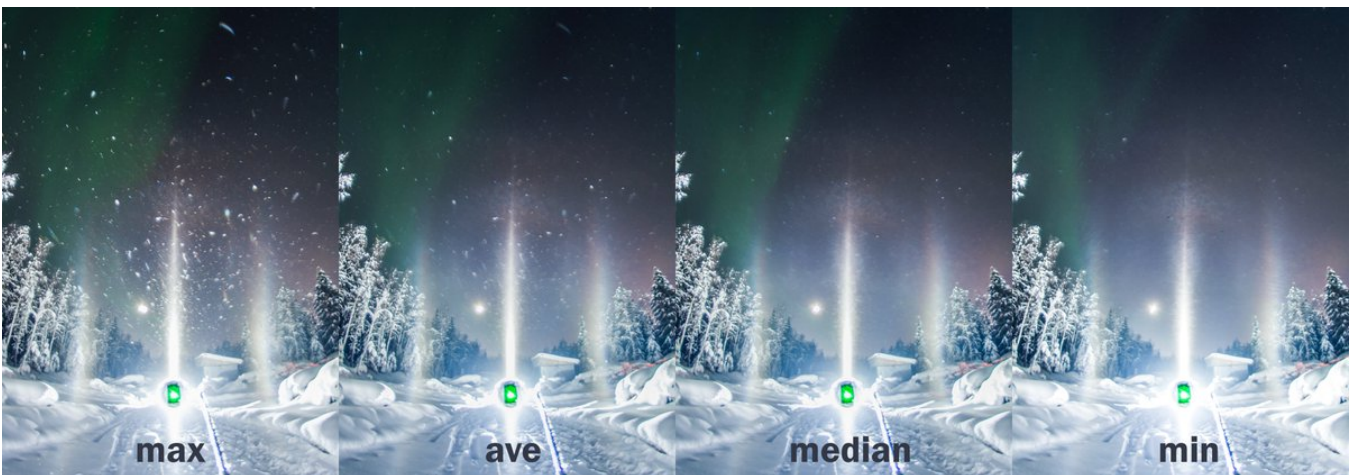






3/32 - 2026-02-26 13:26 UTC

22/23 Feb 2026, Rovaniemi. Comparing Photoshop stacking modes. 6 frames stacked, the same as in the previous post collage except the last frame that was left out.



4/32 - 2026-02-26 13:32 UTC

22/23 Feb 2026, Rovaniemi. Attempt at an alternative interpretation. What if the upper arc is not flip-cna but "flip-46° contact arc"? In that case one population suffices to explain the display. No filterings, the simulation is "realistic".

FileToolsAnalysisInfo

Save work-file

Save light work-file

Open in Analysis Window

Start Simulation

Add to the Simulation

General Parameters

Sun Elevation [°] -4

Sun Diameter True Diameter

Number of rays 80000000

Colour of Light:

☐ Single Wavelength [nm] 600
 ☒ Wavelength Range

From 400 nm to 700 nm

Number of hits allowed: 30

Simulation View

Lens Type Fisheye, equidistant

Focal Length [mm] 8

Sensor Size [mm] W: 24 H: 36

☐ Custom parameters

a: 2 b: 0.5

Center of Image User Defined

Azimuth: 0

Elevation: 35

Portrait

Landscape

Rotation (CW) 90

Appearance

Dots Coloured

Background User Defined

Shade Levels 1

☐ Show Coordinate Grid

Major Step [°] 10

Minor Step [°] 2

Style Faint

☒ Horizon
 ☒ Hide Subhorizon

Ice Crystal Population Parameters

	%	C - axis			Rotation			Prism Aspect Ratio			Upper Pyramid			Lower Pyramid				
		Mean	Std	Distribution	Mean	Std	Distribution	Mean	Std	Distribution	Apex	h	Std	Distribution	Apex	h	Std	Distribution
☐ 1	1	0	1	Gaussian	90	360	Uniform	0	0	Gaussian	62	1	0	Gaussian	56,142	0	0	Gaussian
☐ 2	1	0	1	Gaussian	90	360	Uniform	0	0	Gaussian	63	1	0	Gaussian	56,142	0	0	Gaussian
☐ 3	1	0	2	Gaussian	90	0.5	Uniform	0	0	Gaussian	64	1	0	Gaussian	56,142	0	0	Gaussian
☒ 4	50	0	9	Gaussian	90	2	Gaussian	0.6	0.05	Uniform	65	0	0	Gaussian	41	0	0	Gaussian
☐ 5	50	0	13	Gaussian	90	1	Gaussian	0.1	0.04	Uniform	66	0	0	Gaussian	56,142	0	0	Gaussian
☐ 6	12	0	11	Gaussian	90	2	Uniform	0.05	0.04	Uniform	67	0	0	Gaussian	56,142	0	0	Gaussian
☐ 7	6	0	8	Gaussian	90	2	Uniform	0.05	0.04	Uniform	68	0	0	Gaussian	56,14	0	0	Uniform
☐ 8	3	0	5	Gaussian	90	2	Uniform	0.05	0.04	Uniform	56,142	0	0	Gaussian	56,142	0	0	Gaussian
☐ 9	3	0	2	Gaussian	90	2	Uniform	0.05	0.04	Uniform	56,142	0	0	Gaussian	56,142	0	0	Gaussian
☐ 10	3	0	15	Gaussian	90	1	Uniform	0.05	0	Gaussian	56,142	0	0	Gaussian	56,142	0	0	Gaussian

Ice Crystal Visualisation

View Angle Control

Reset

y

z

Line Width, Front 2

Line Width, Back 1

Dash Length (Back) 5

Dash Spacing (Back) 5

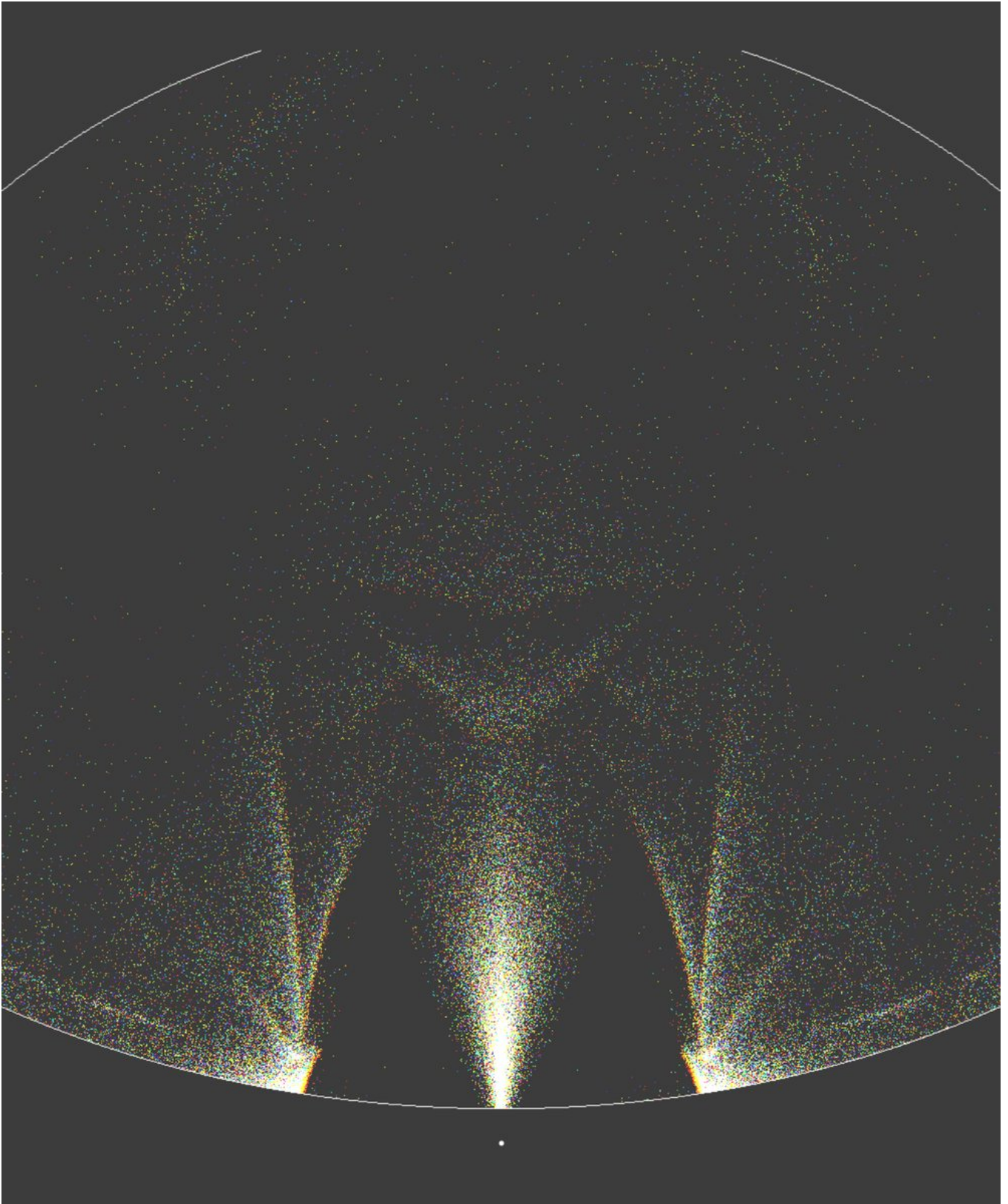
Back Line Colour Gray

Ice Crystal Population 4

Draw Selected Crystal

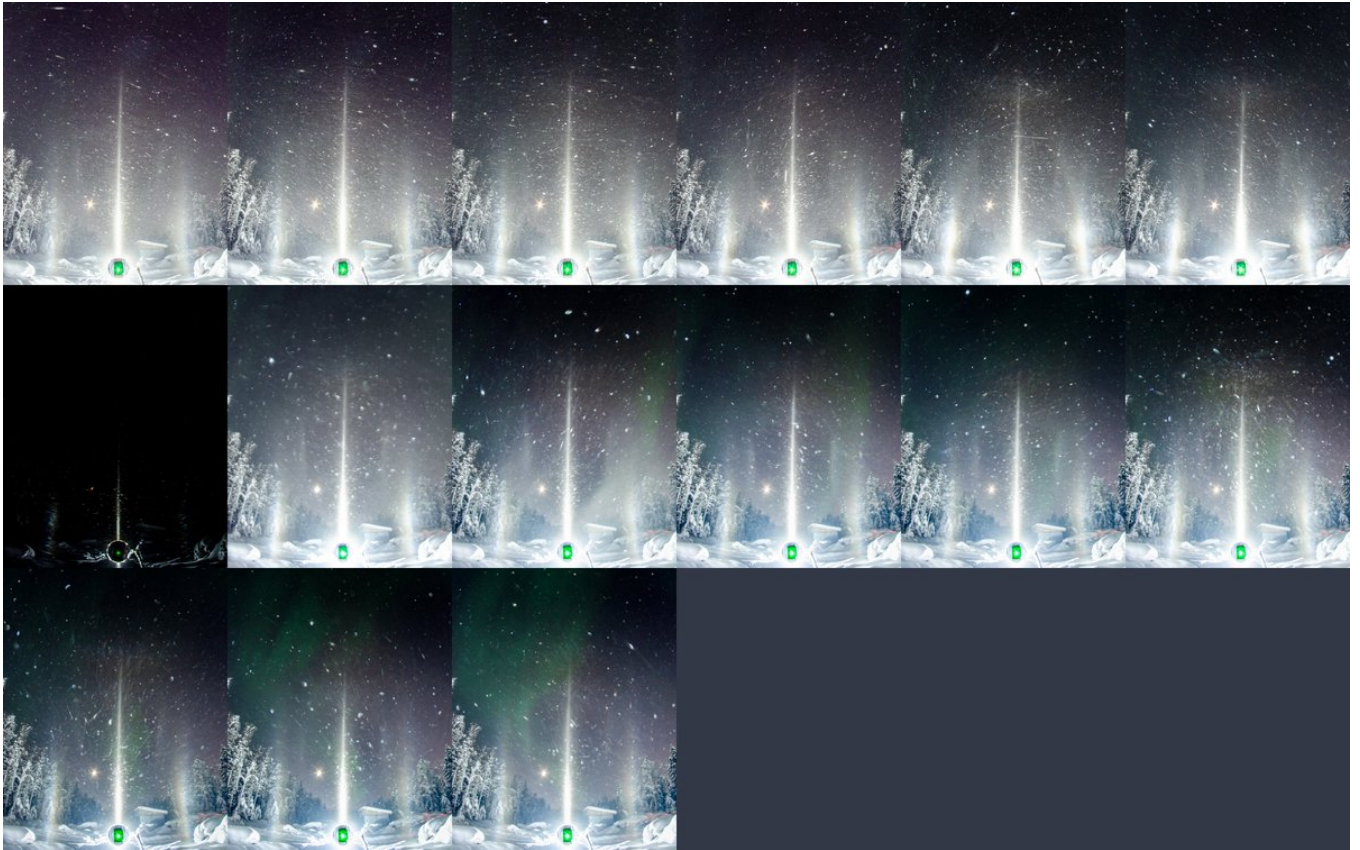
Prism Face Distances

Apply	Face Distance from c-axis								Std of Distance								Sync	Distribution
	3	4	5	6	7	8	3	4	5	6	7	8						
☒ 1	0.1	1	1	0.1	1	1	0	0	0	0	0	0	☐ Gaussian					
☒ 2	0.1	1	1	0.1	1	1	0	0	0	0	0	0	☐ Gaussian					
☒ 3	0.1	1	1	0.1	1	1	0	0	0	0	0	0	☐ Uniform					
☐ 4	1	1	1	1	1	1	0.1	0	0	0.1	0	0	☐ Uniform					
☐ 5	0.1	1	1	0.1	1	1	0	0	0	0	0	0	☐ Uniform					
☐ 6	0.1	1	1	0.1	1	1	0	0	0	0	0	0	☐ Gaussian					
☐ 7	0.1	1	1	0.1	1	1	0	0	0	0	0	0	☐ Gaussian					
☐ 8	0.3	1	1	0.3	1	1	0.1	0.1	0.1	0.1	0.1	0.1	☐ Gaussian					
☐ 9	2	1	2	1	2	1	0	0	0	0	0	0	☐ Gaussian					
☐ 10	1	2	1	1	2	1	0	0	0	0	0	0	☐ Gaussian					



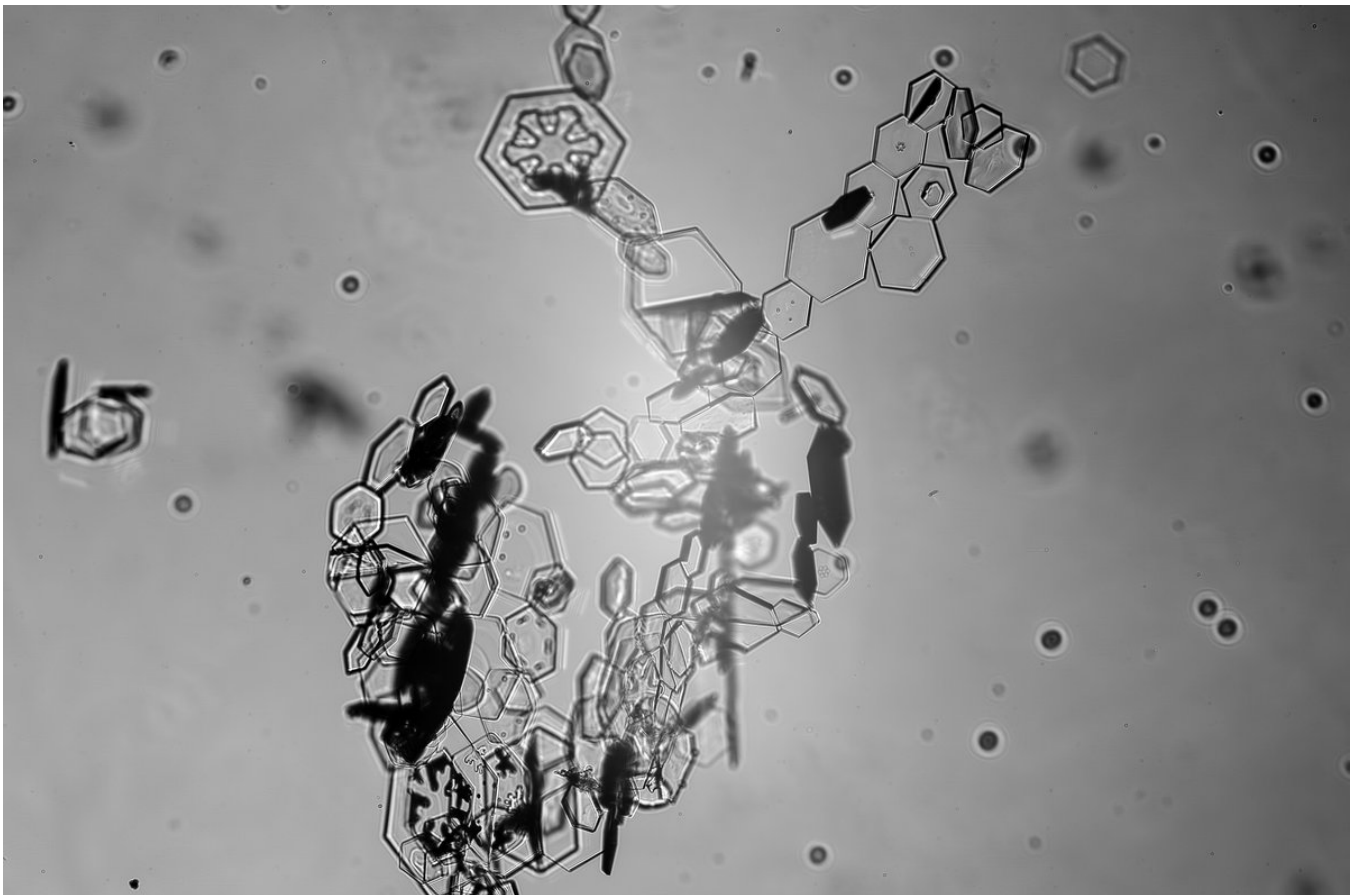
5/32 - 2026-02-26 14:07 UTC

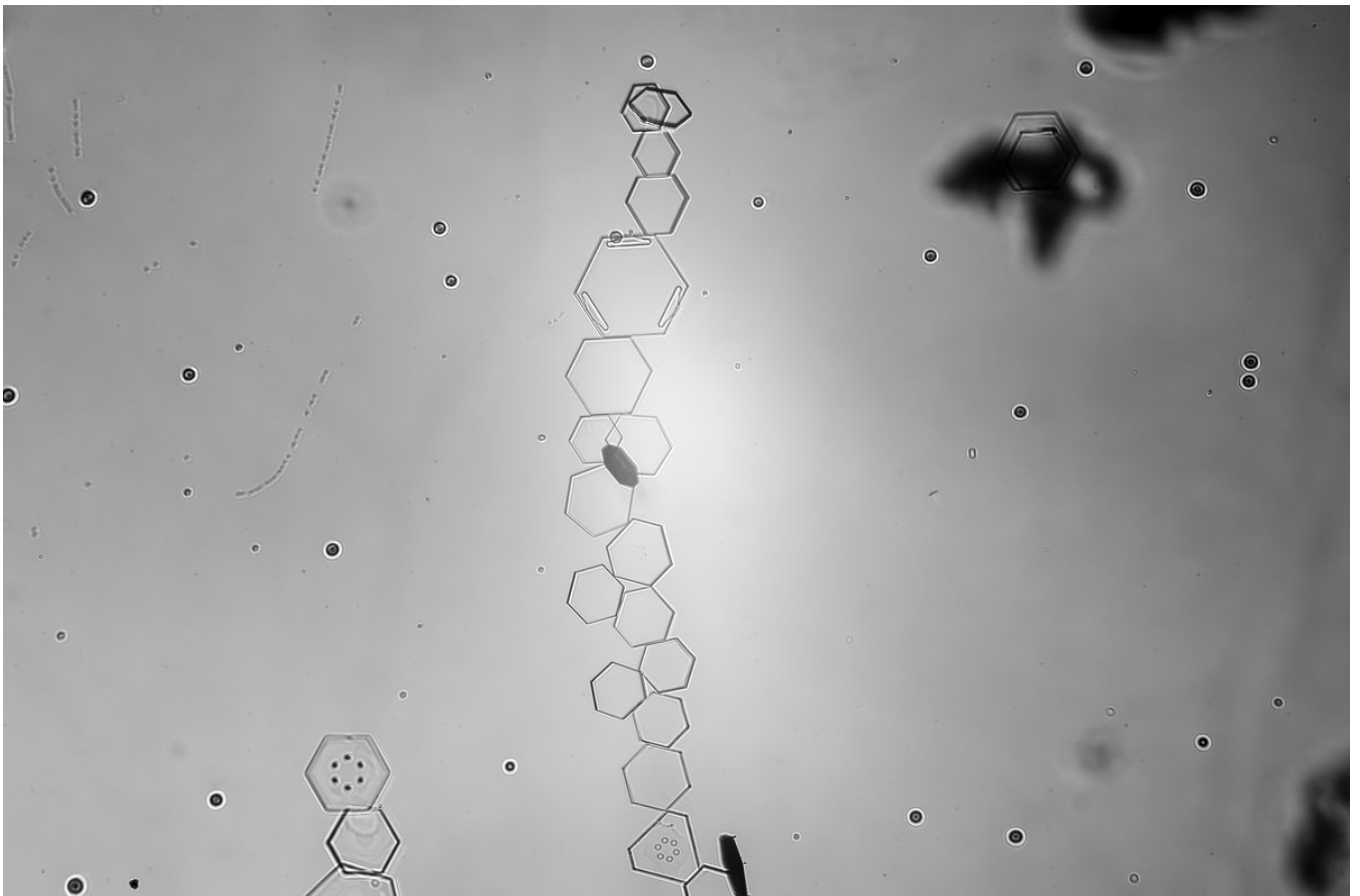
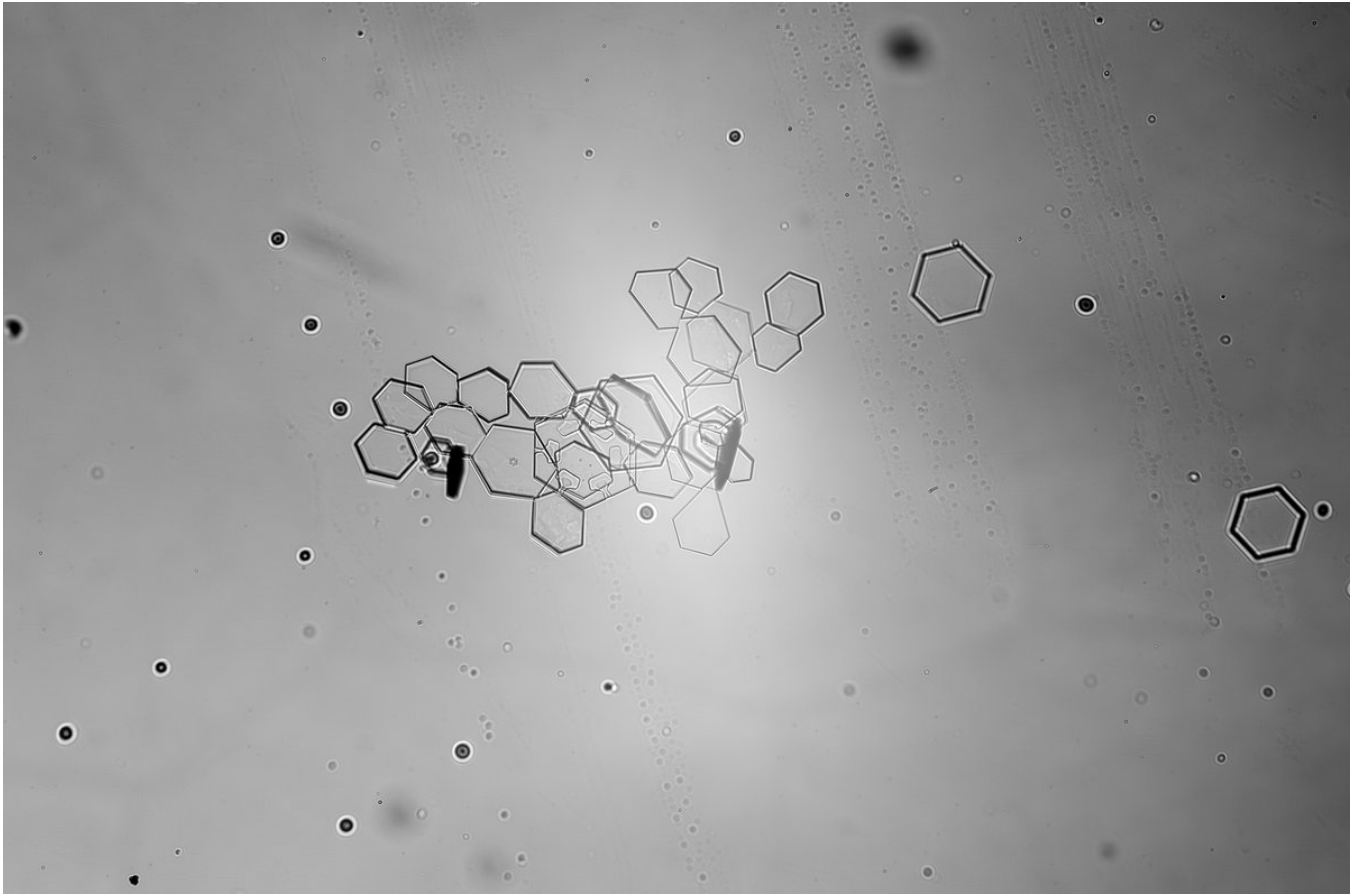
22/23 Feb 2026, Rovaniemi. Collage of photos from the start of crystal sampling. The earlier 7 frame collage with the best 46° contact arc continues from this. Here 46° contact arc is present perhaps only in the last three frames. It's appearance was thus a short-lived affair just before the display vanished completely.

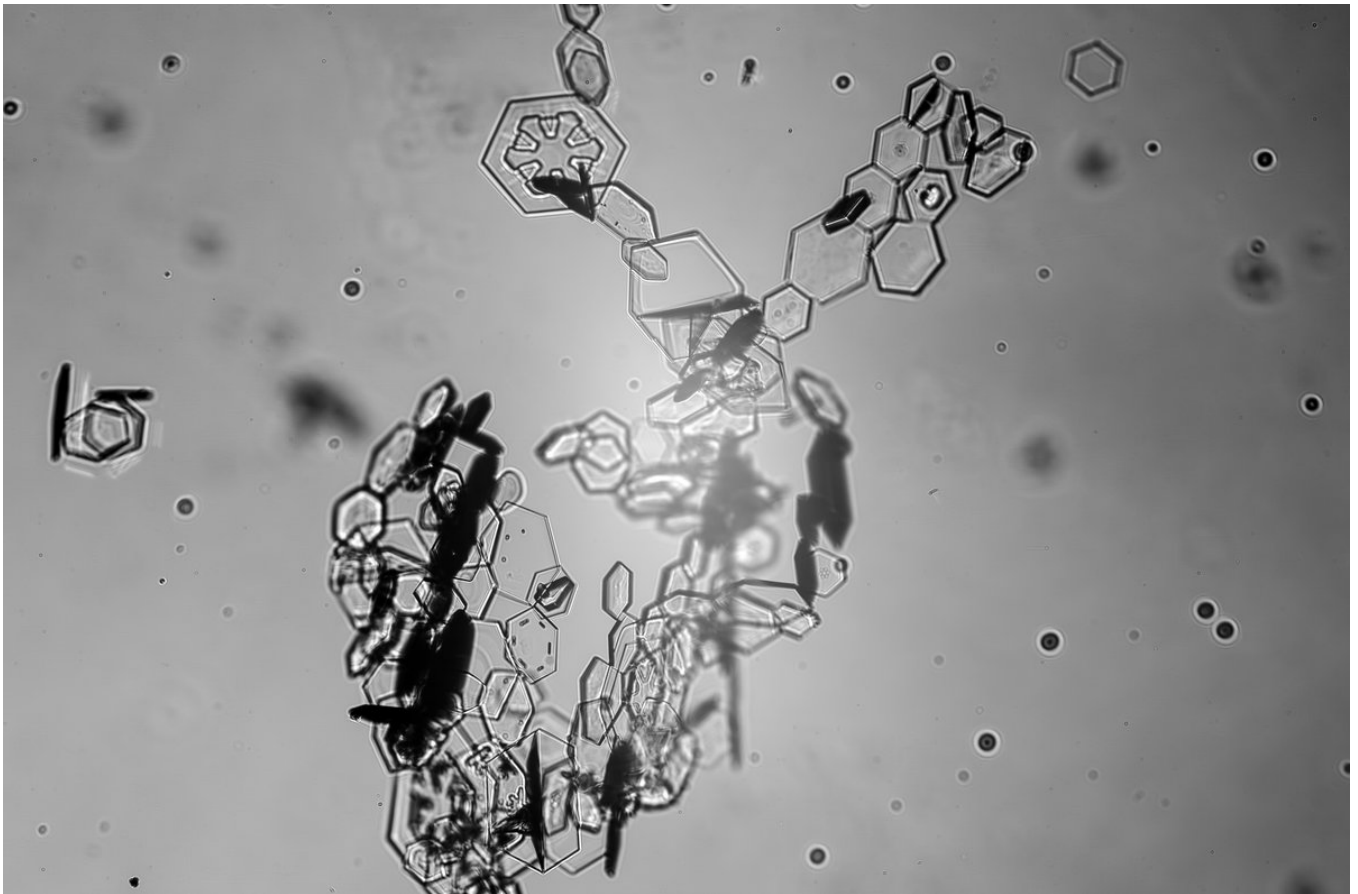


6/32 - 2026-02-26 14:32 UTC

22/23 Feb 2026, Rovaniemi. Crystals for the stage that involved 46° contact arc.



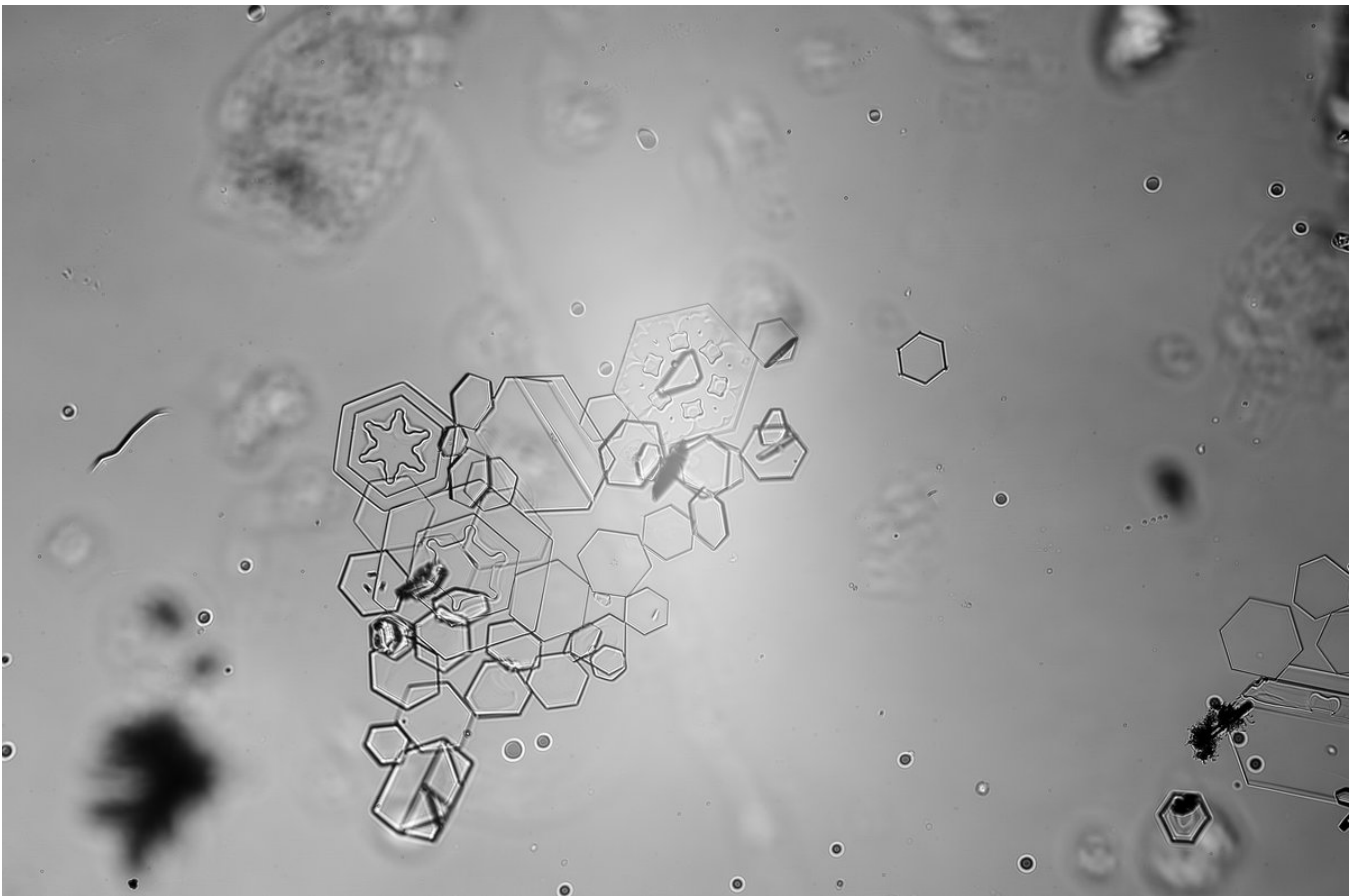
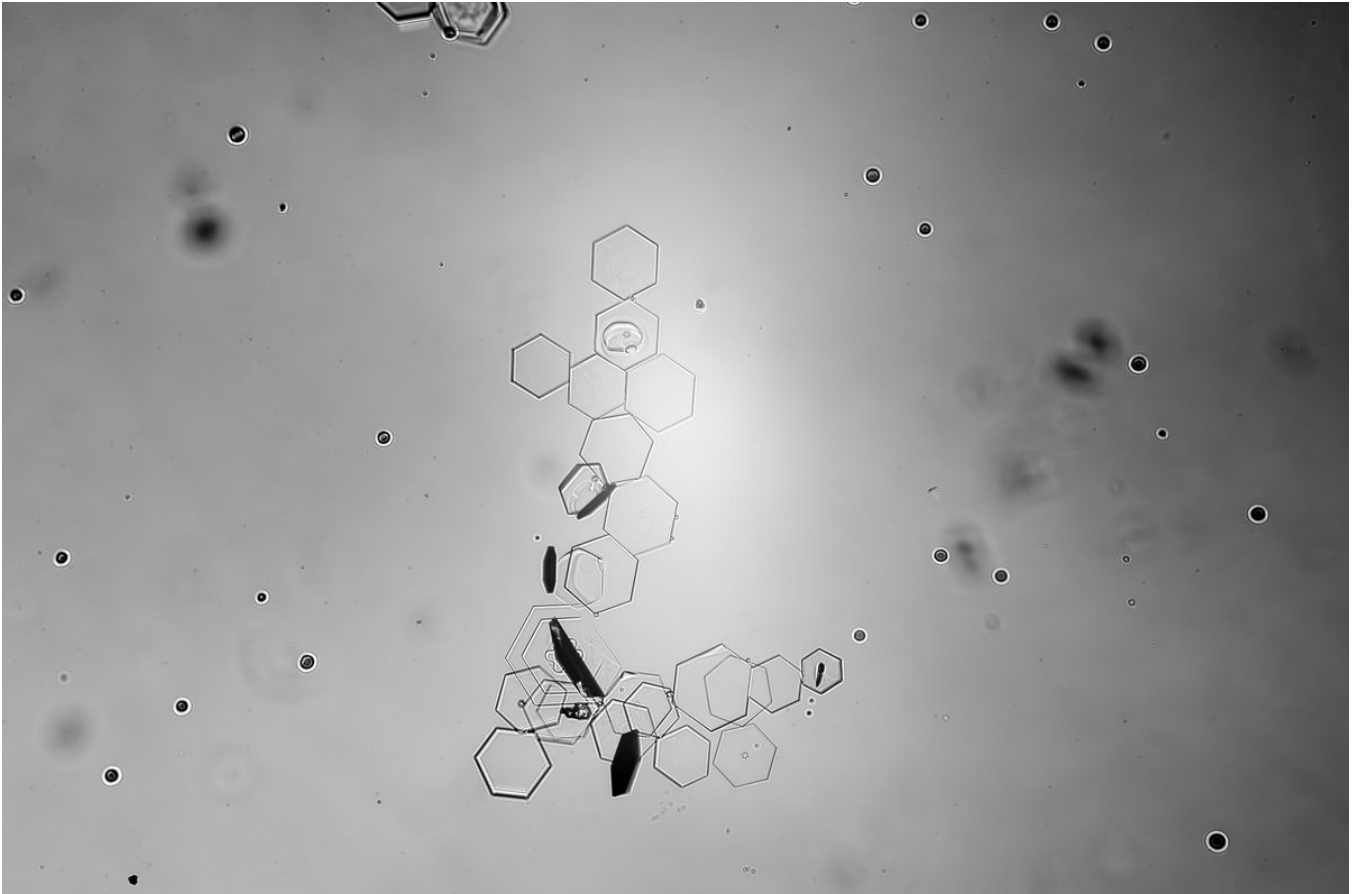


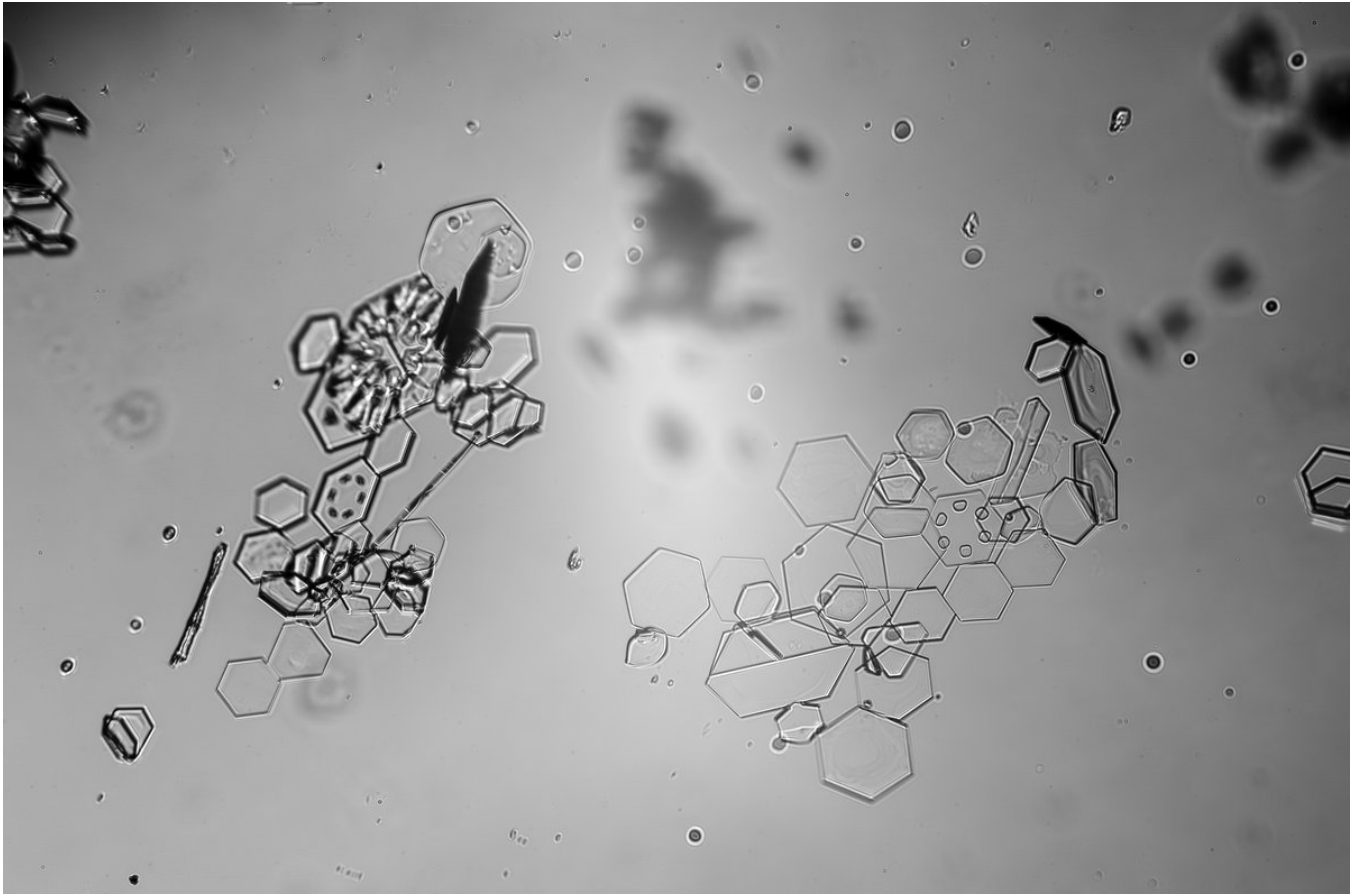


7/32 - 2026-02-26 14:33 UTC

22/23 Feb 2026, Rovaniemi. I was surprised to get this much crystals from the emaciated swarm.

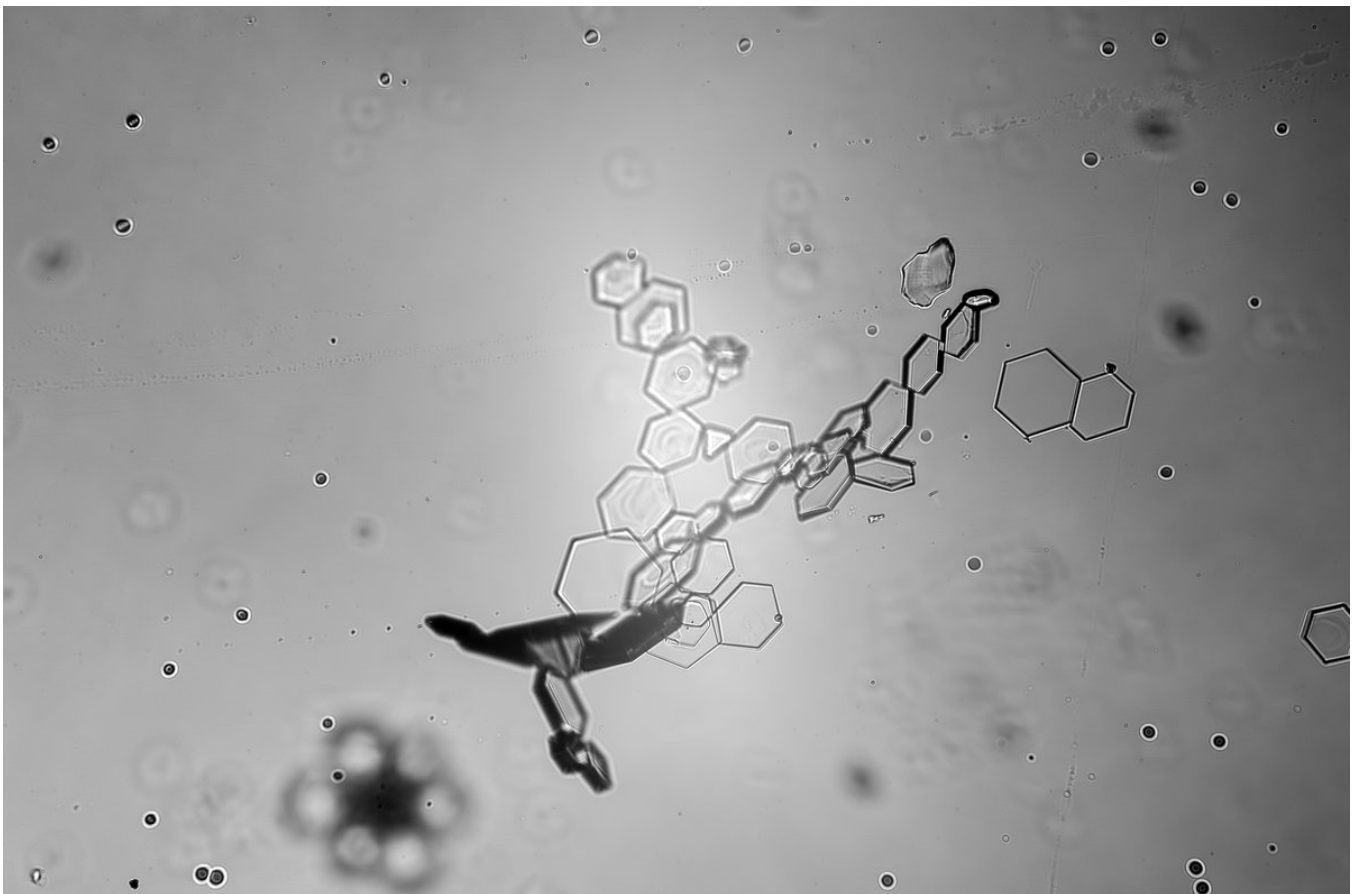


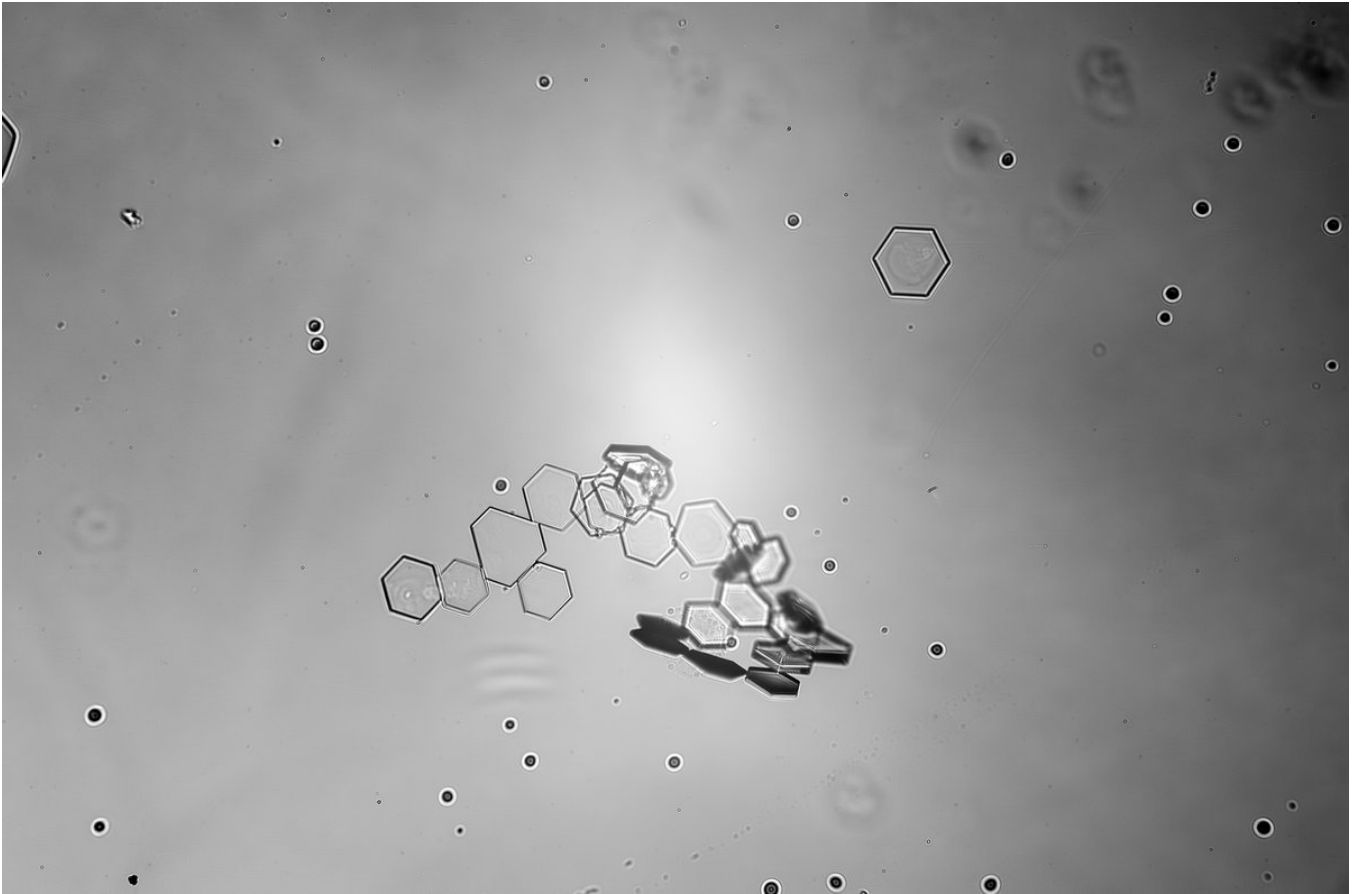


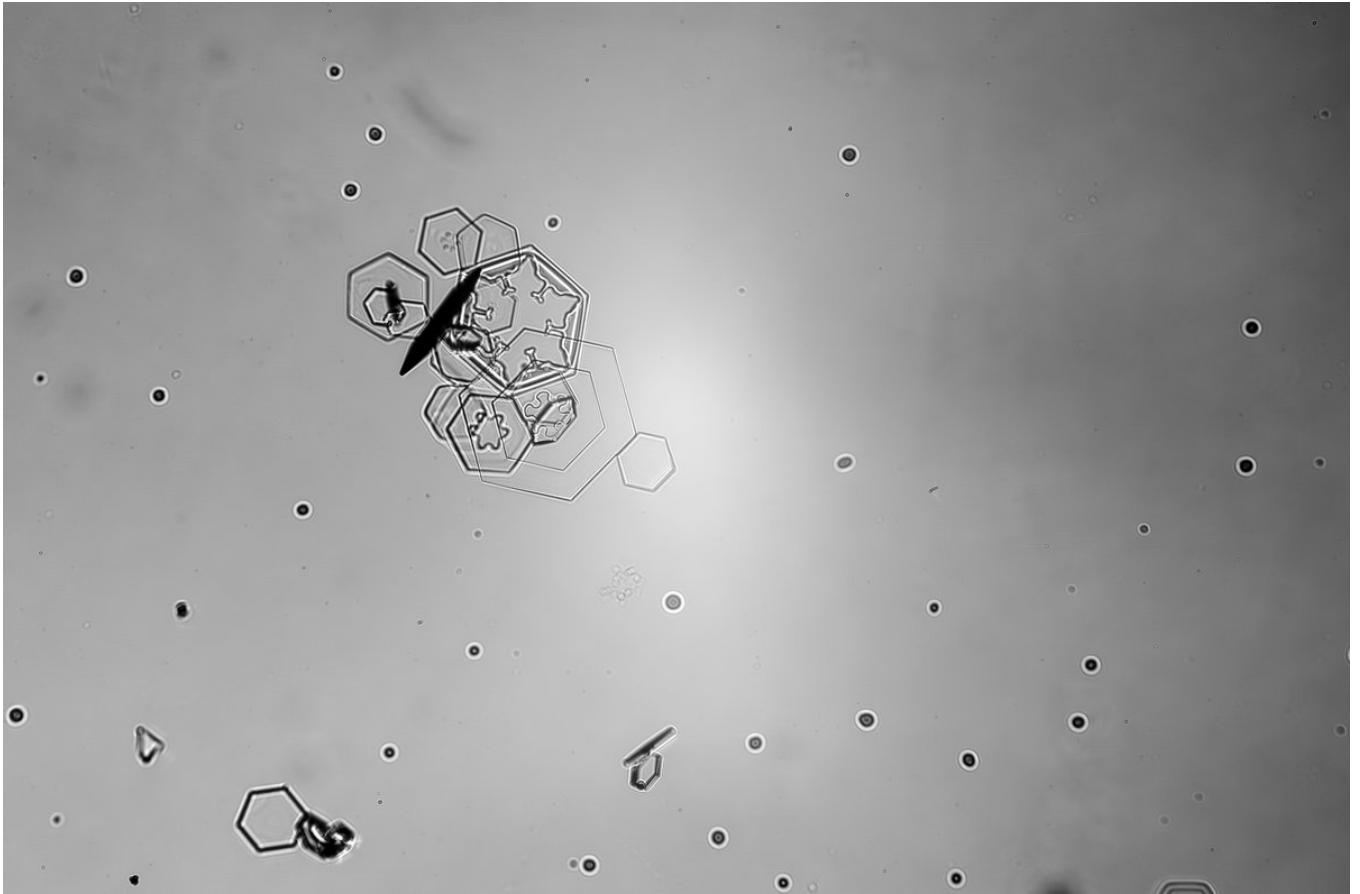


8/32 - 2026-02-26 14:52 UTC

22/23 Feb 2026, Rovaniemi. More crystals of the 46° contact arc involving stage.

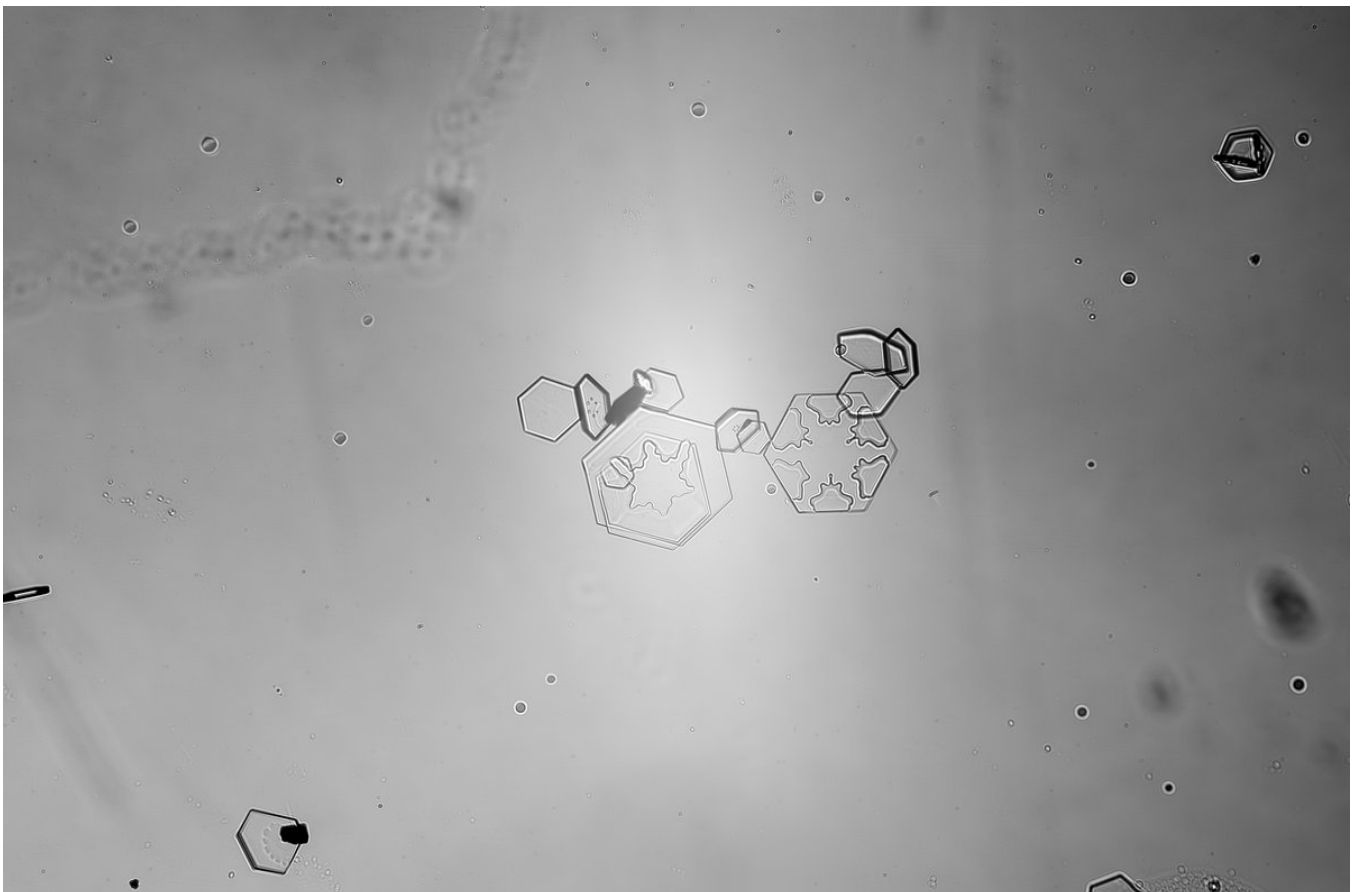


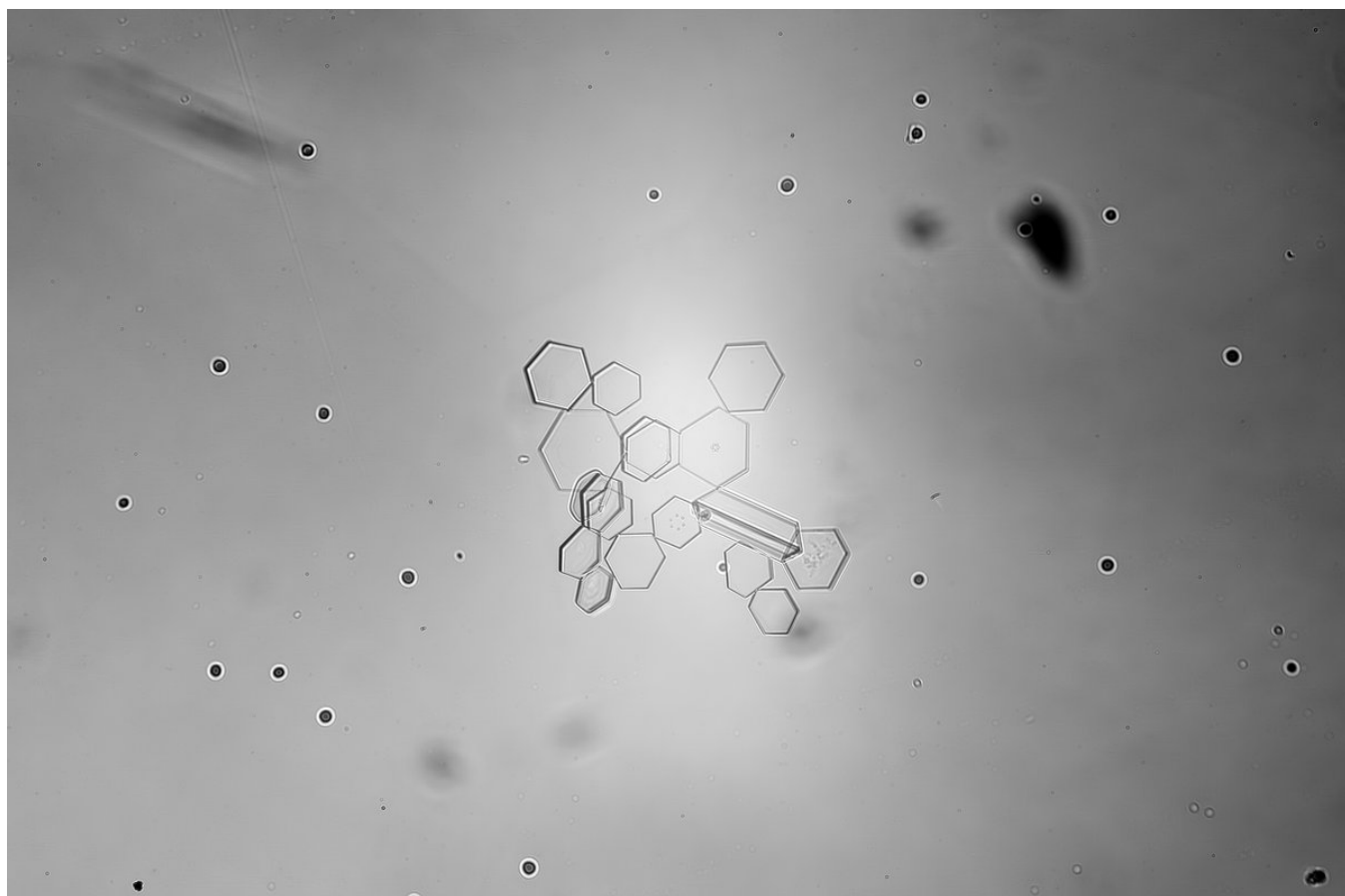
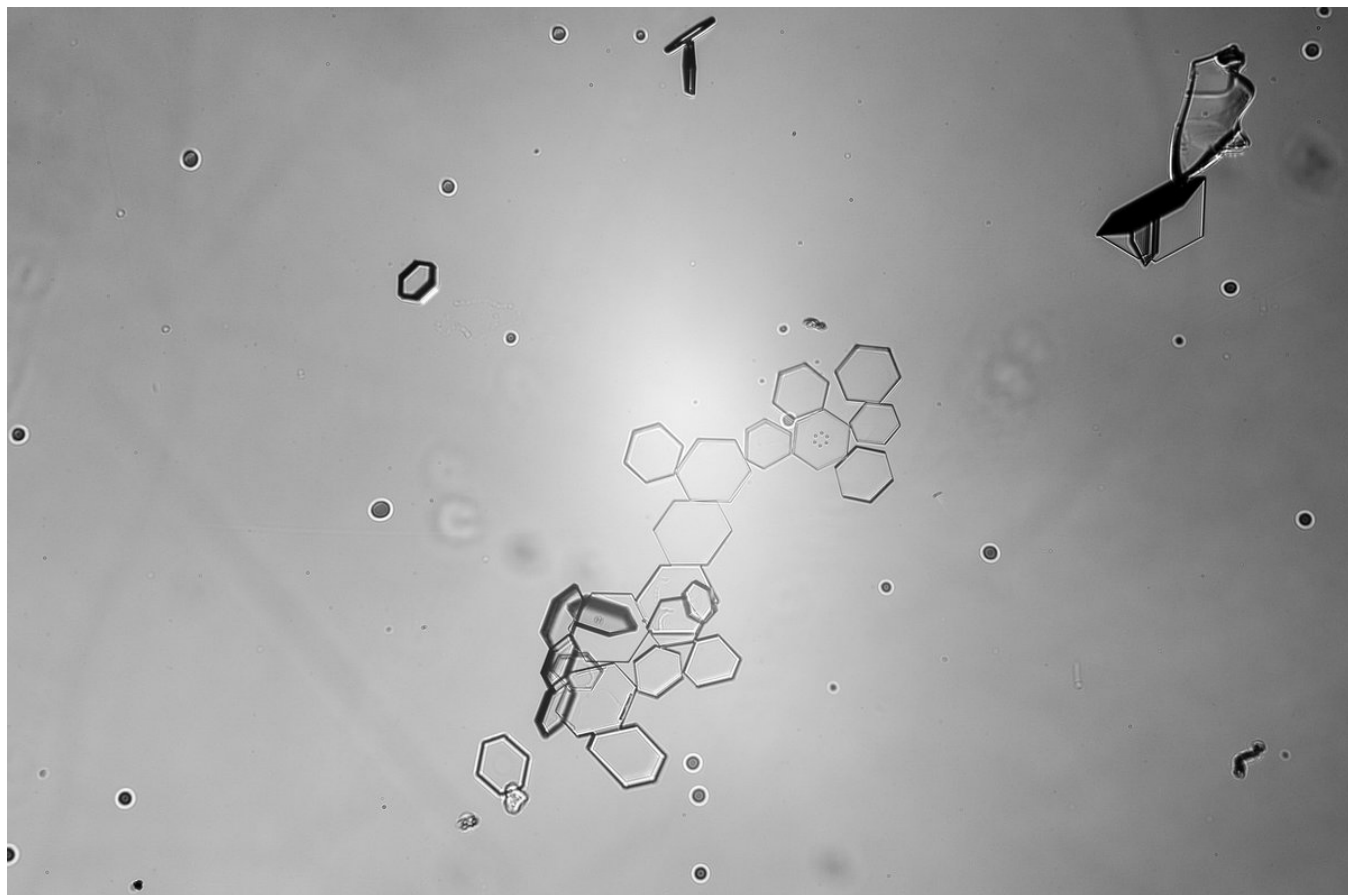


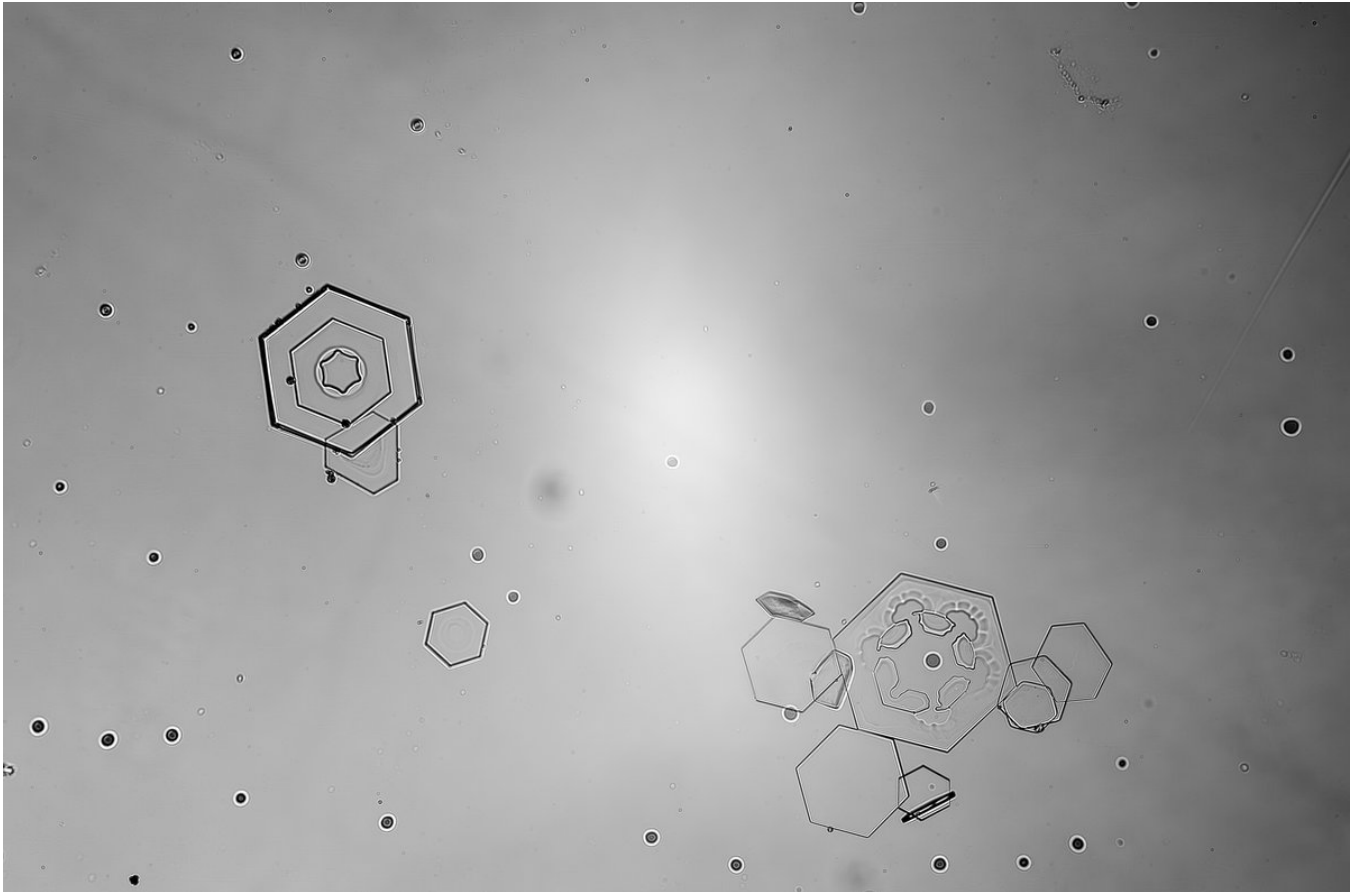


9/32 - 2026-02-26 14:53 UTC

22/23 Feb 2026, Rovaniemi. Even more crystals of the 46° contact arc involving stage.

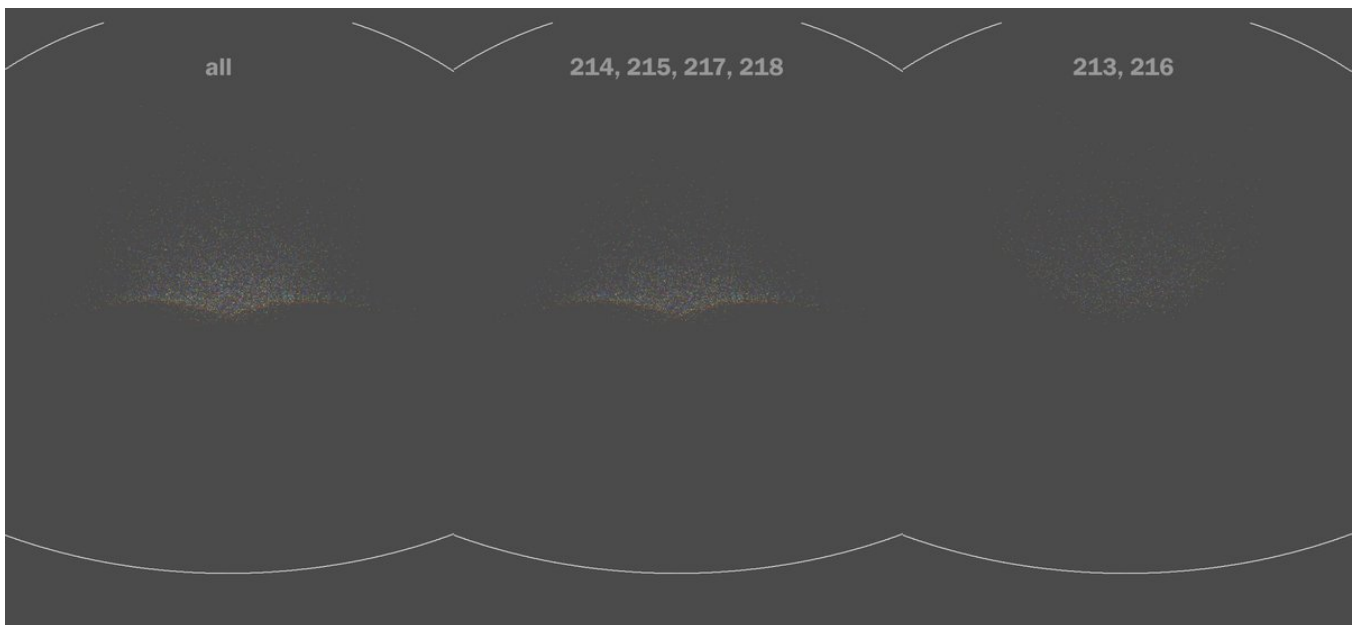






10/32 - 2026-02-26 14:59 UTC

22/23 Feb 2026, Rovaniemi. The "flip" 46° contact arc dissected to its two components. In the display the feature looks rather like solar vertical centered 213–216 arc, but it is not possible to enhance it even by using highly tabular crystals – the "side arcs" always win.



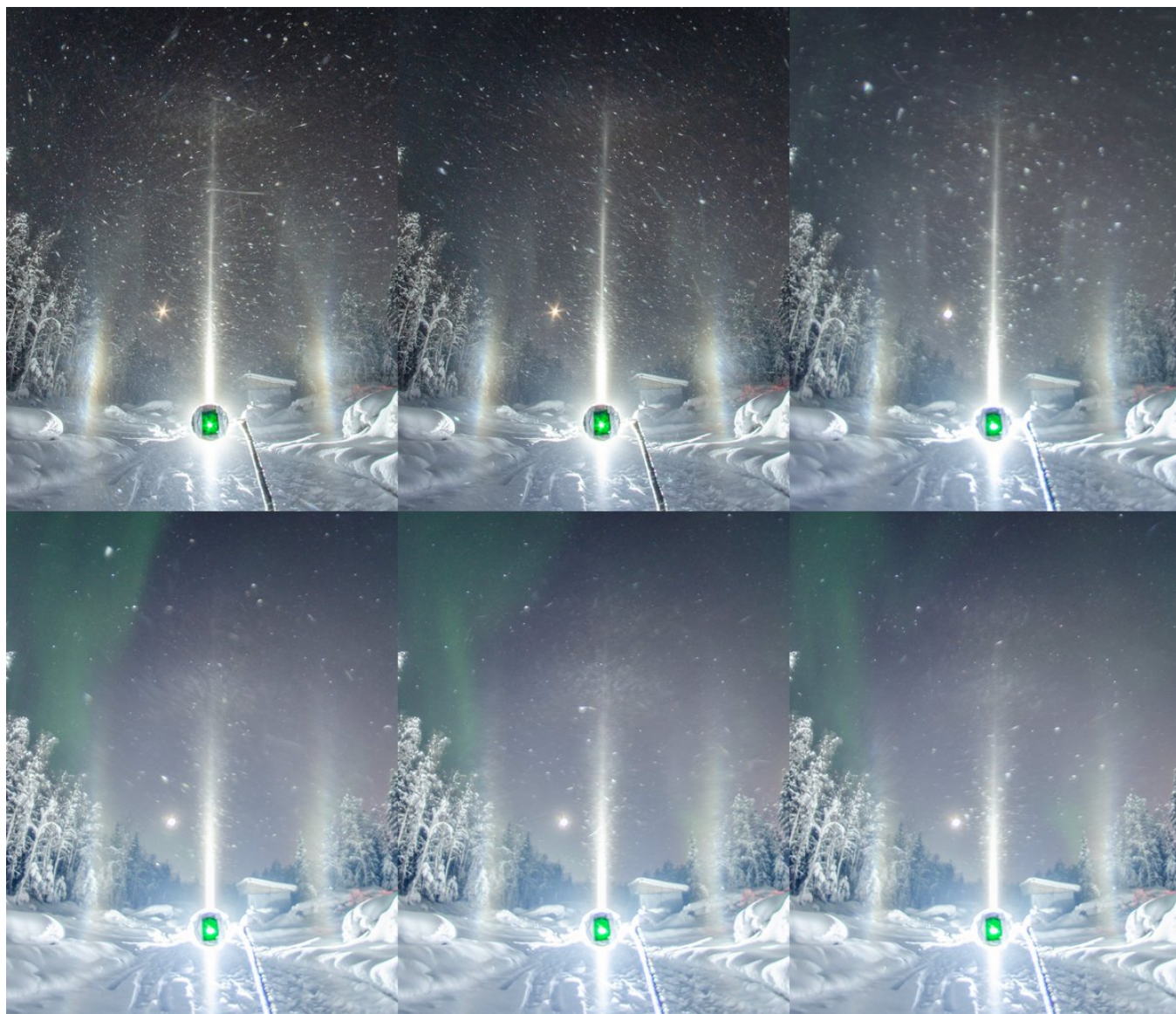
11/32 - 2026-02-27 12:58 UTC

22/23 Feb 2026, Rovaniemi. Comparing the "flip-cna – no 46° contact arc" and "flip-cna – yes 46° contact arc" stages, upper and lower row respectively. In the former stage flip-parhelia are more

prominent and Liljequist arcs feel like a separate feature. In the latter stage the Liljequist arc (it's really just one arc now) extend smoothly the flip-parhelia – if there is flip-parhelia to talk about at all.

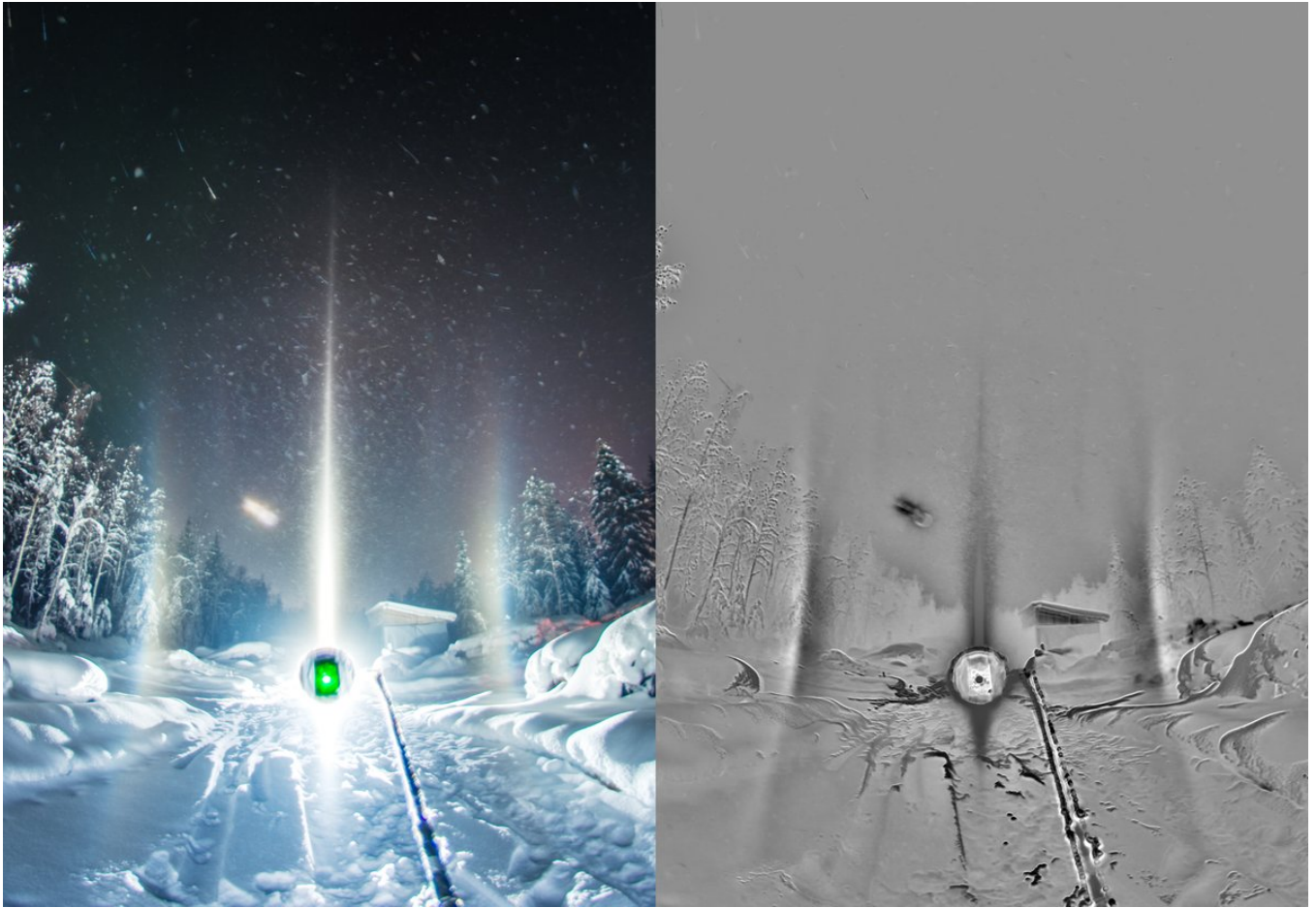
Needless to say, this is quite confusing from the perspective of simulations.

I used HID and it's higher flood component has caused also clear "superparhelia" (divergent flip-parhelia) in the upper row frames.



12/32 - 2026-02-27 13:29 UTC

22/23 Feb 2026, Rovaniemi. All frames ave stacked (31 x 30s).



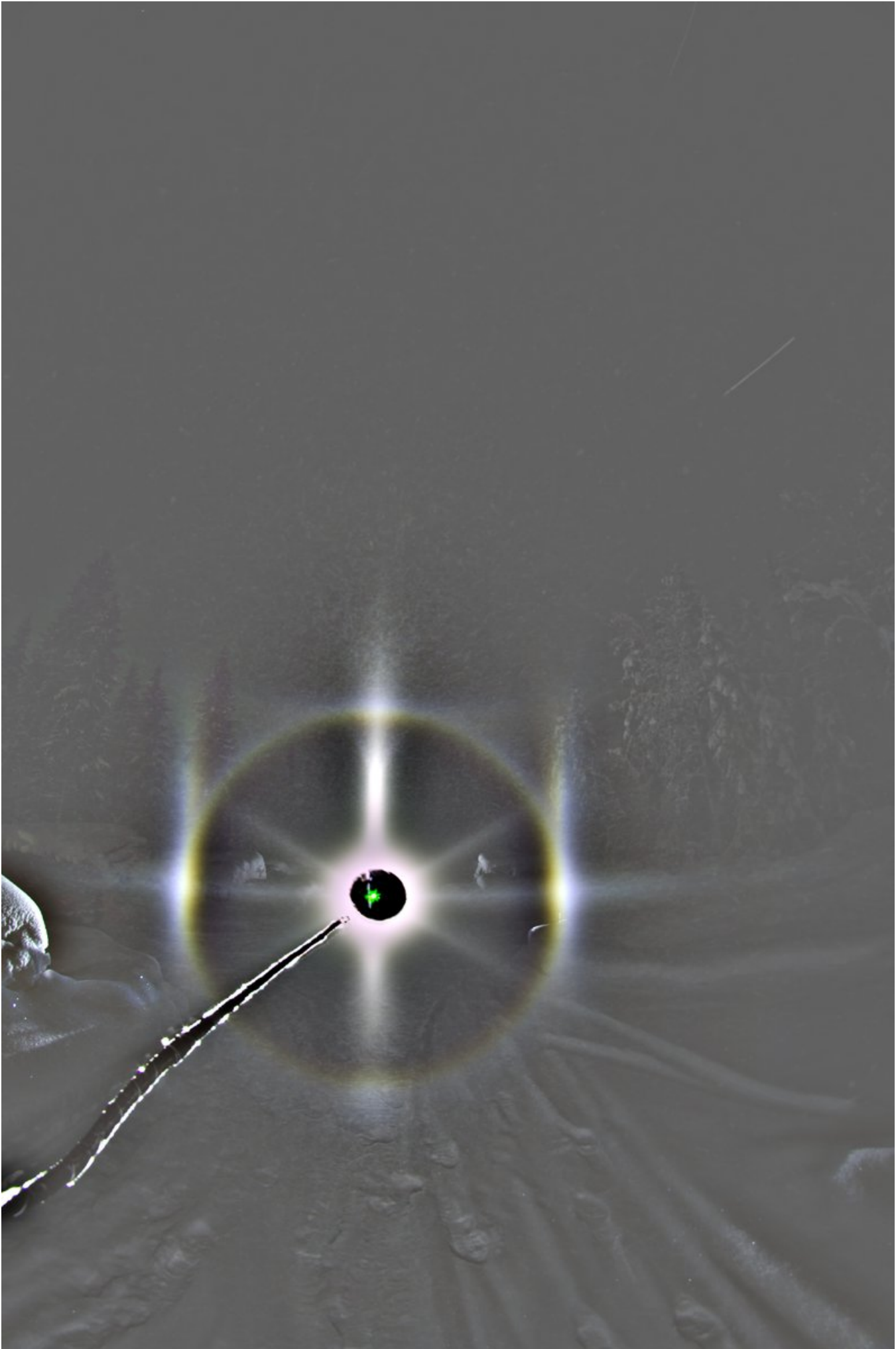
13/32 - 2026-02-27 13:46 UTC

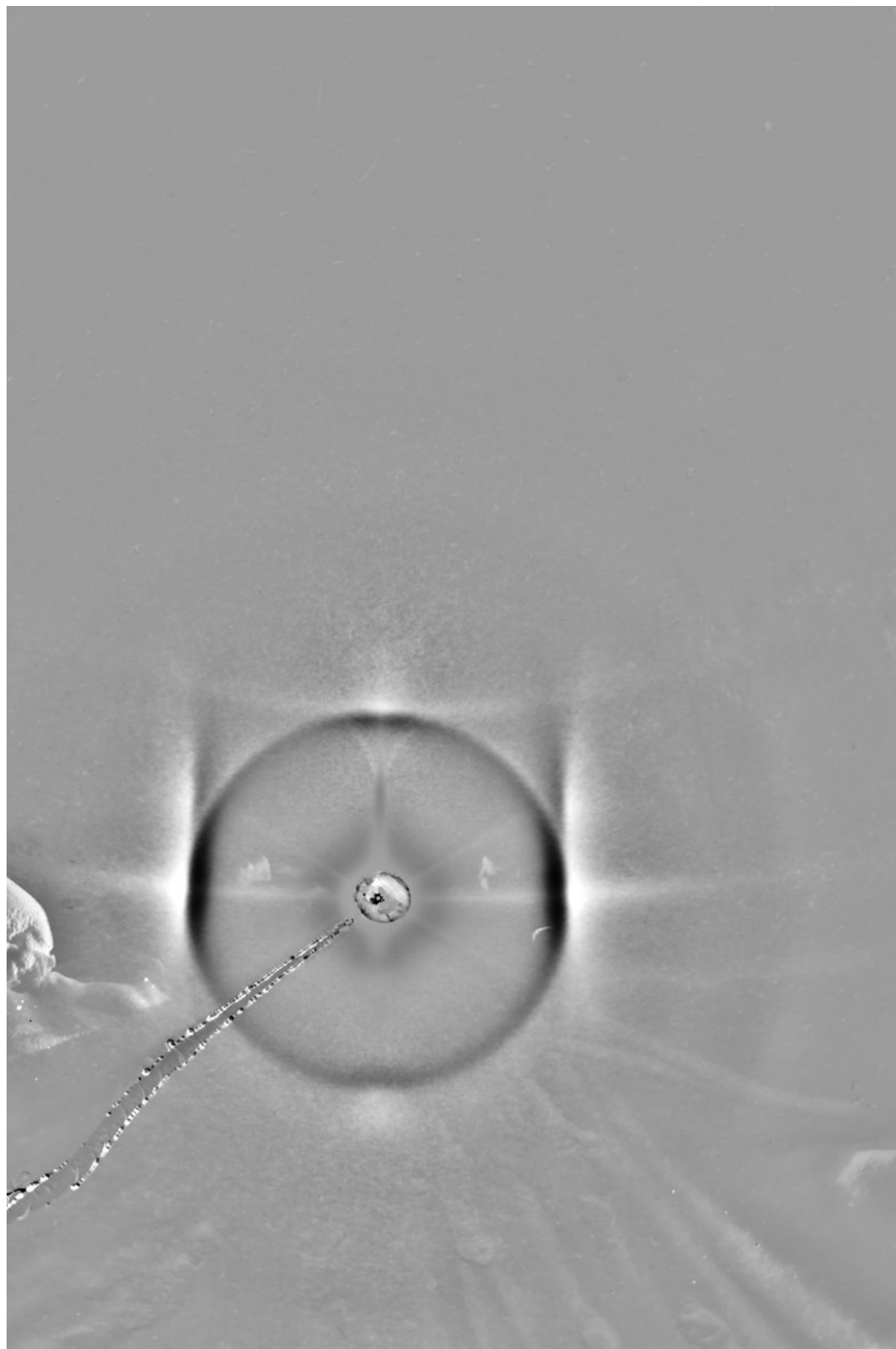
22/23 Feb 2026, Rovaniemi. An early stage 8f and late stage 7f b-r stacks flashed.

[video: 2027379763023663561-HCKy-ljXwAAaalc.mp4]

14/32 - 2026-02-28 10:32 UTC

22/23 Feb 2026, Rovaniemi. The first stage, 91 x 25s ave. It is quite different to second stage of the above posts with the 46° contact arc. Can't simulate this – Hastings/Wegener is too well defined to account for lacking of V-shaped Parry/tanarc. I have also the crystals.





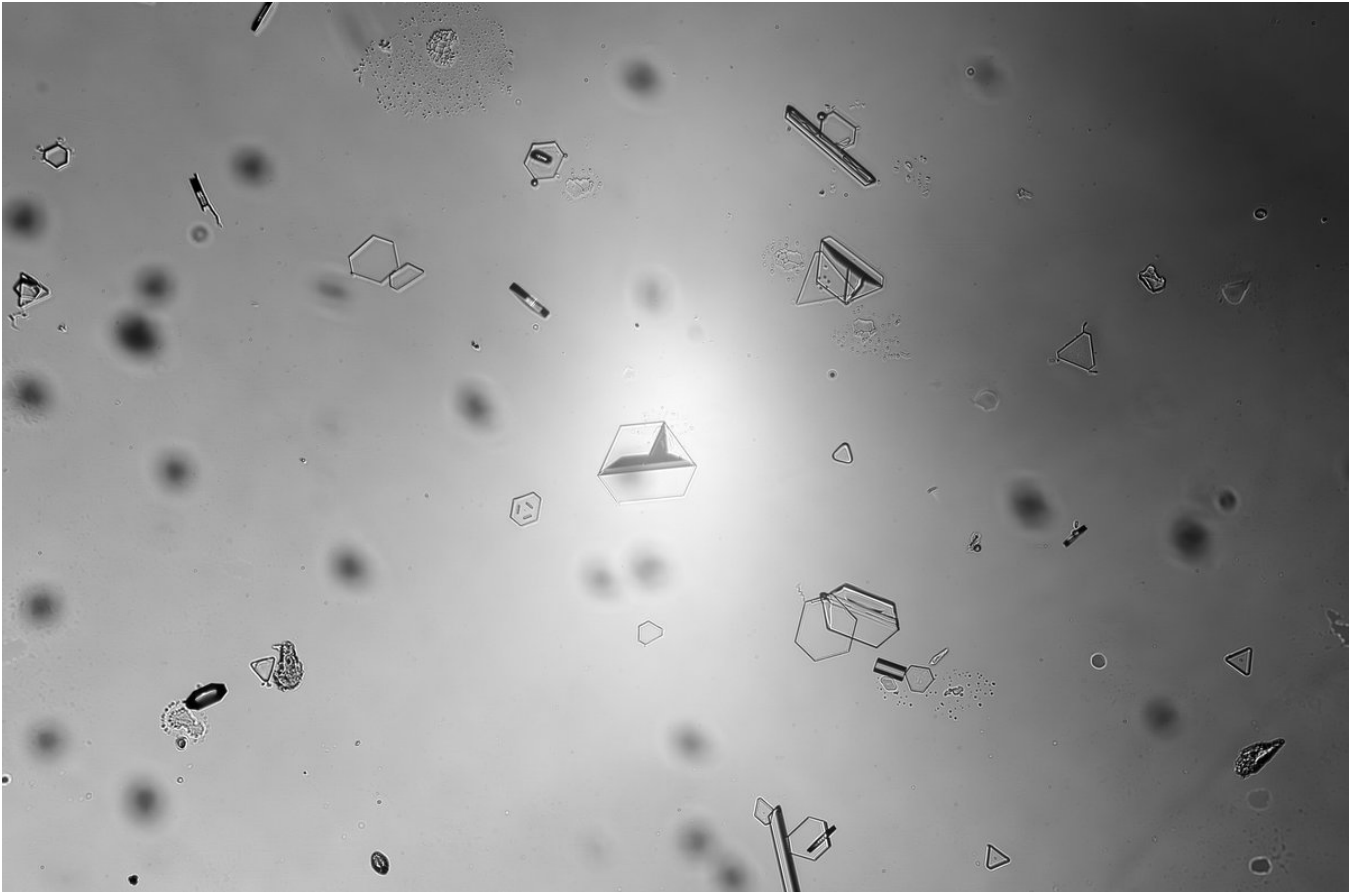


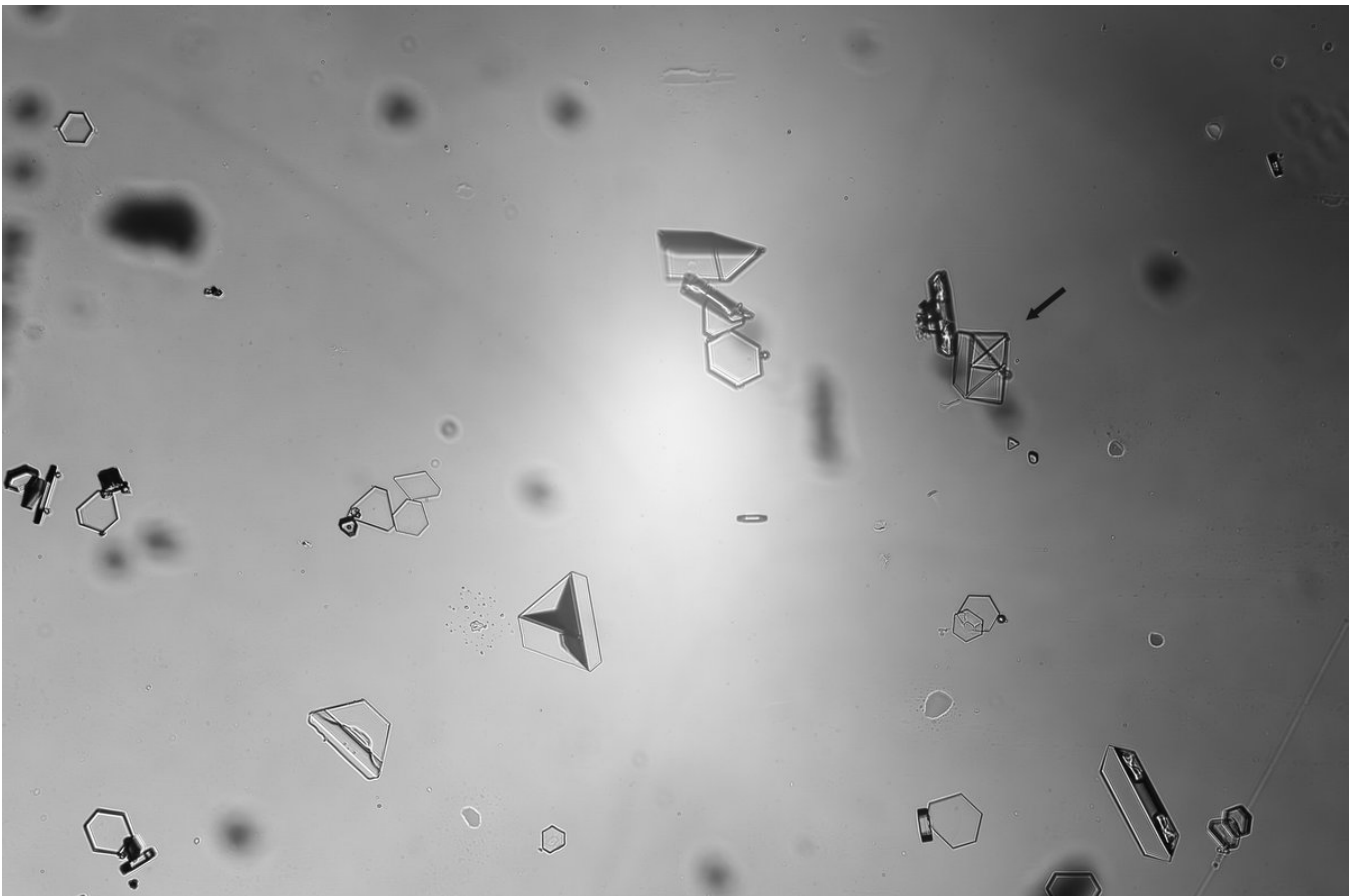
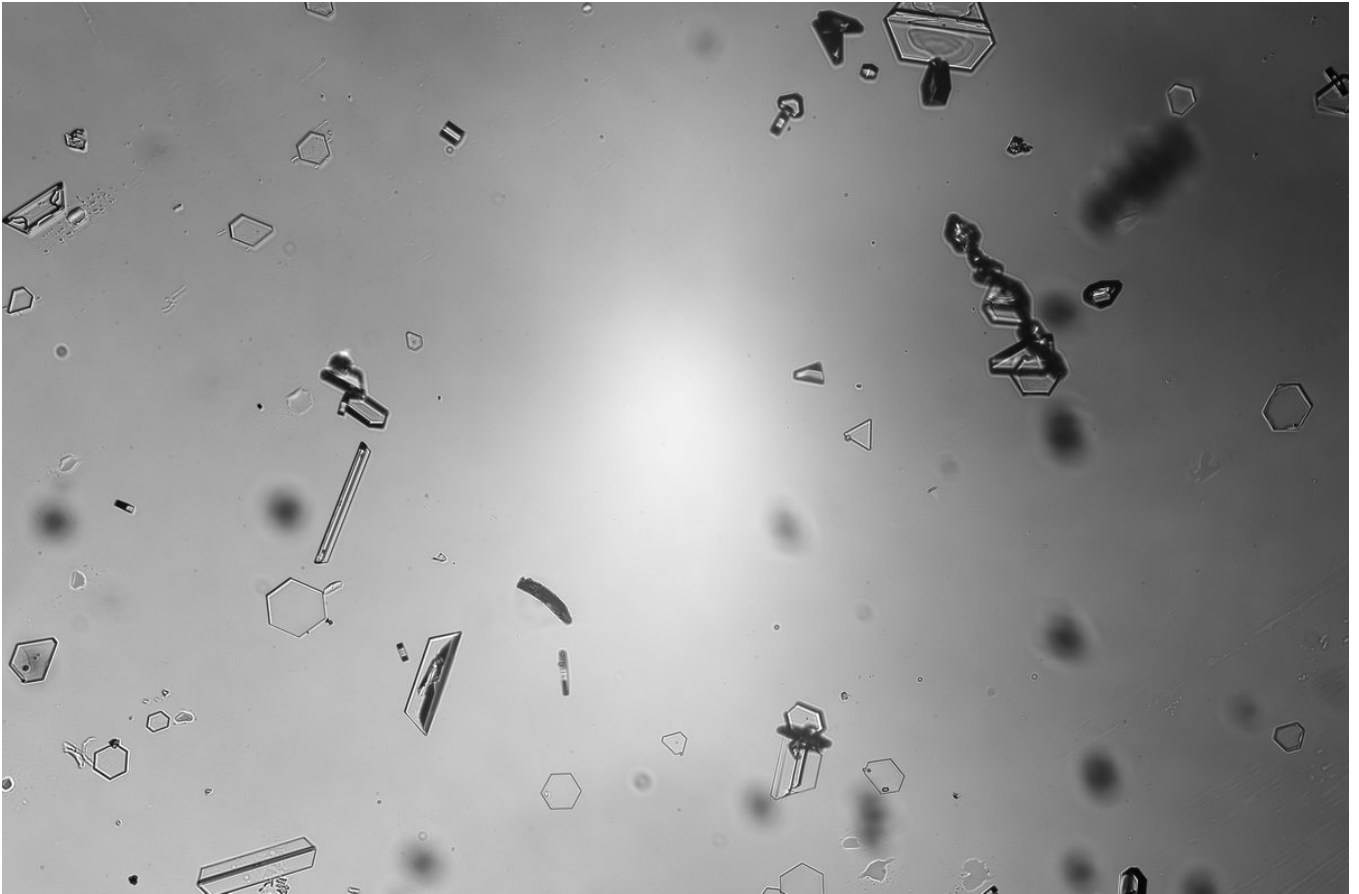
15/32 - 2026-02-28 11:12 UTC

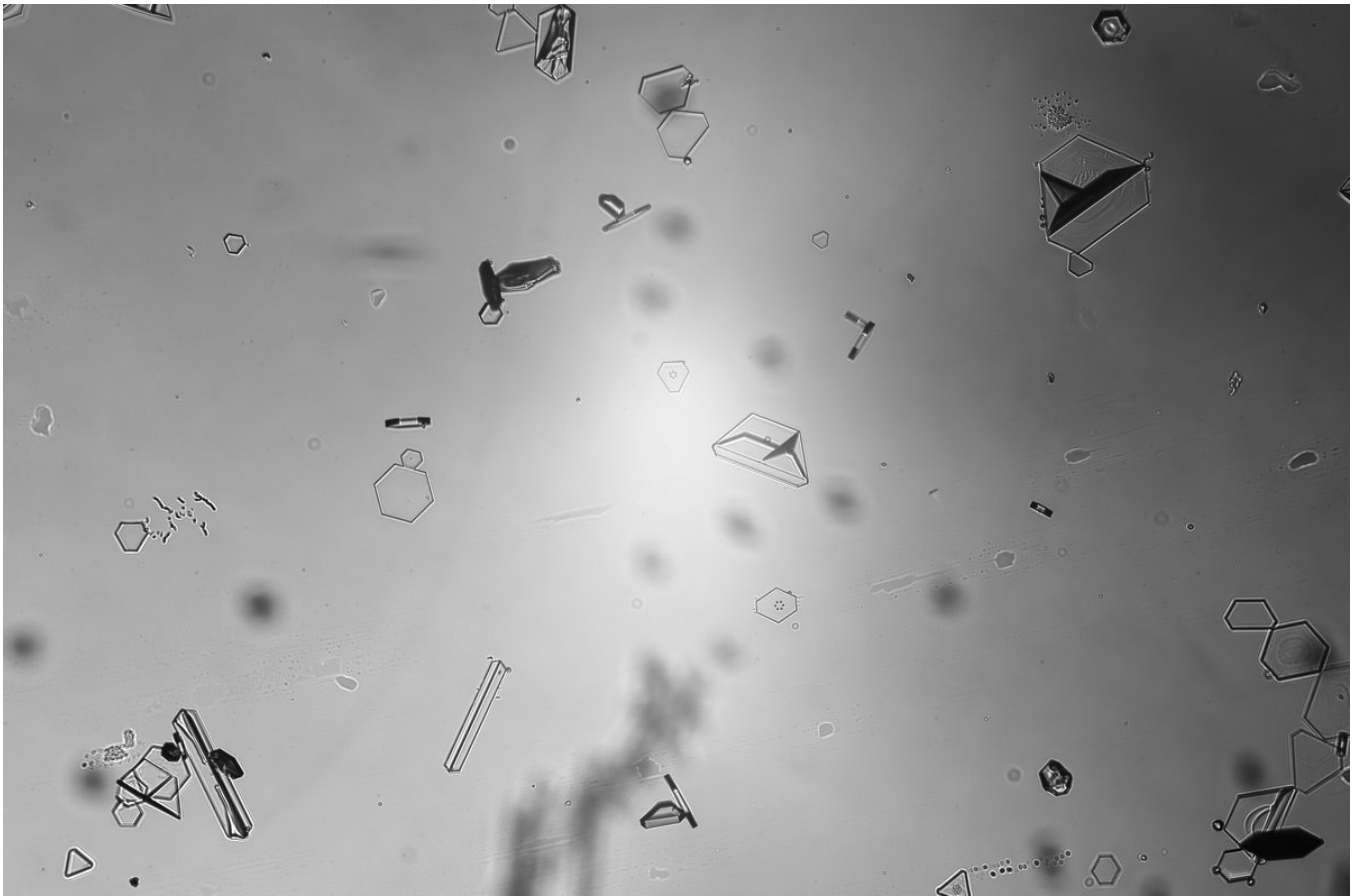
22/23 Feb 2026, Rovaniemi. 1st stage crystals. An abundance of the usual elongated Moilanen suspects, an abundance of less elongated triple crossed plates. There are also perfectly normal looking columns. So why we do not have V-shaped tanarc/Parry? Why the helic arc is so strong?

As to the triple crossed plates, they do not seem to be consistently parallel to a-axis in each (see TapeMoilanen page 184).

I can't wrap my head around the arrow-denoted crystal of the third image. It is not the first such crystal seen in the samples. Is there more than just three plates crossed?







16/32 - 2026-02-28 11:31 UTC

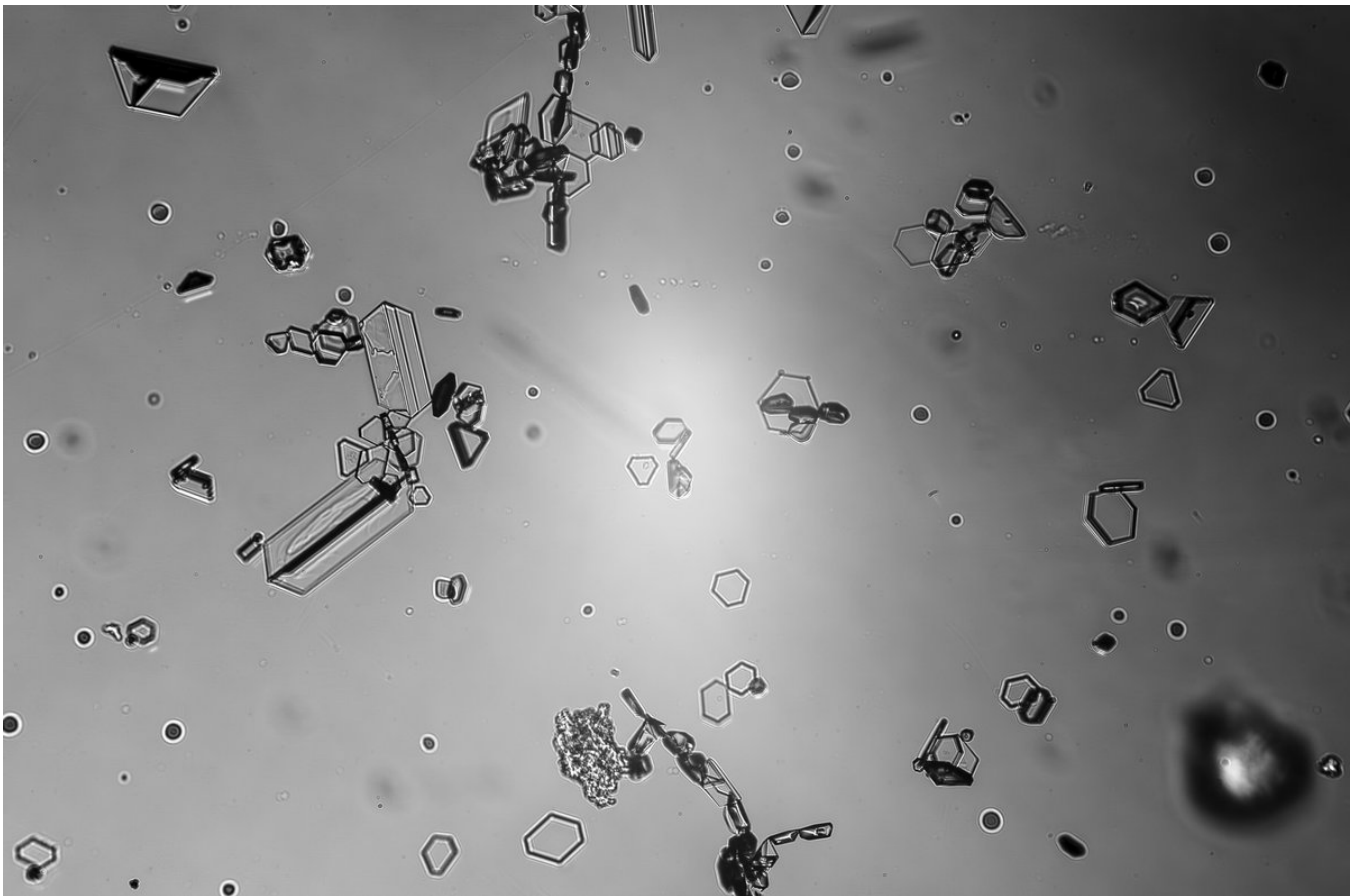
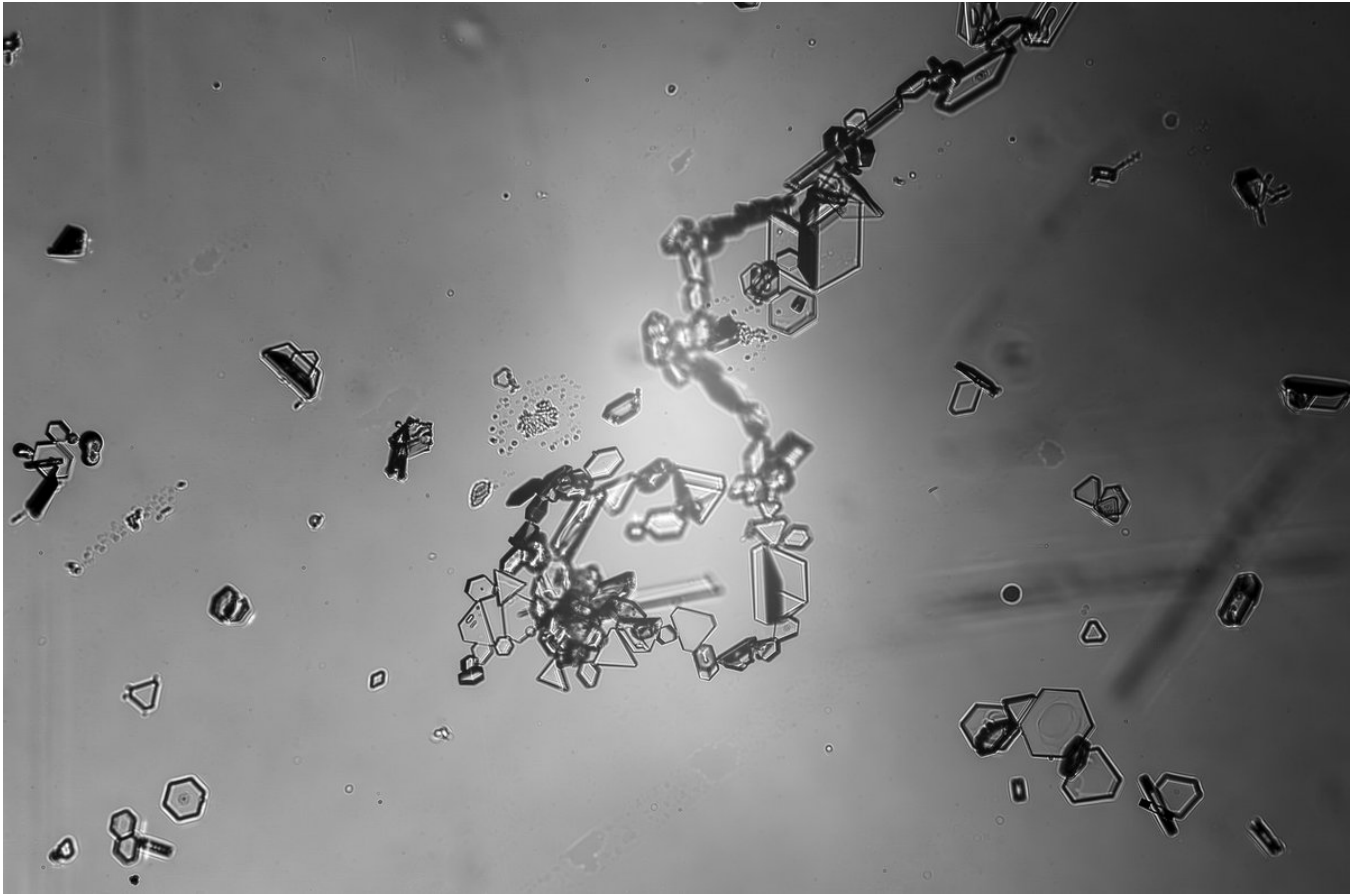
22/23 Feb 2026, Rovaniemi. 1st stage crystals. Hasn't the Moilanen arc already mocked us enough? All these decades it's been glittering in front of our noses and we still don't know how it is formed. Moilanen arc will never be solved by taking more light microscope photos. The resolution is not sufficient. We need ESEM. In Fairbanks Moilanen arcs are abundant in the power plant plumes and there is also ESEM. Someone needs to apply for a grant for a winter in Fairbanks to take those ESEM photos. Barring an ingenious theoretical breakthrough on crystal growth, it's our best chance at cracking the Moilanen mystery.

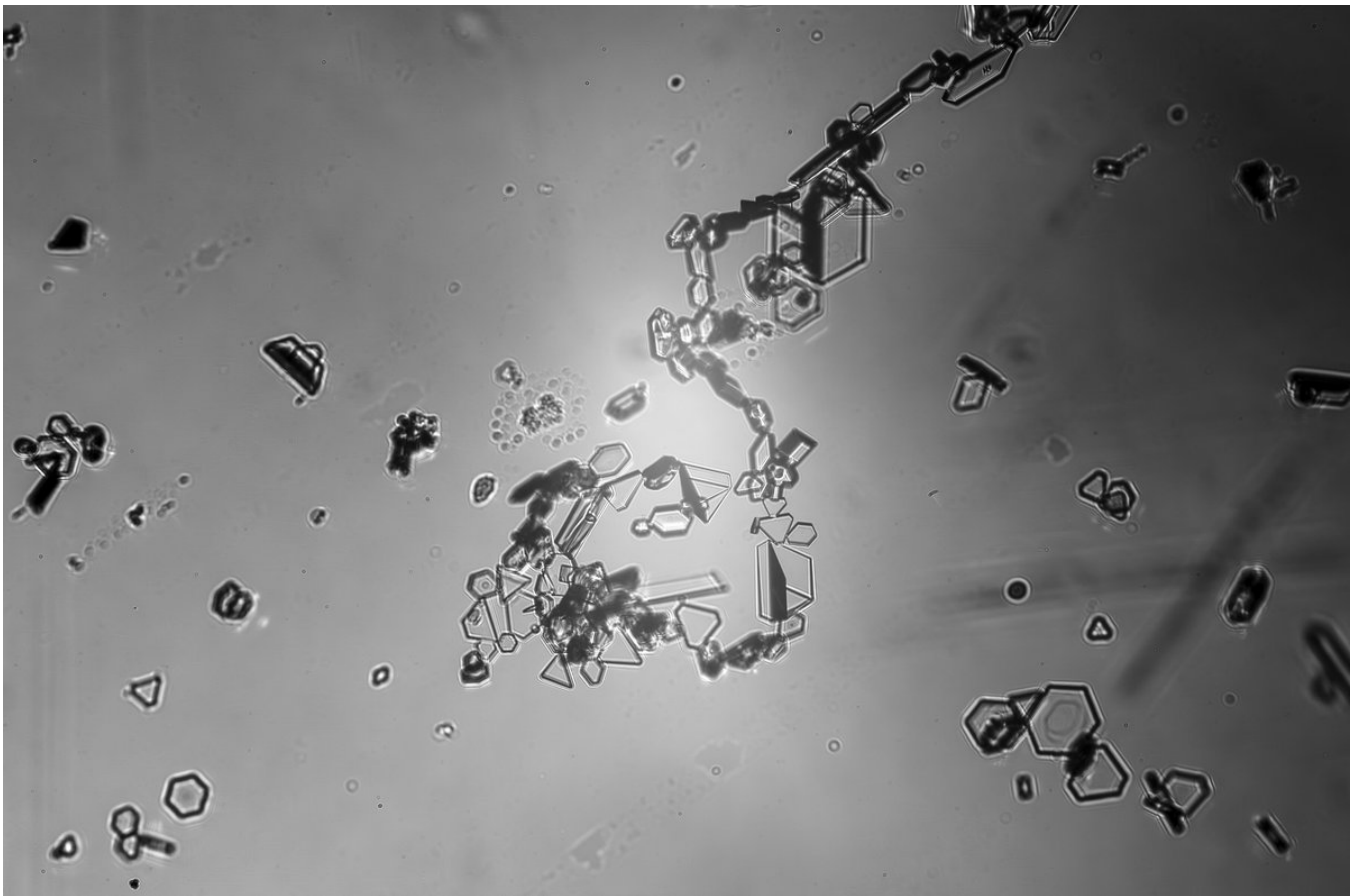
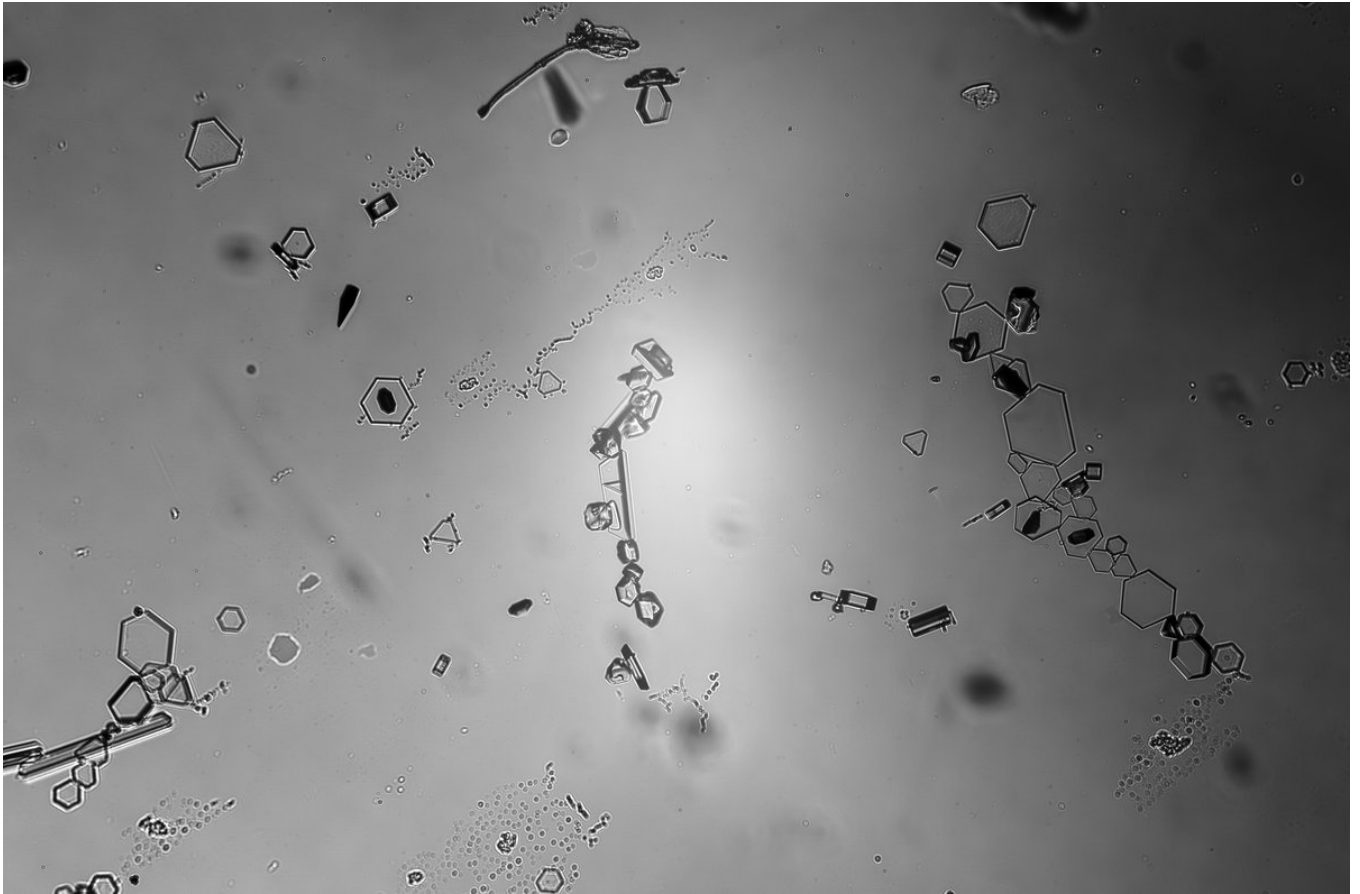




17/32 - 2026-02-28 11:41 UTC

22/23 Feb 2026, Rovaniemi. More 1st stage crystals.

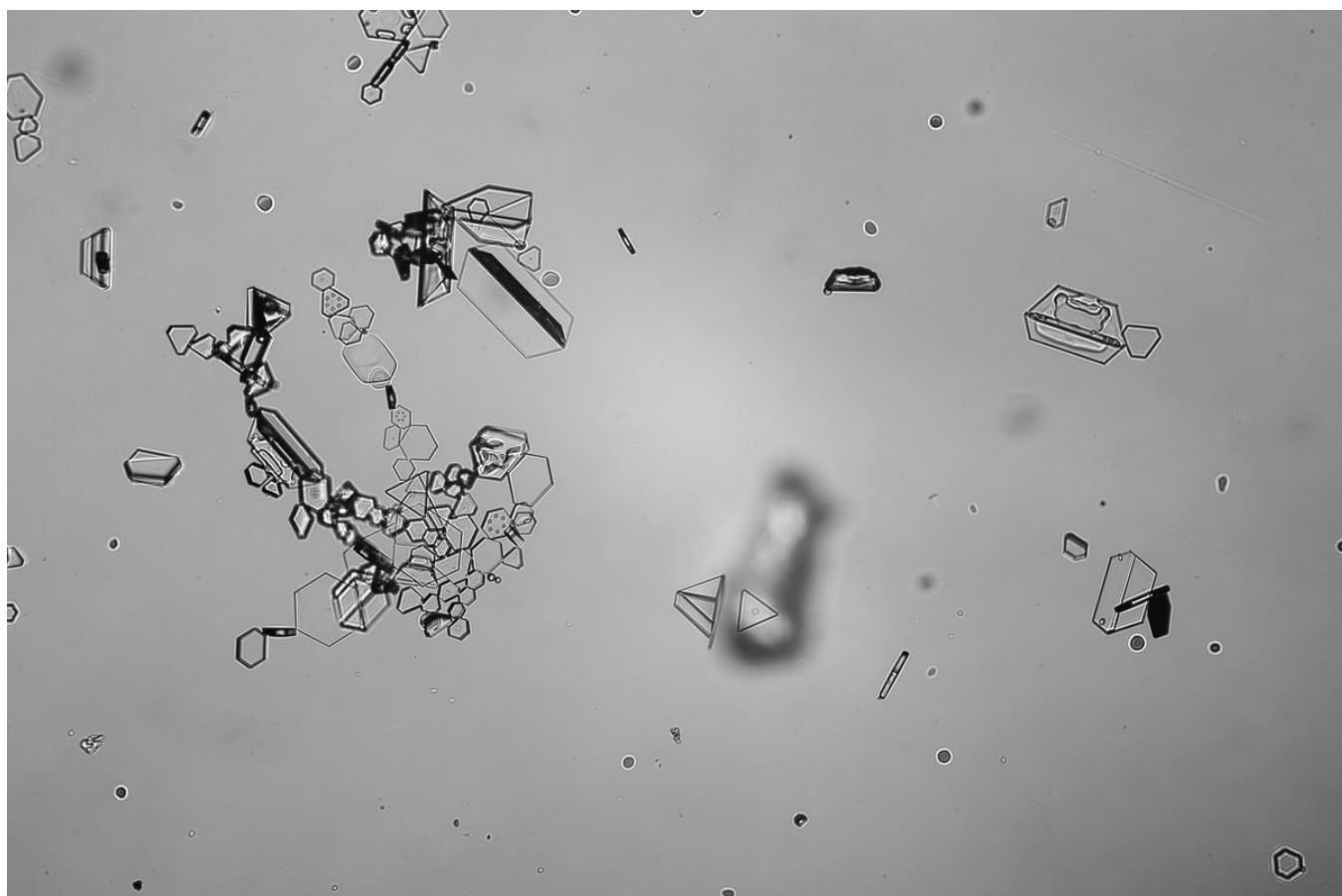
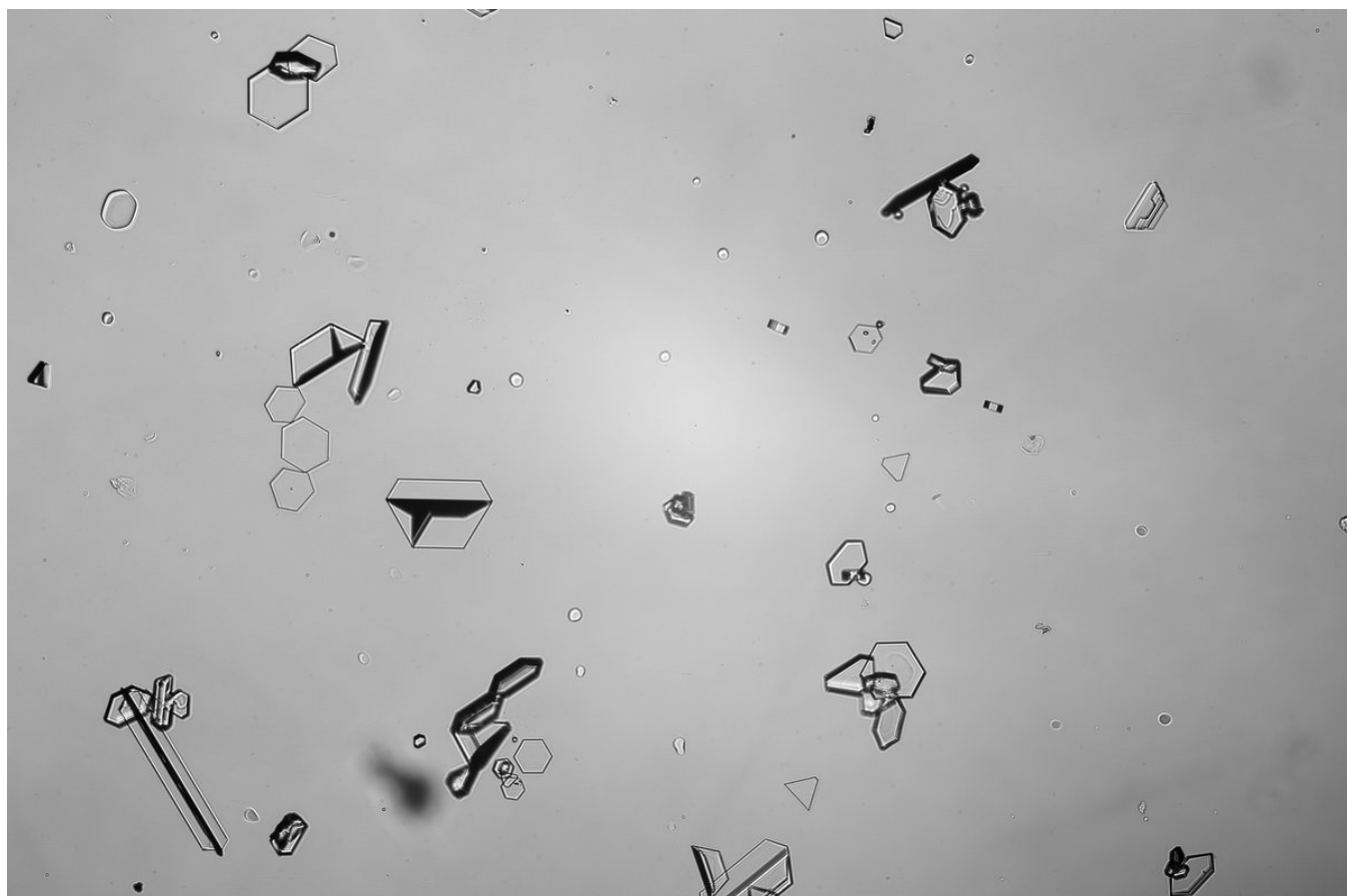


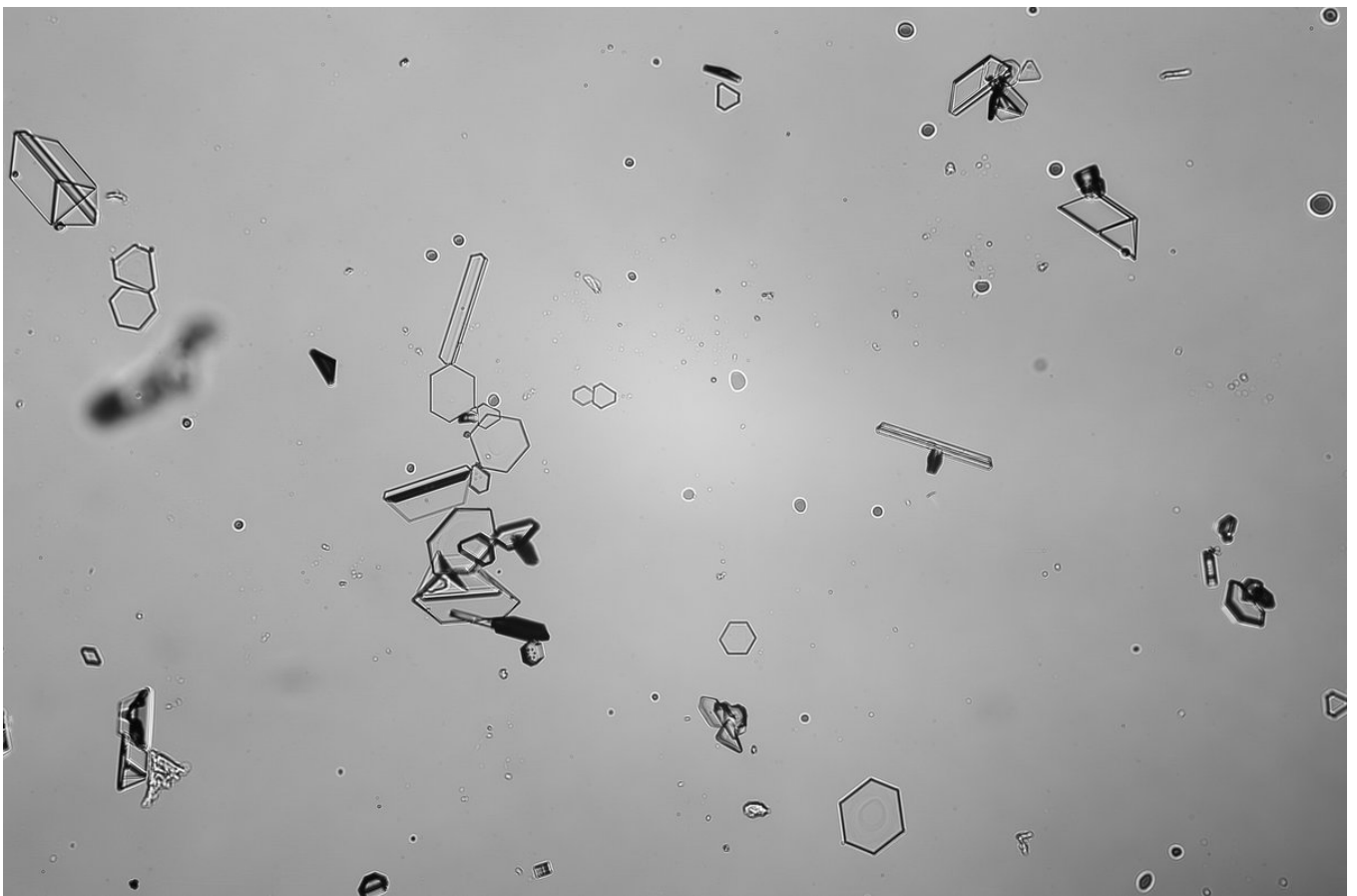


18/32 - 2026-02-28 11:51 UTC

22/23 Feb 2026, Rovaniemi. More 1st stage crystals. I have gotten sharper crystal photos now that I

discovered that you can magnify the live view in D7000.

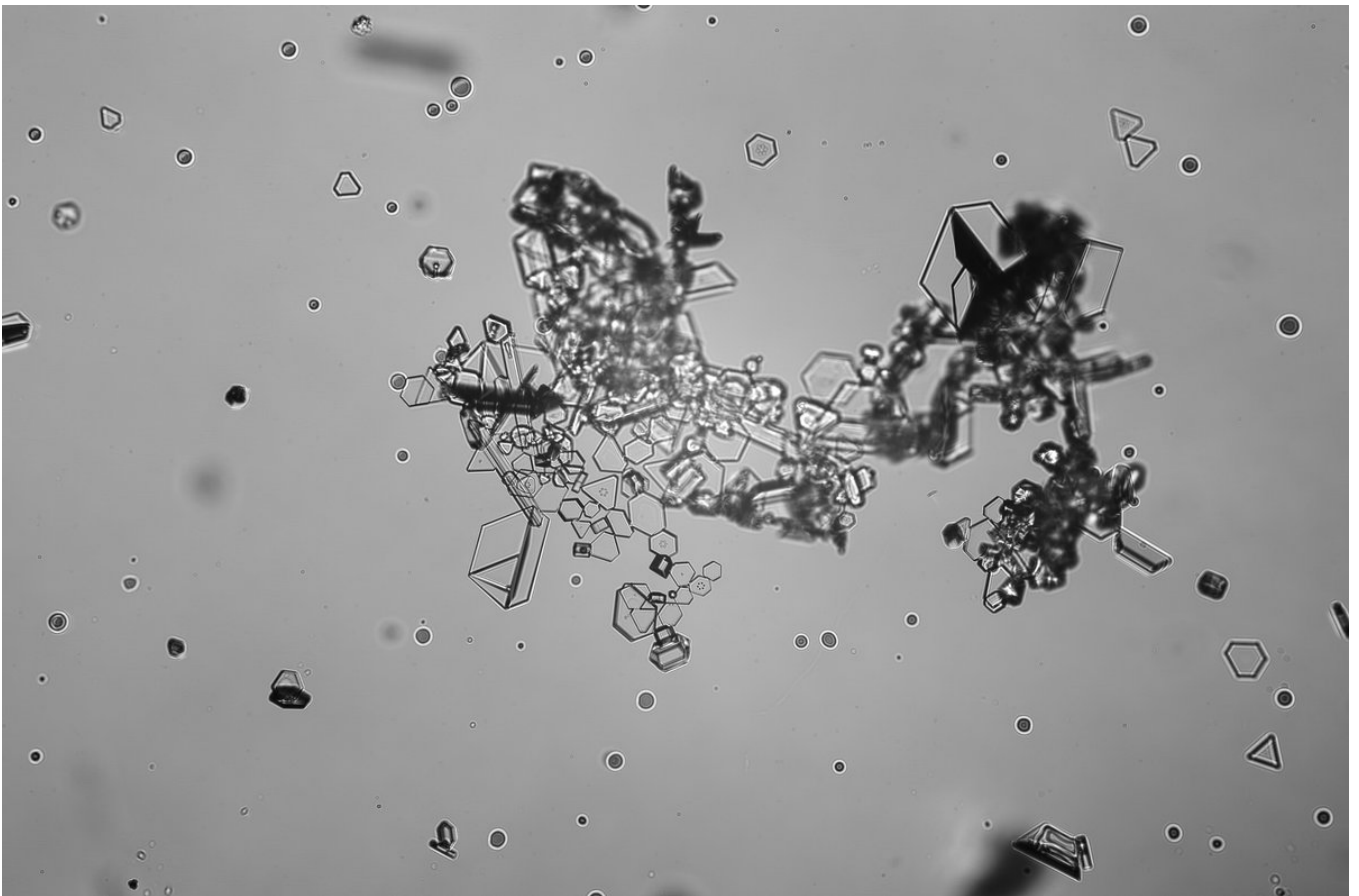
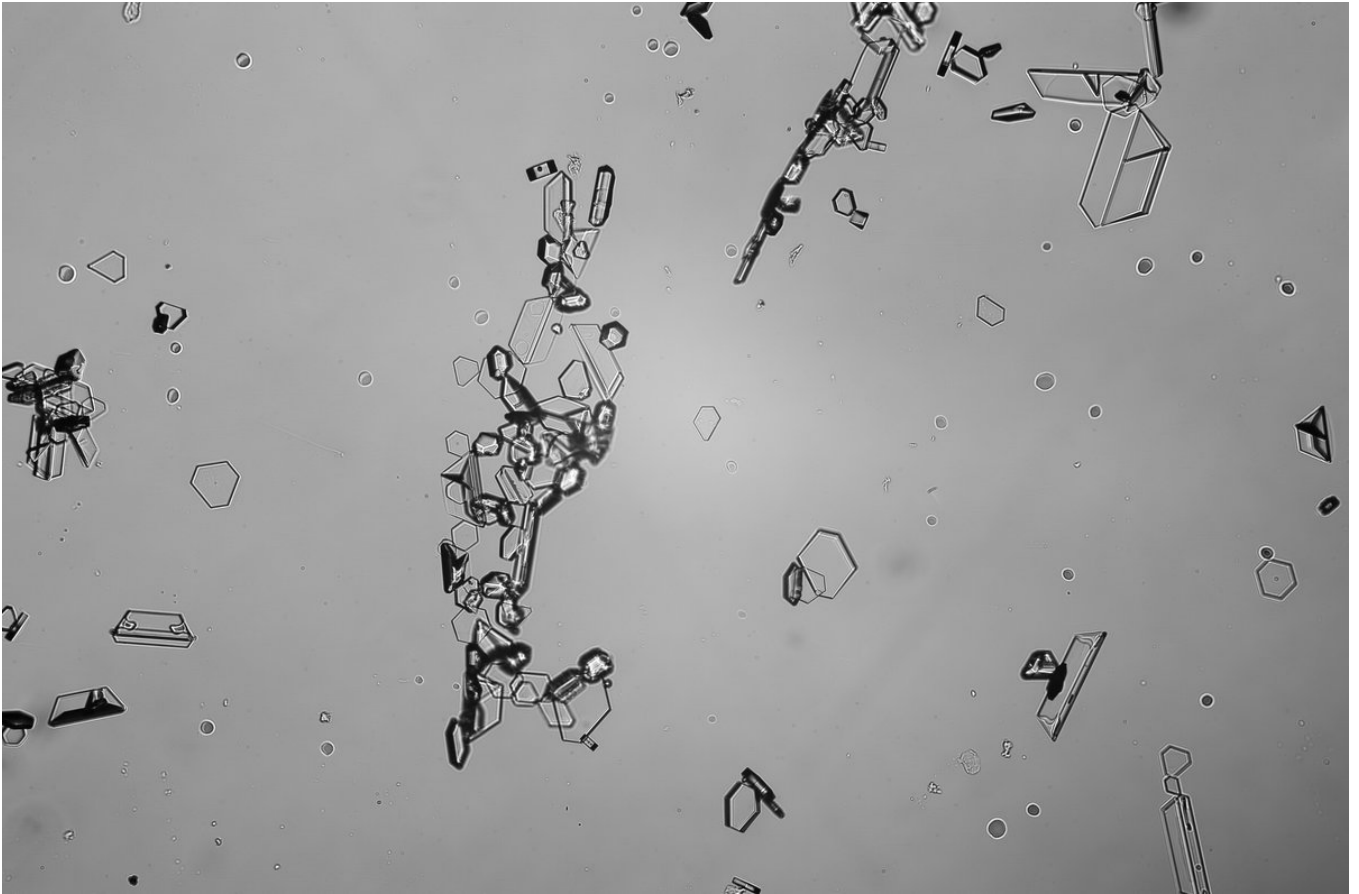




19/32 - 2026-02-28 11:55 UTC

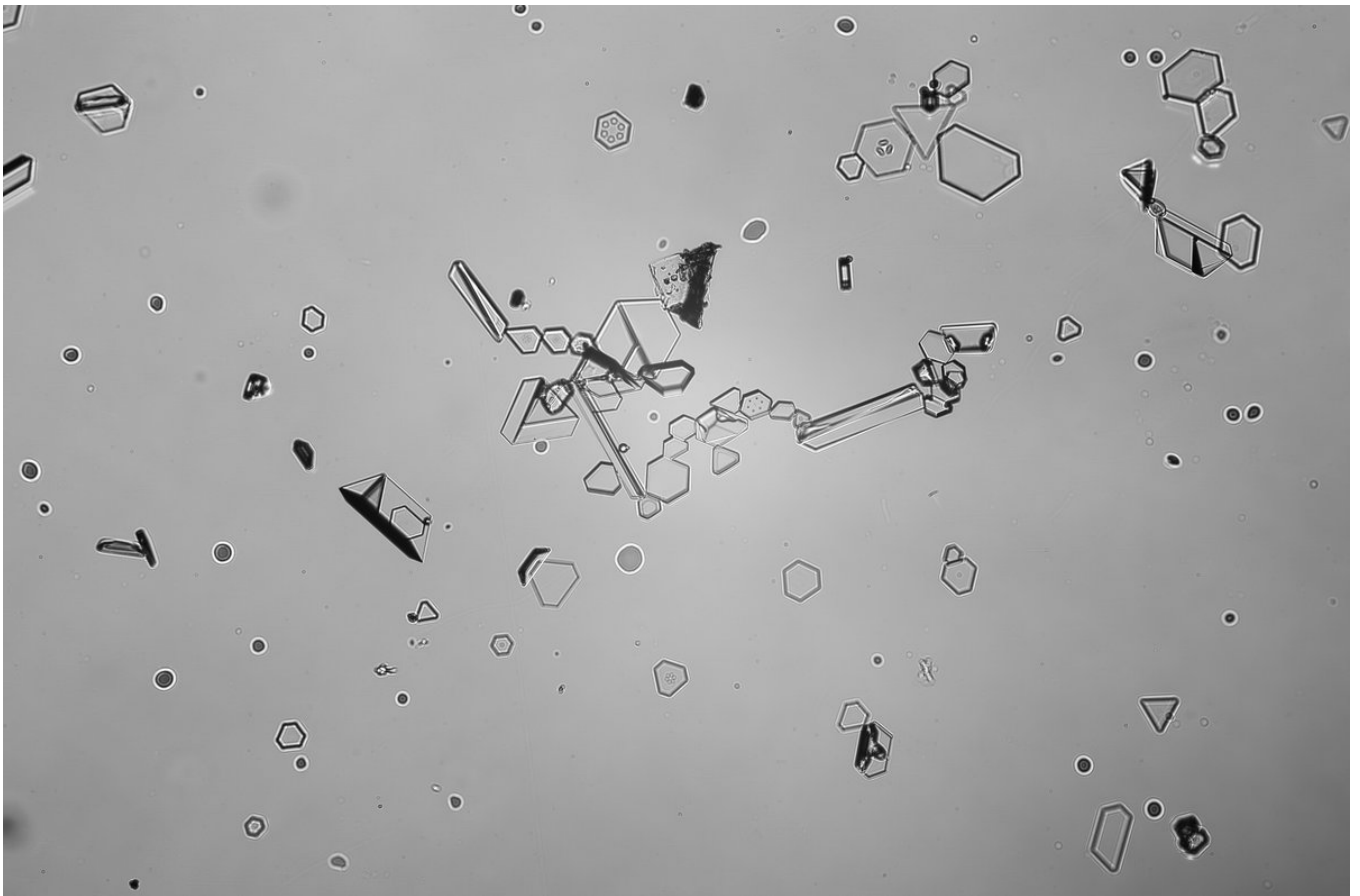
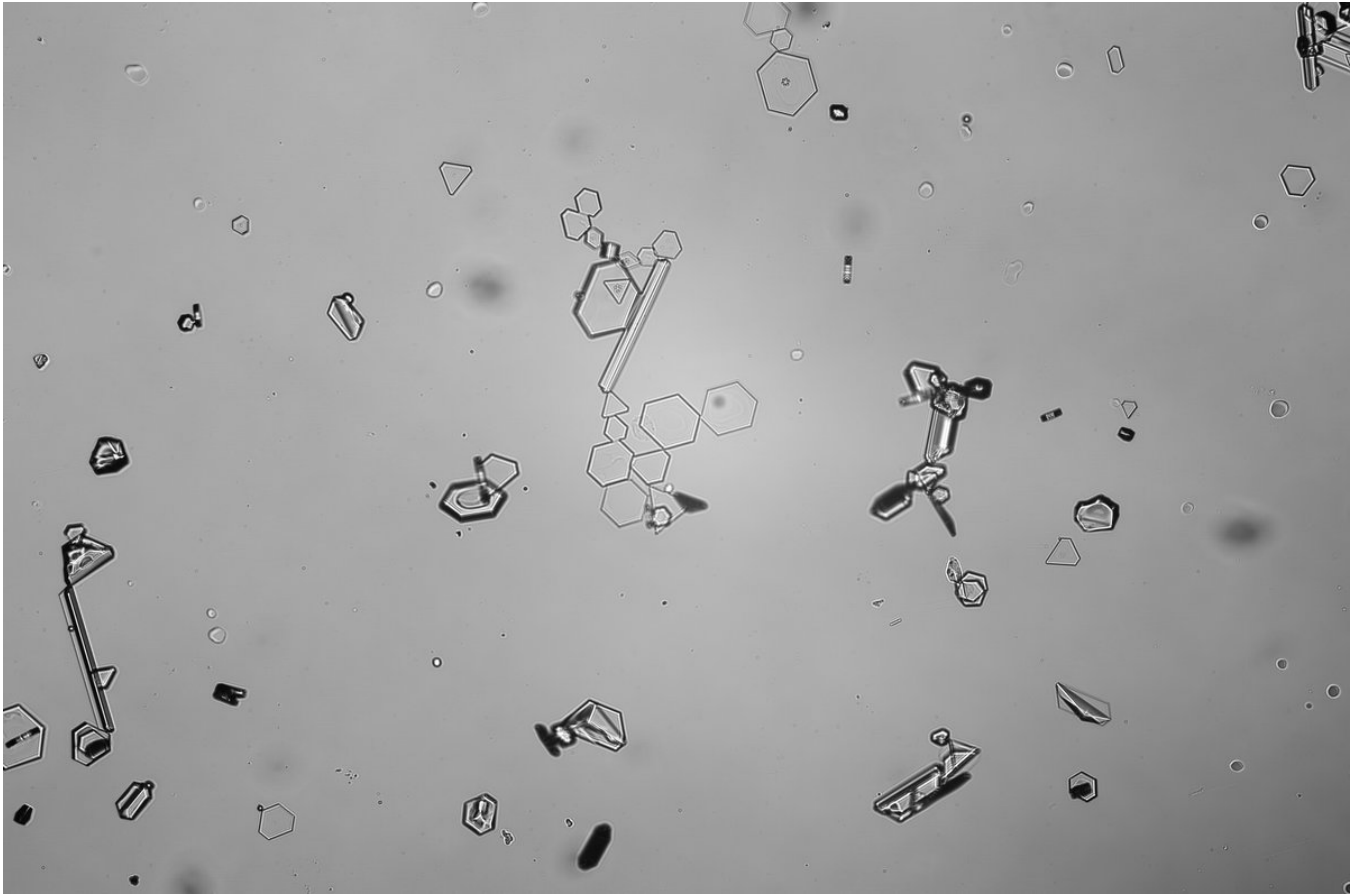
22/23 Feb 2026, Rovaniemi. Even more 1st stage crystals.

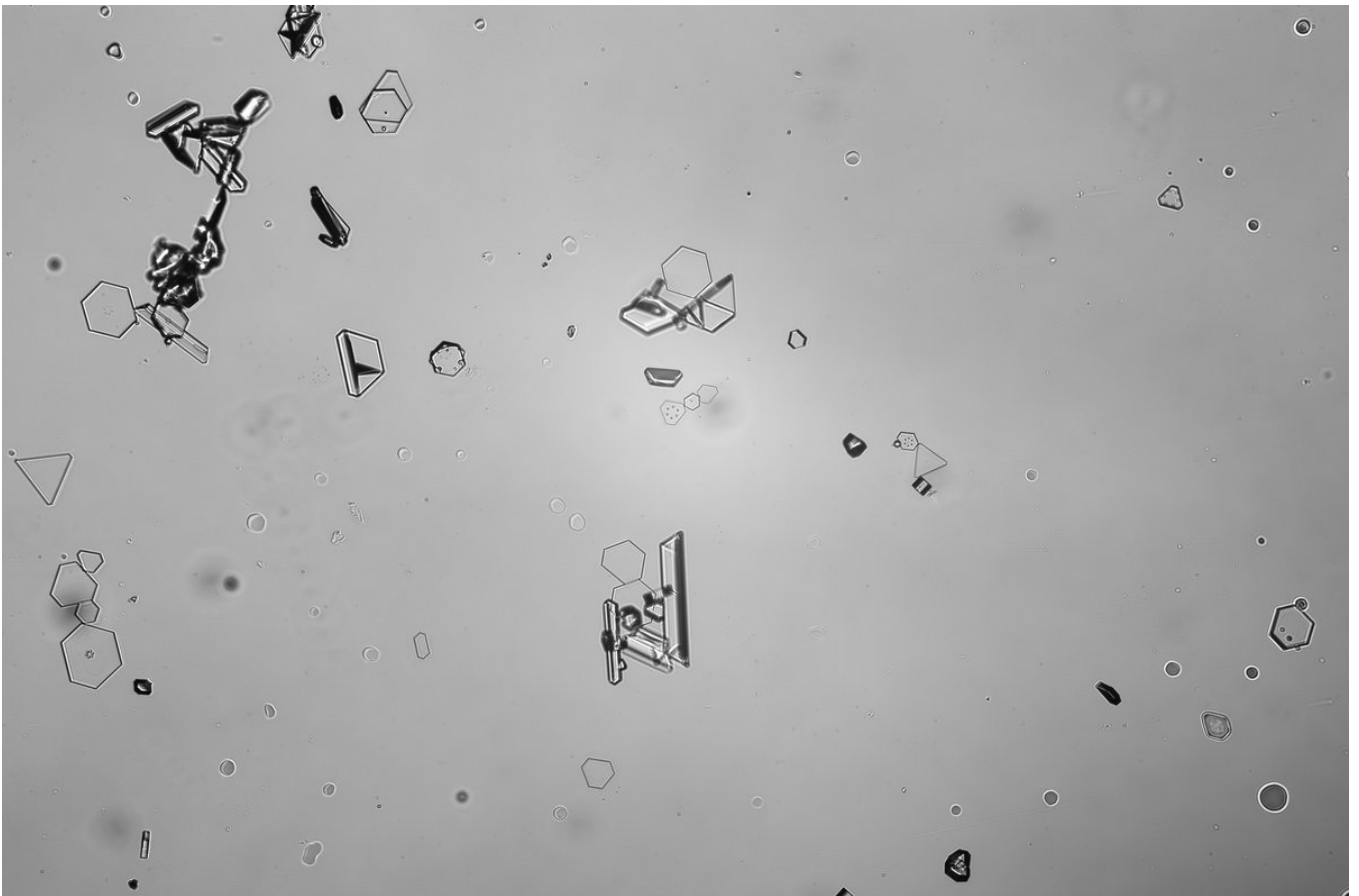
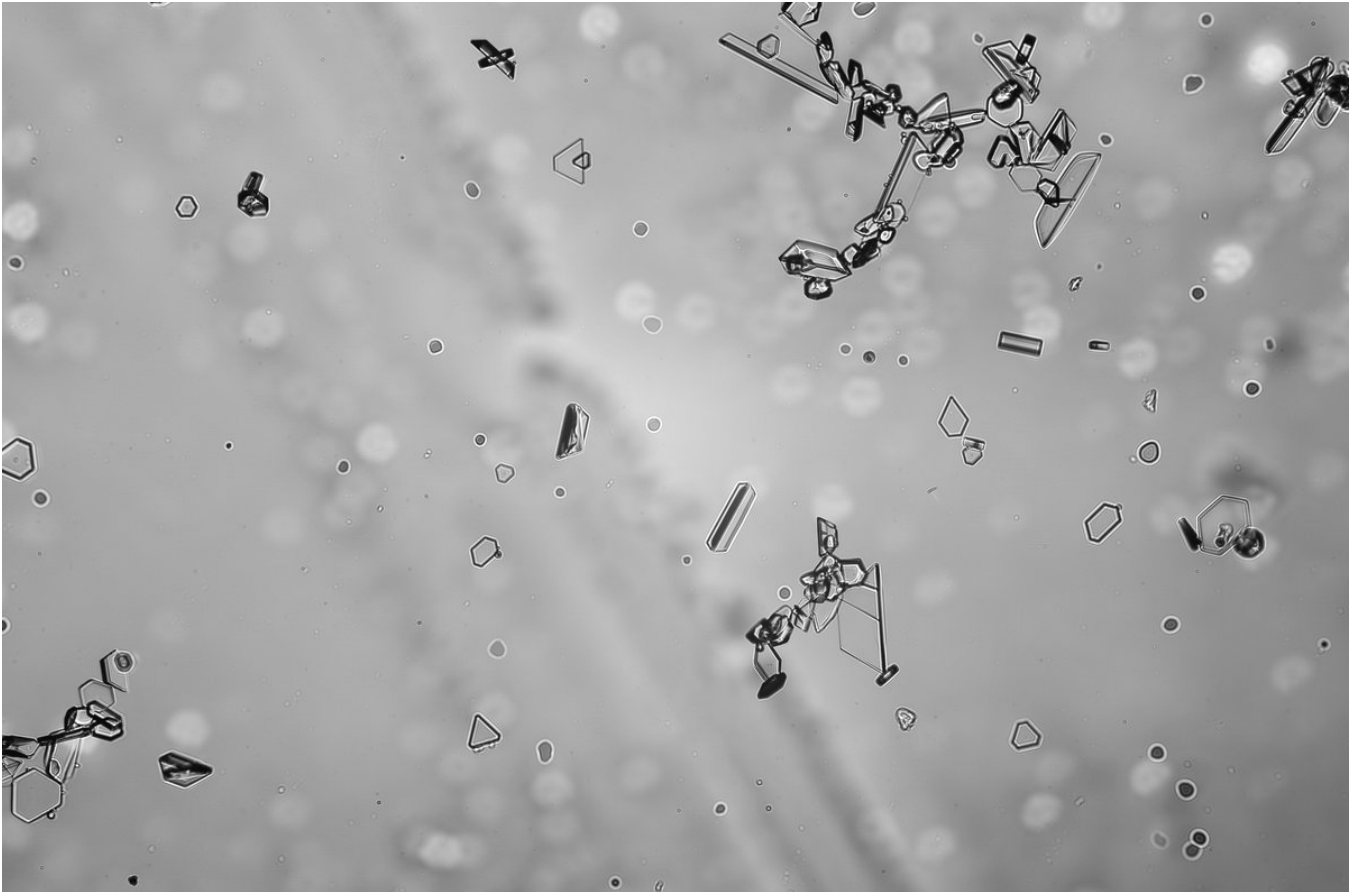




20/32 - 2026-02-28 12:02 UTC

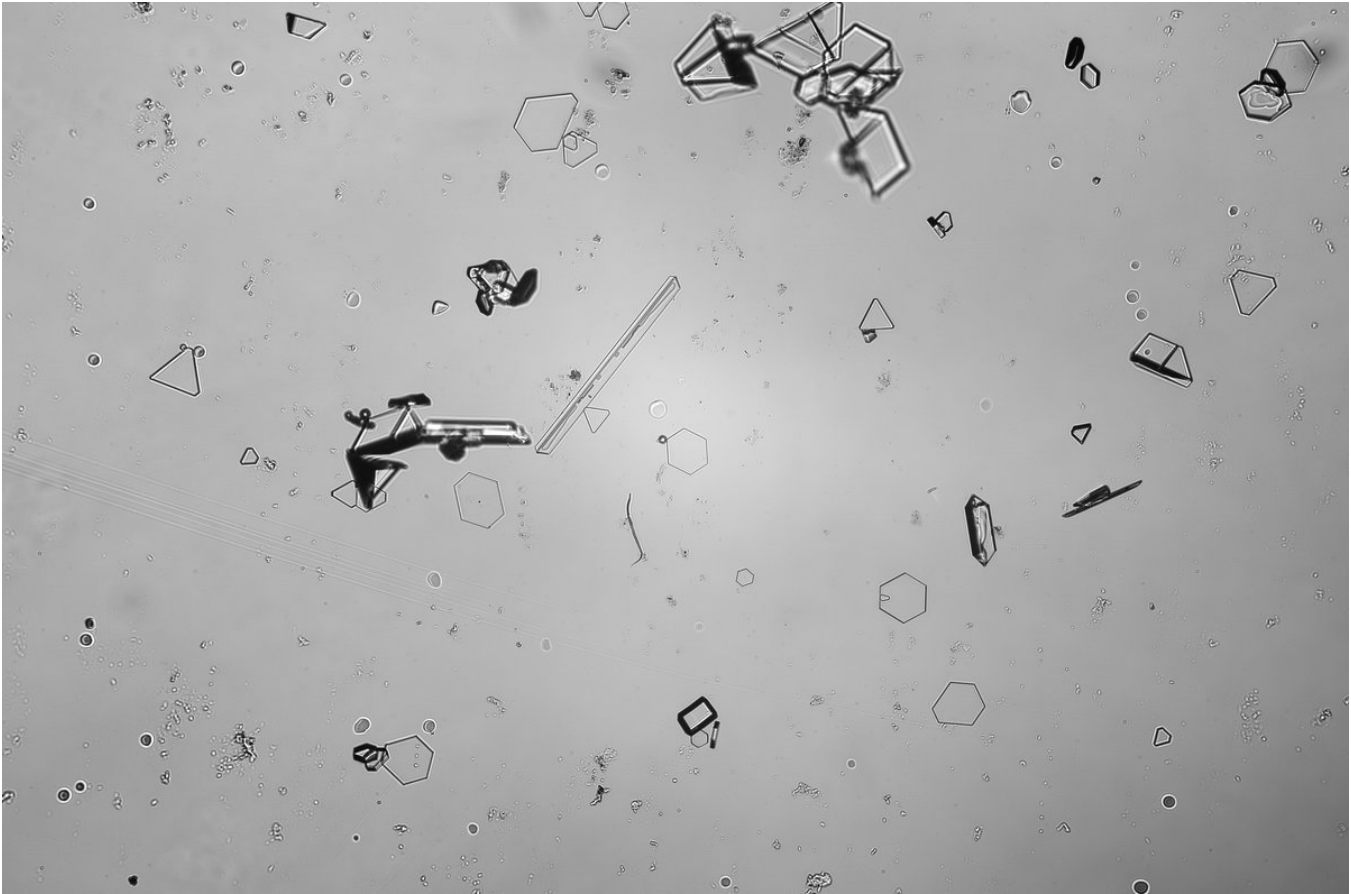
22/23 Feb 2026, Rovaniemi. More 1st stage...

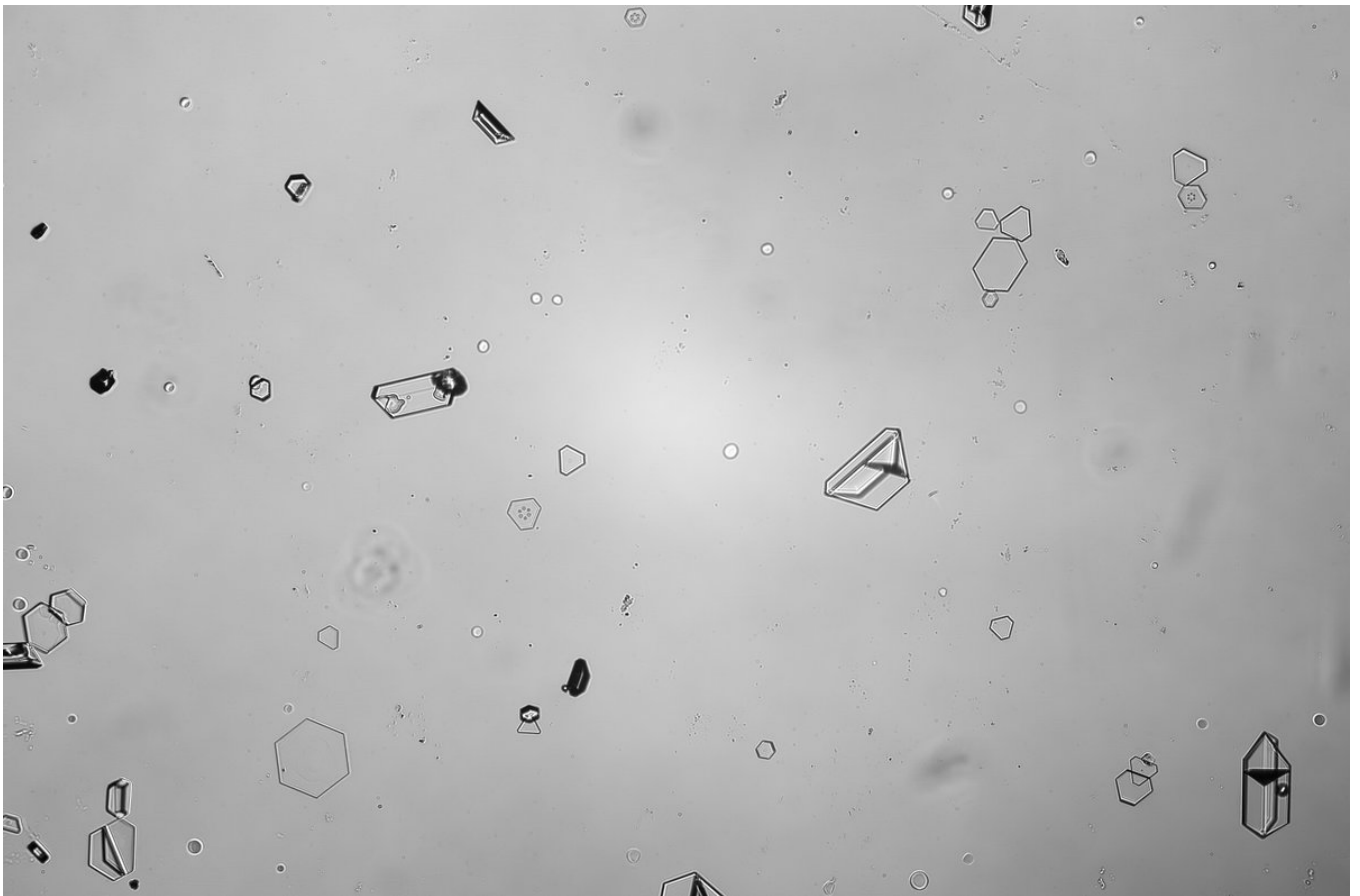
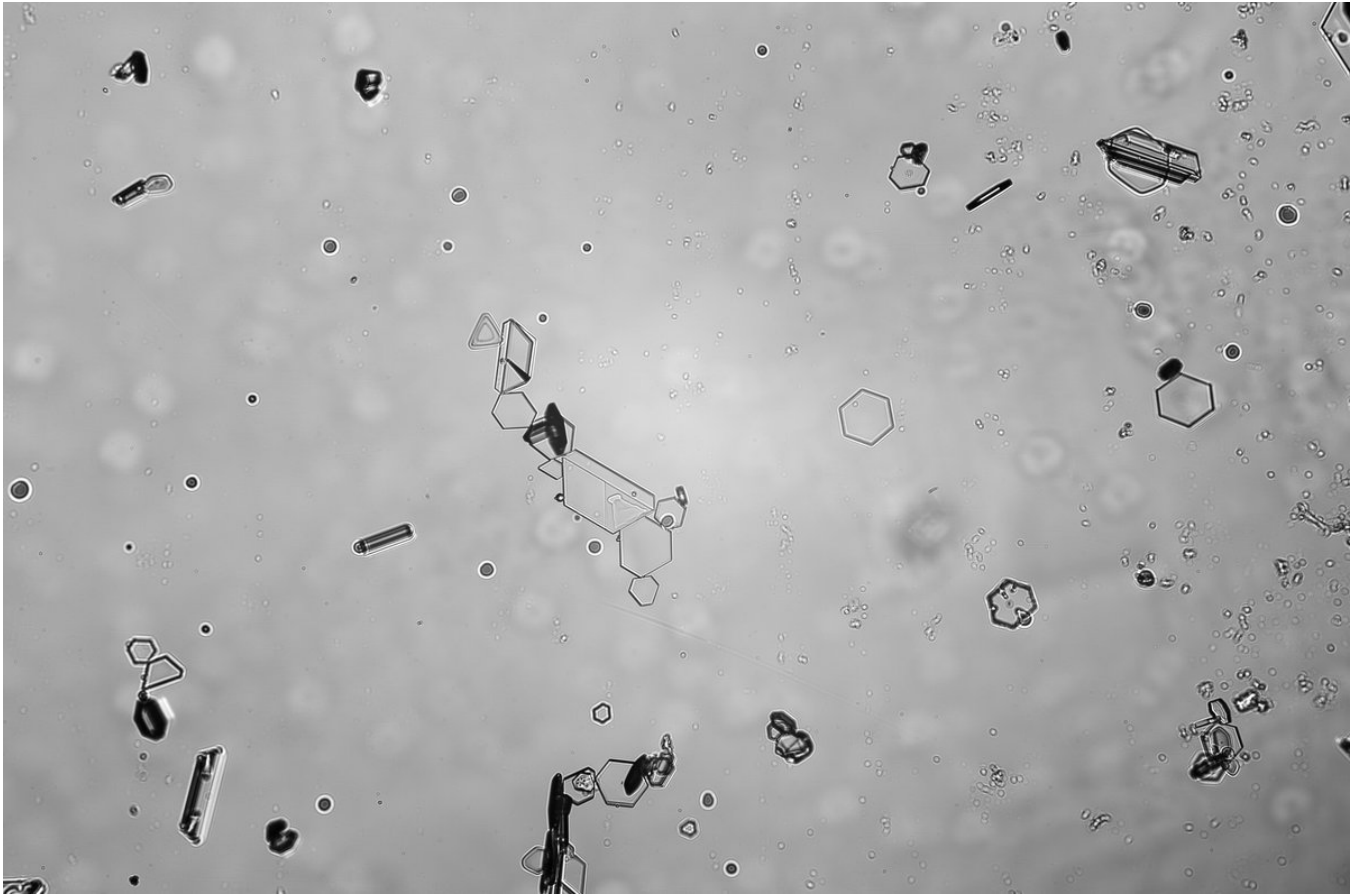




21/32 - 2026-02-28 12:07 UTC

22/23 Feb 2026, Rovaniemi. More 1st stage... ..

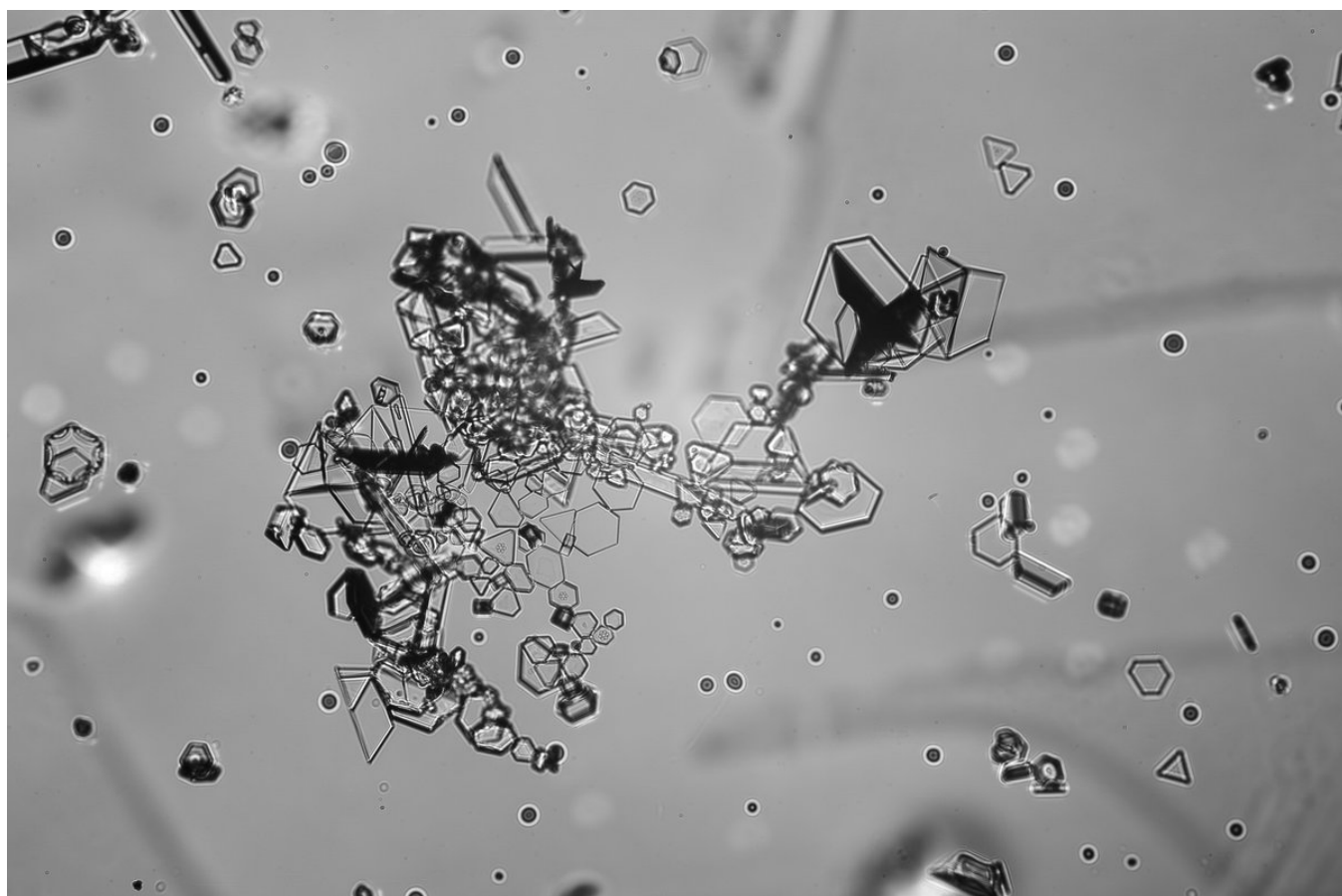


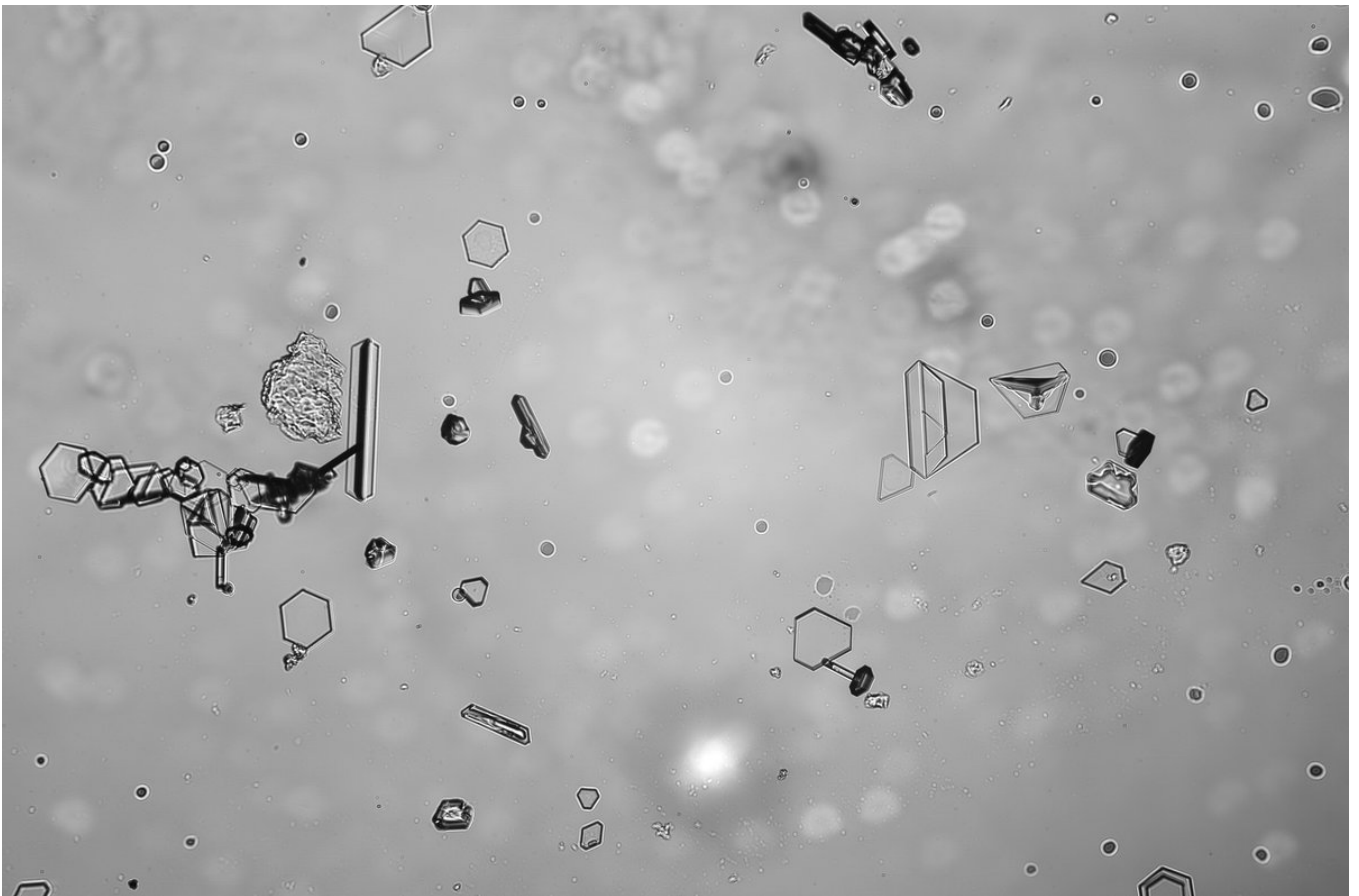


22/32 - 2026-02-28 12:24 UTC

22/23 Feb 2026, Rovaniemi. More 1st stage. This is really a supersample of the triple-crosseds and

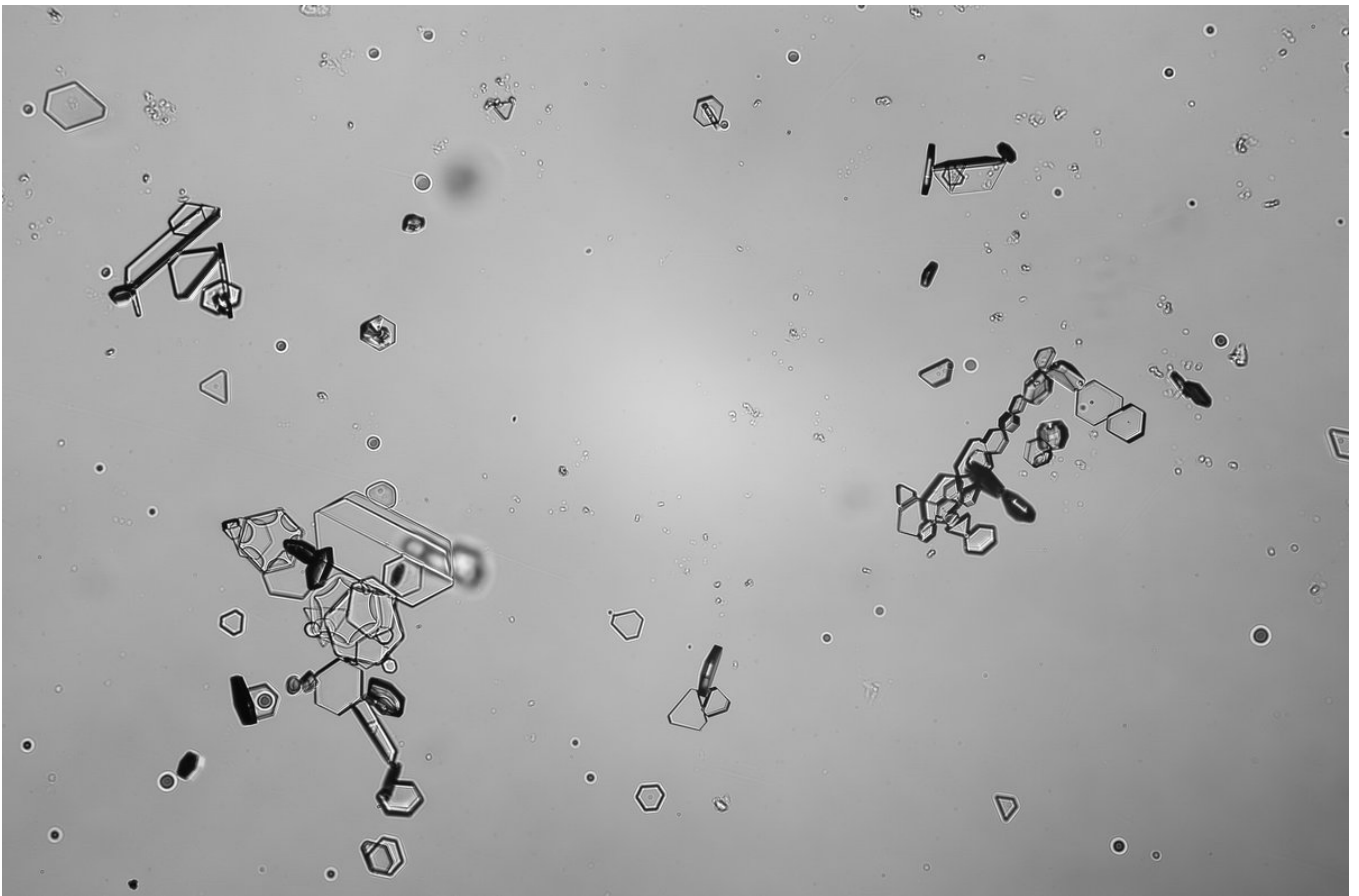
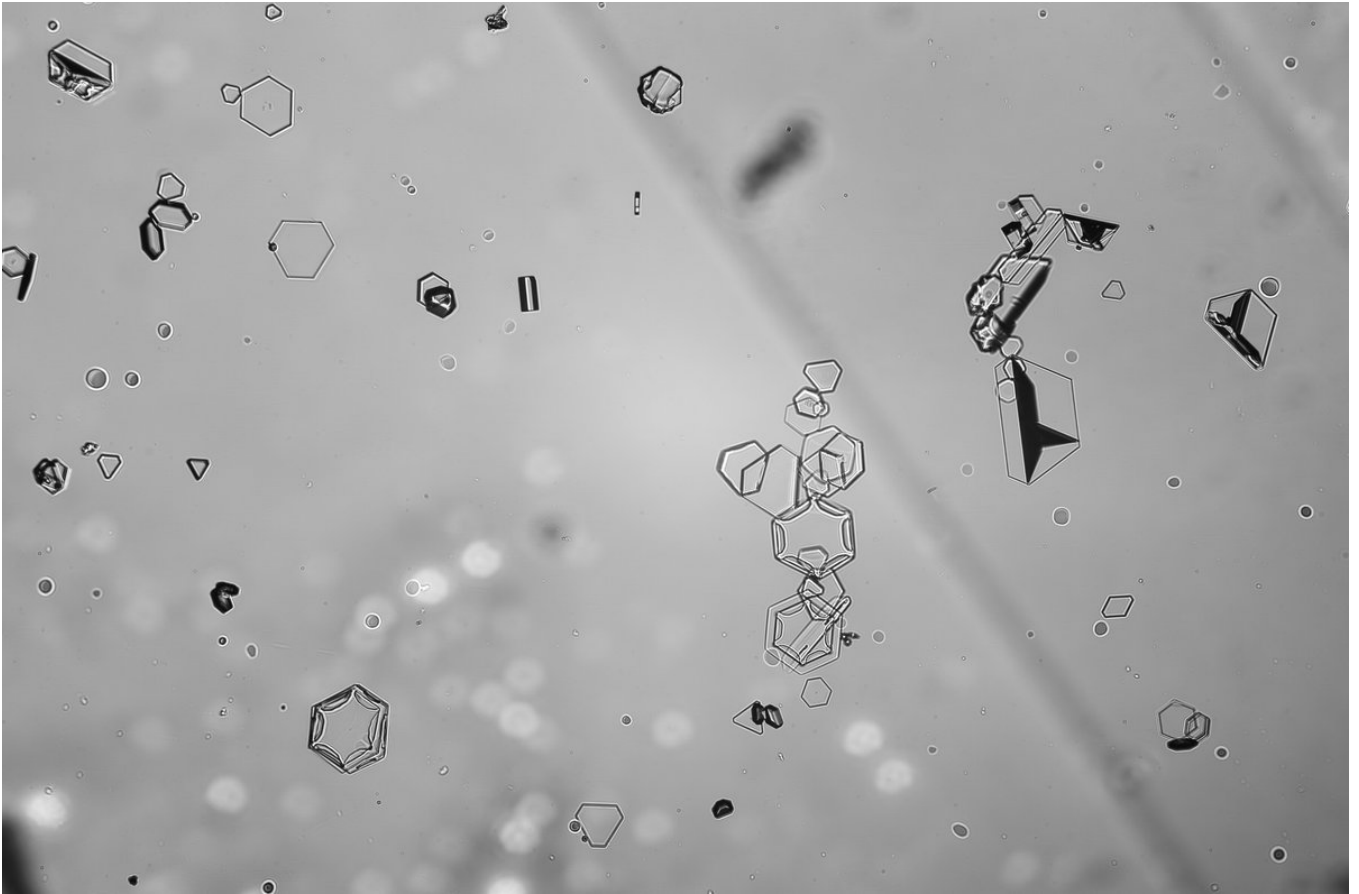
the more elongated double-crossed Moilanen suspects.





23/32 - 2026-02-28 12:27 UTC

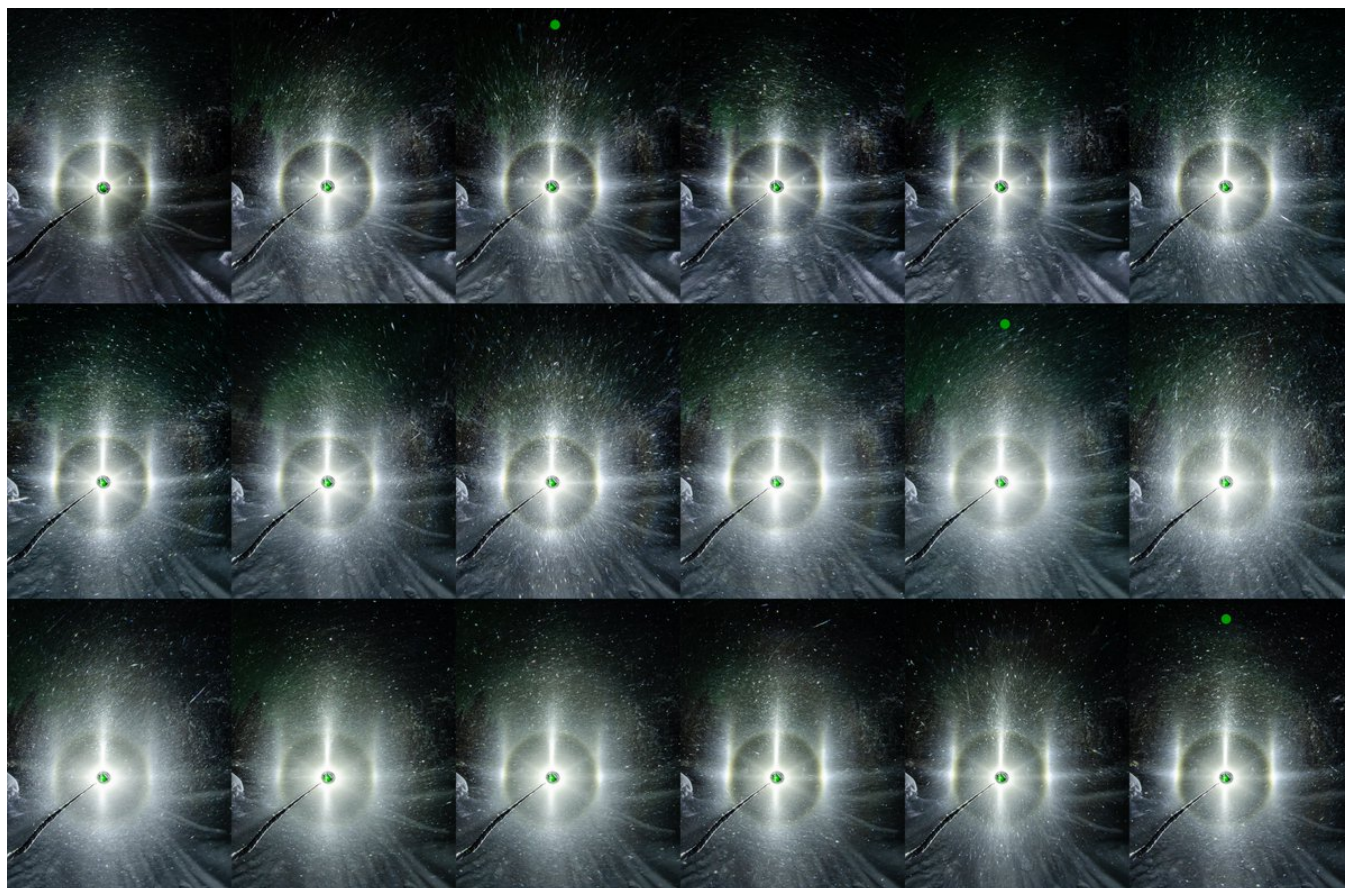
22/23 Feb 2026, Rovaniemi. And this is the last of the 1st stage crystals.



24/32 - 2026-02-28 13:35 UTC

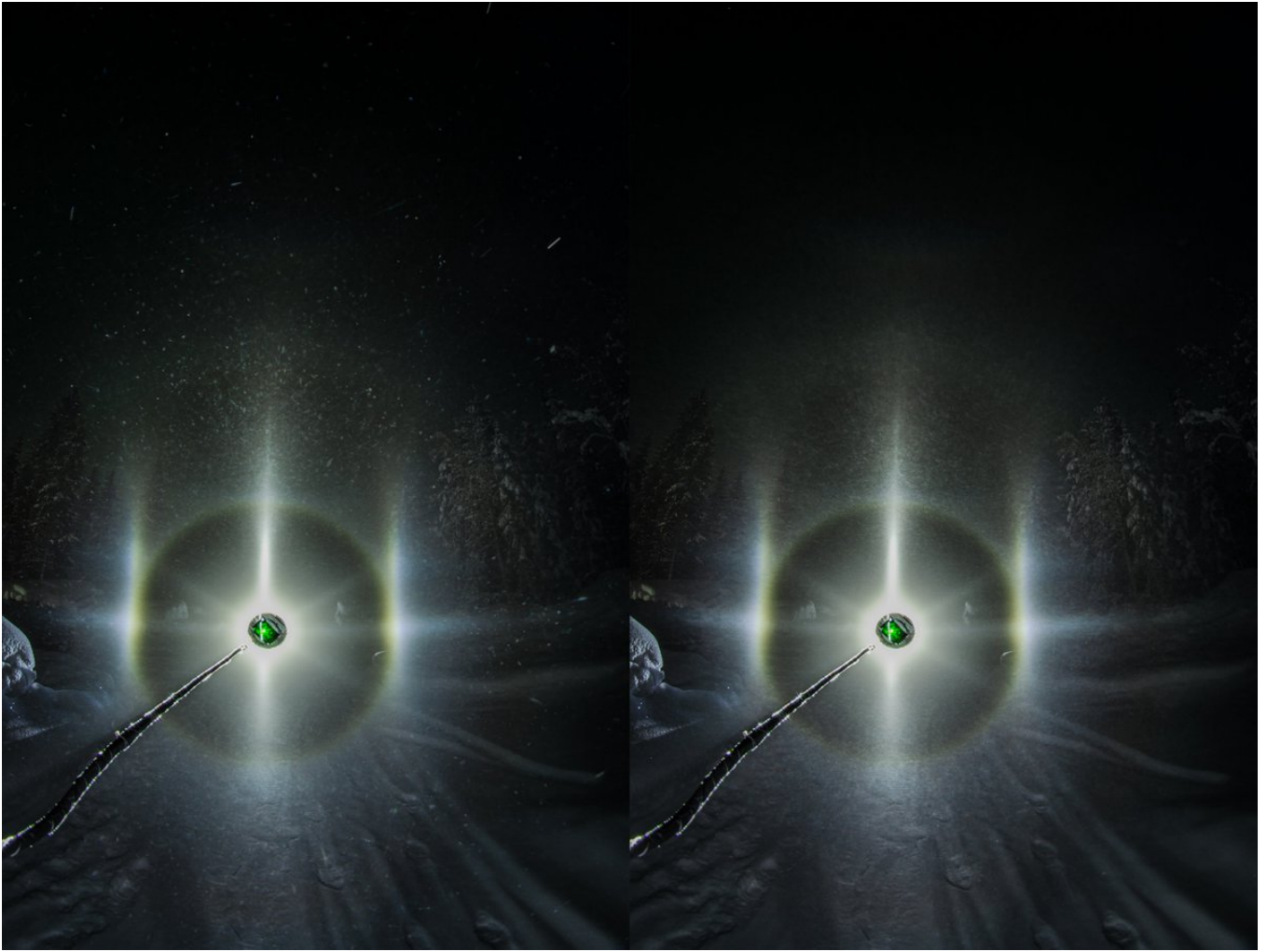
22/23 Feb 2026, Rovaniemi. This collage has every 5th photo of the full set. The first green dot marks

the time when I put the dish out, second when I started to photograph the crystals and third the end of crystal photography. New crystals were falling continuously on the dish as I photographed them.



25/32 - 2026-02-28 14:00 UTC

22/23 Feb 2026, Rovaniemi. 23 frames stacked from the end, ave and median. The strong helic arc stands out given there is nothing indicative of Parry (above or below the lamp). The second image is b-r of the ave.



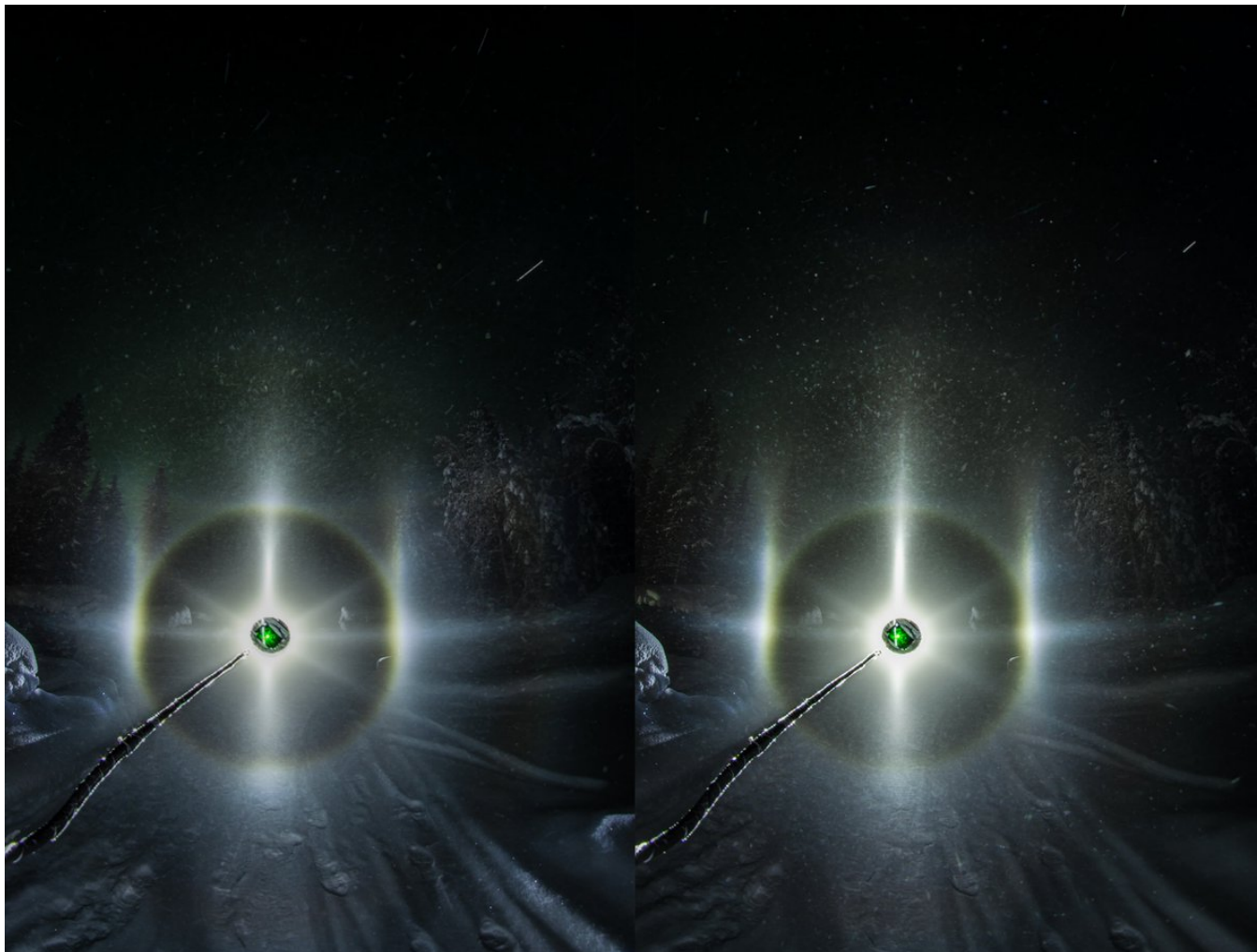


22/23 Feb 2026, Rovaniemi. Flashing the above post's stack with a 46 frame stack from the beginning of the set. Helic arc is impervious to the changes in the display.

[video: 2027756477943812480-HCQInsiXYAAEekR.mp4]

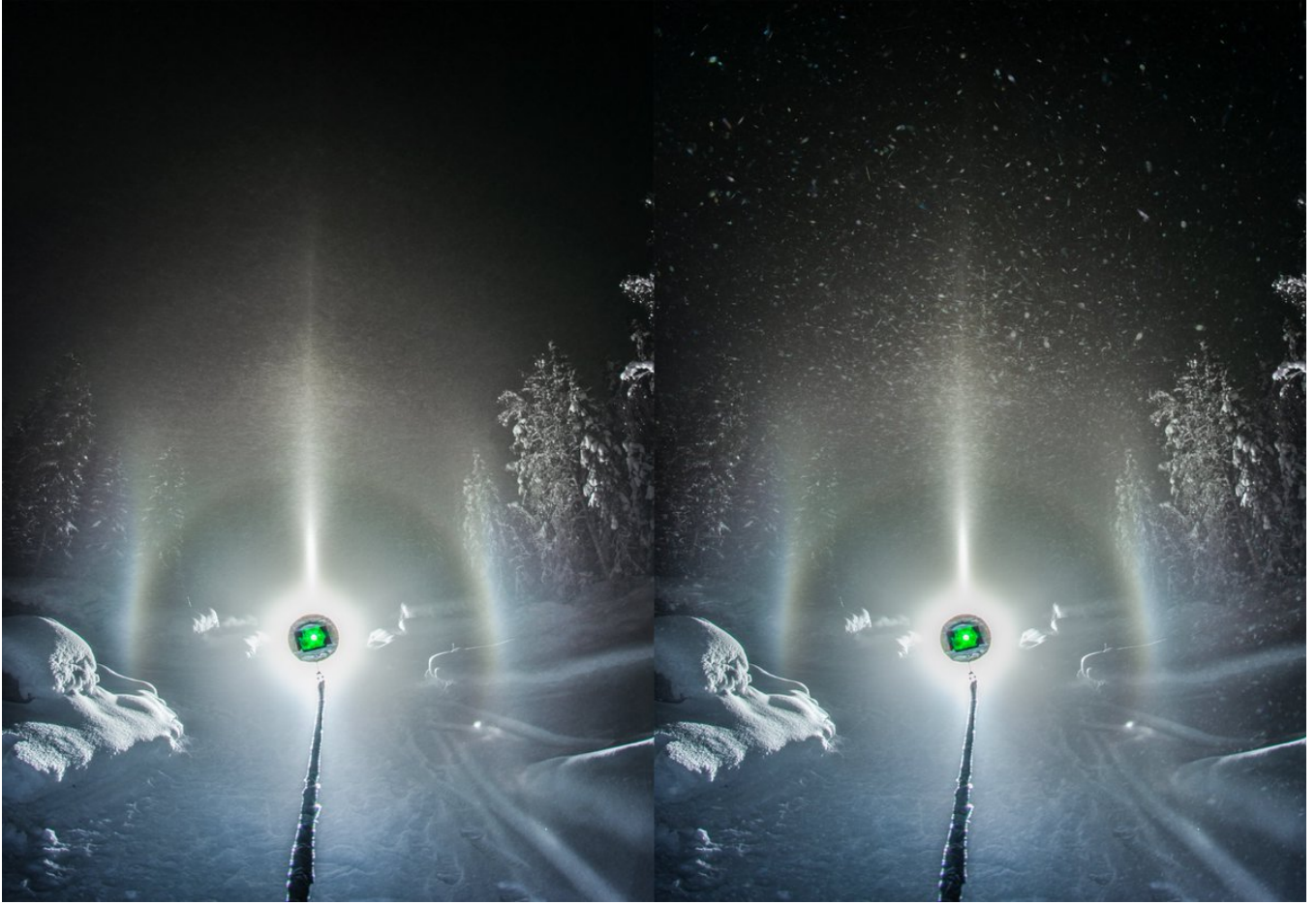
27/32 - 2026-02-28 14:45 UTC

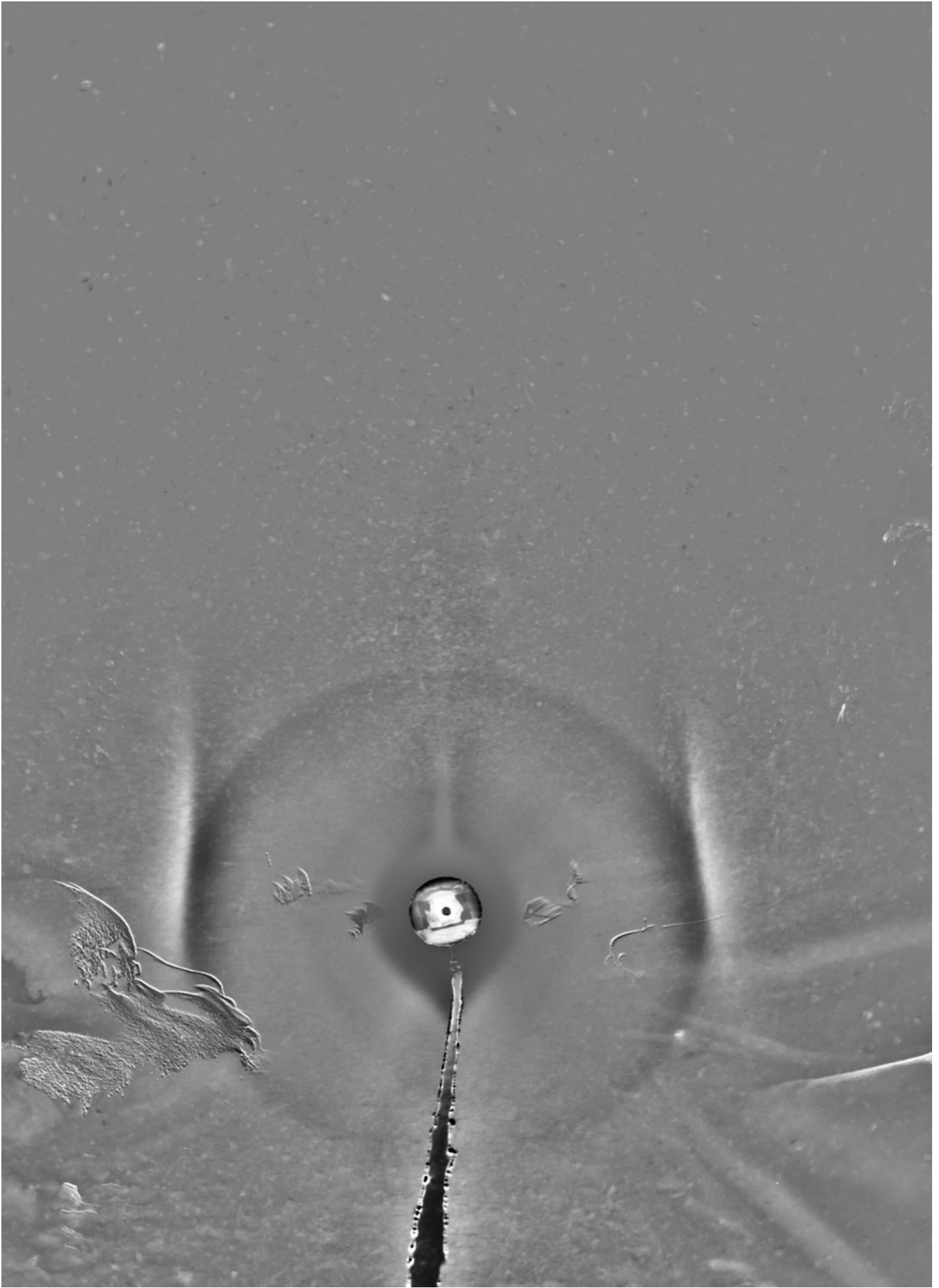
22/23 Feb 2026, Rovaniemi. Side-by-side comparison of the above post's images.

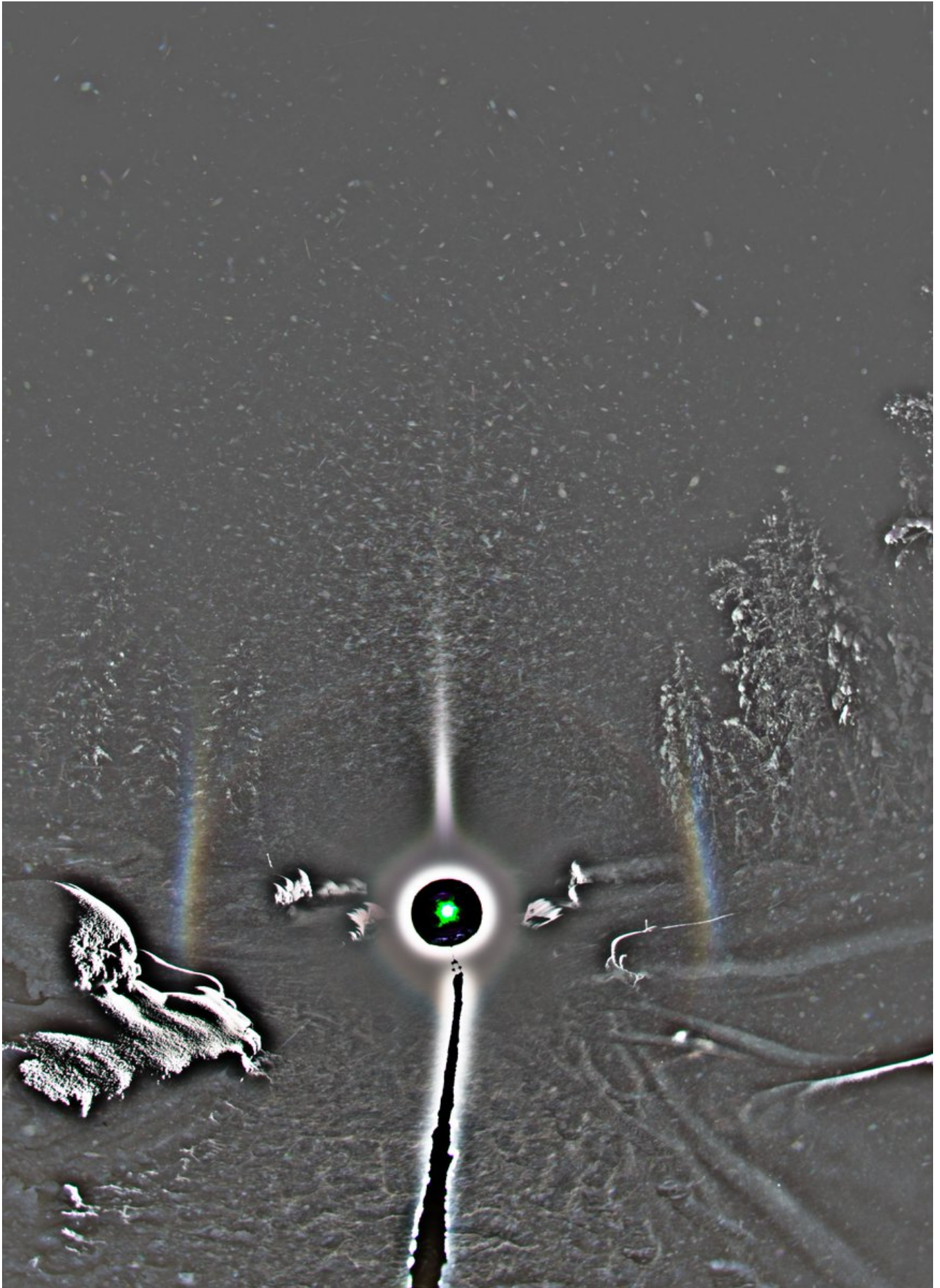


28/32 - 2026-03-01 11:24 UTC

22/23 Feb 2026, Rovaniemi. The 3rd and last swarm. 28 x 15s stack with frames starting from the beginning of crystal collecting. First median-ave comparison using only Camera Raw adjustments, the other two images are b-r and bgr of ave. I saw Moilanen arc just before putting the dish cooking, but no more when I came back behind the camera right after. Maybe it is weakly present in this stack, though. Helic arc certainly. But no brightening of any kind at the top or bottom of 22° halo.

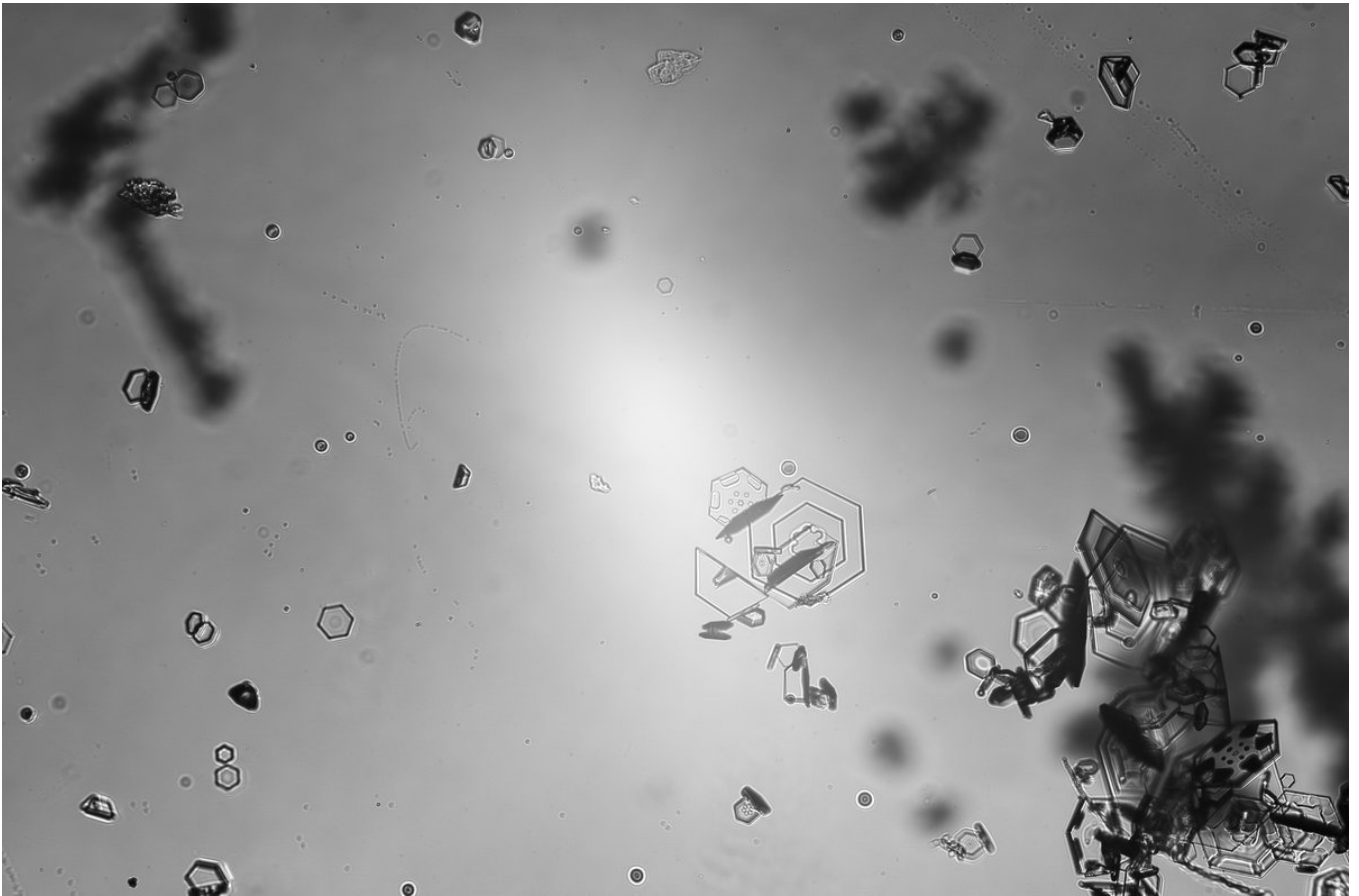


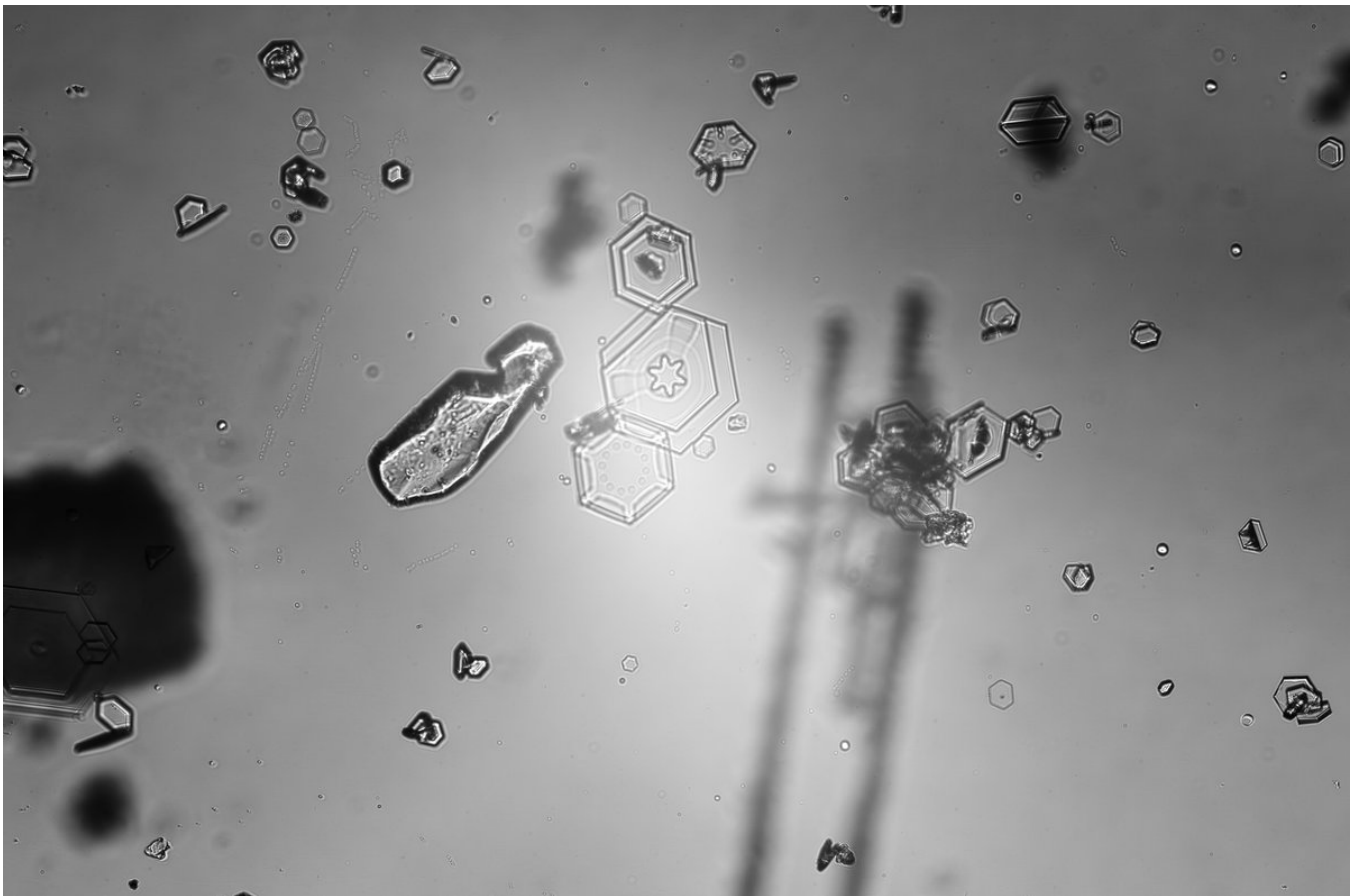
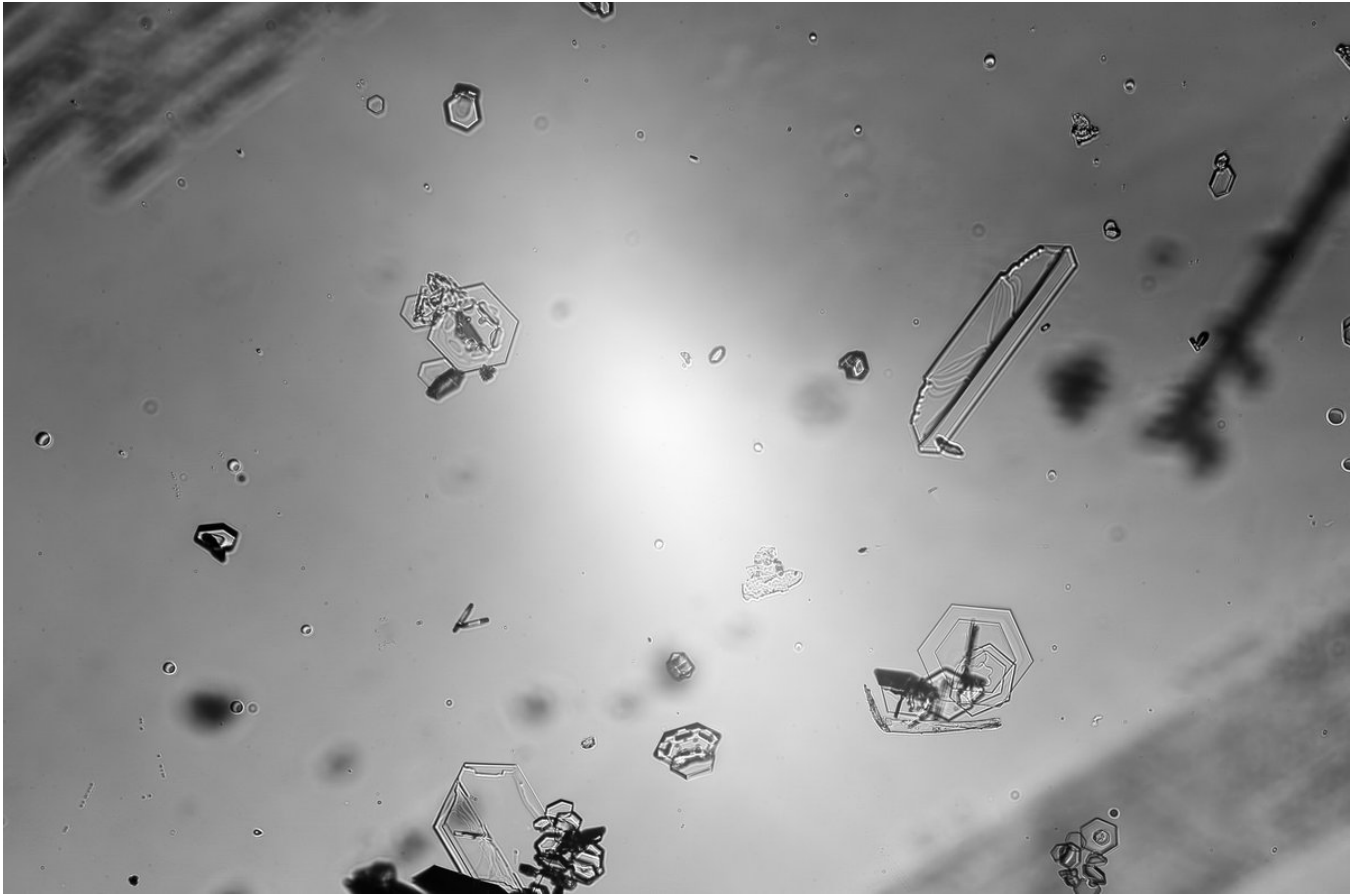


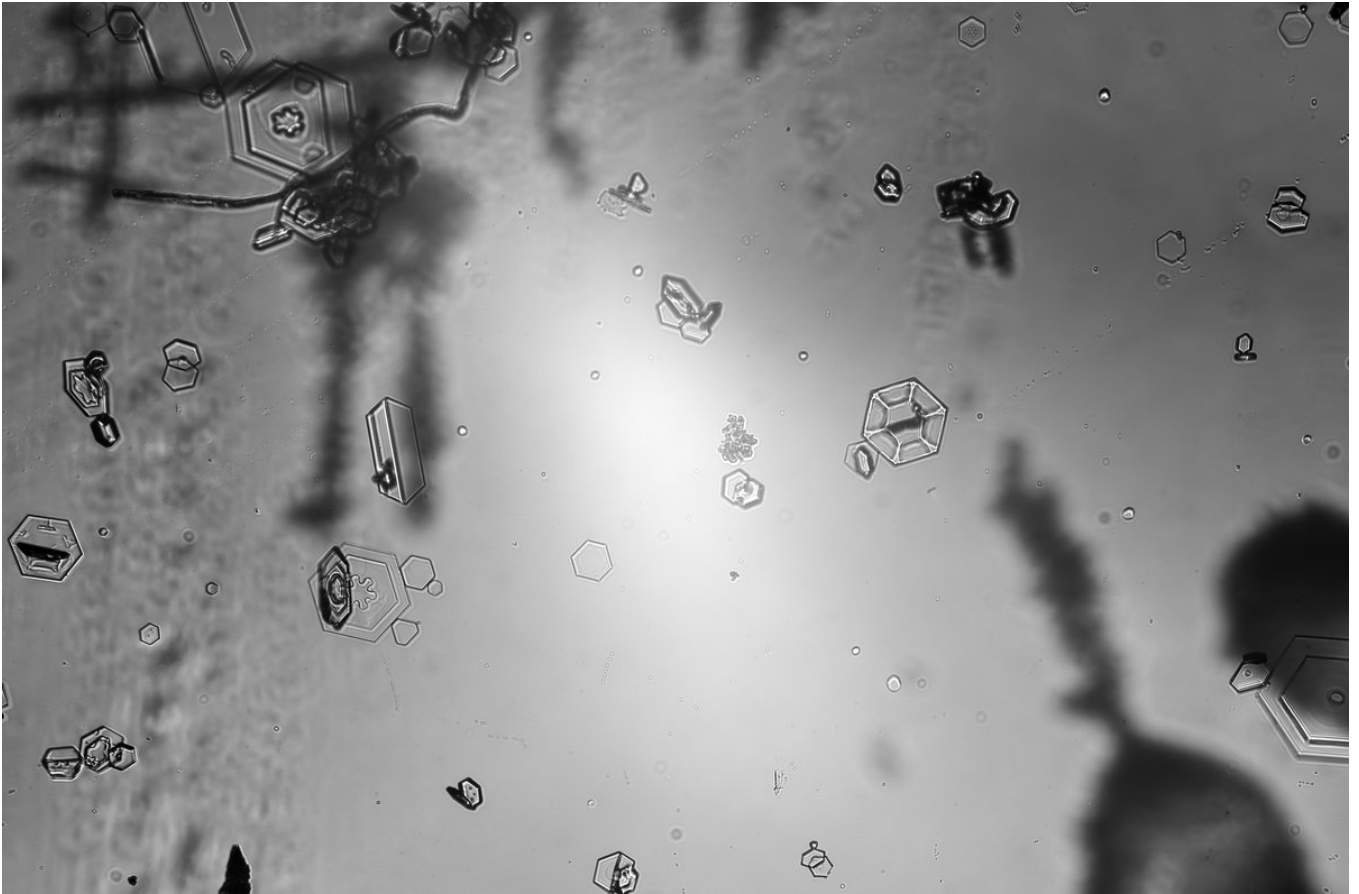


22/23 Feb 2026, Rovaniemi. Here follows a few posts presenting the crystals of the 3rd stage (above post). What comes to the weak helic arc in the display, I see not one column in the photos. Not even anything nearing equidimensional. All crystals are firmly in the plate land.

I see I was a infatuated by those large decorated plates. Should have set the focus more often on the small stuff.



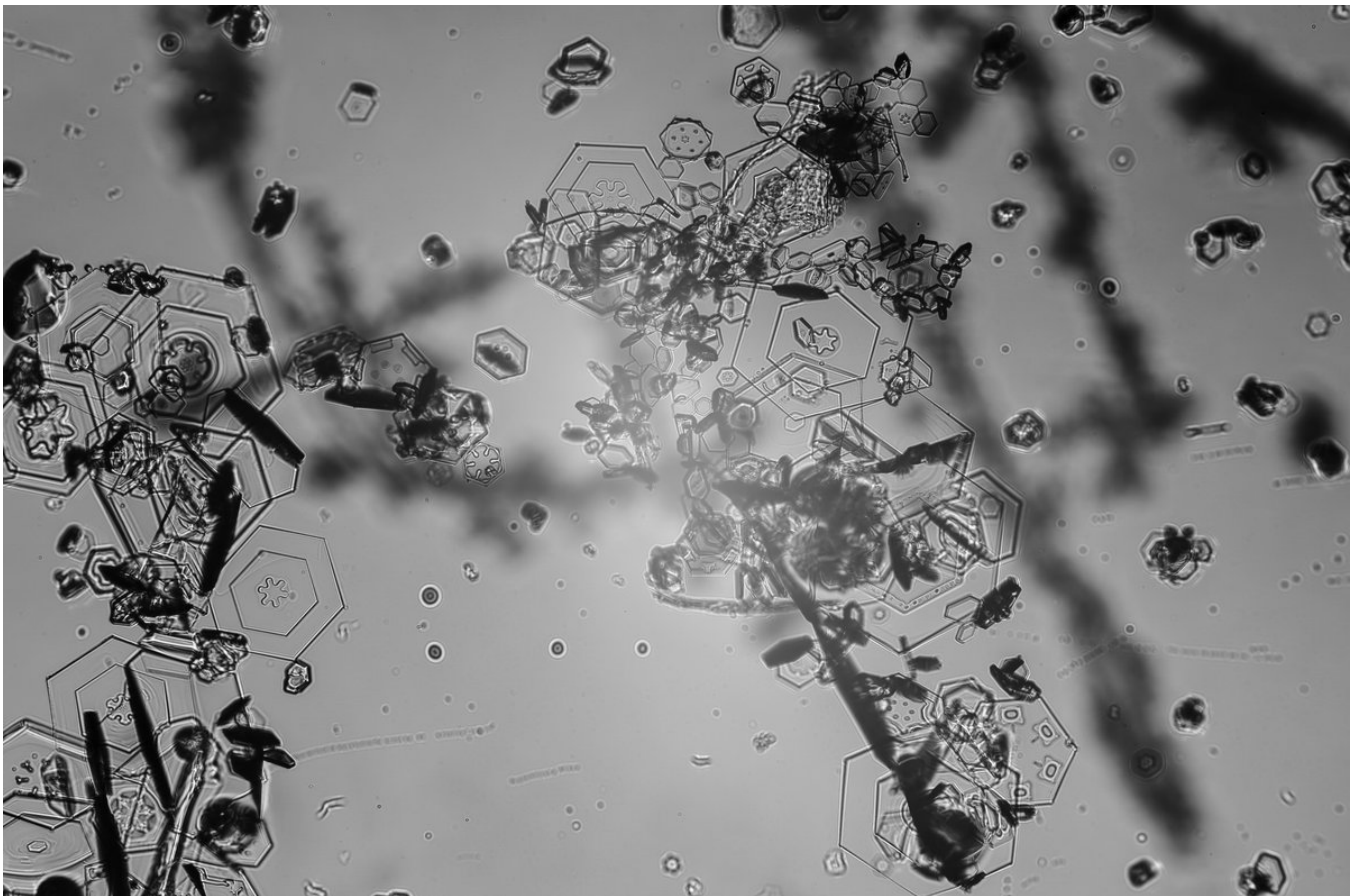
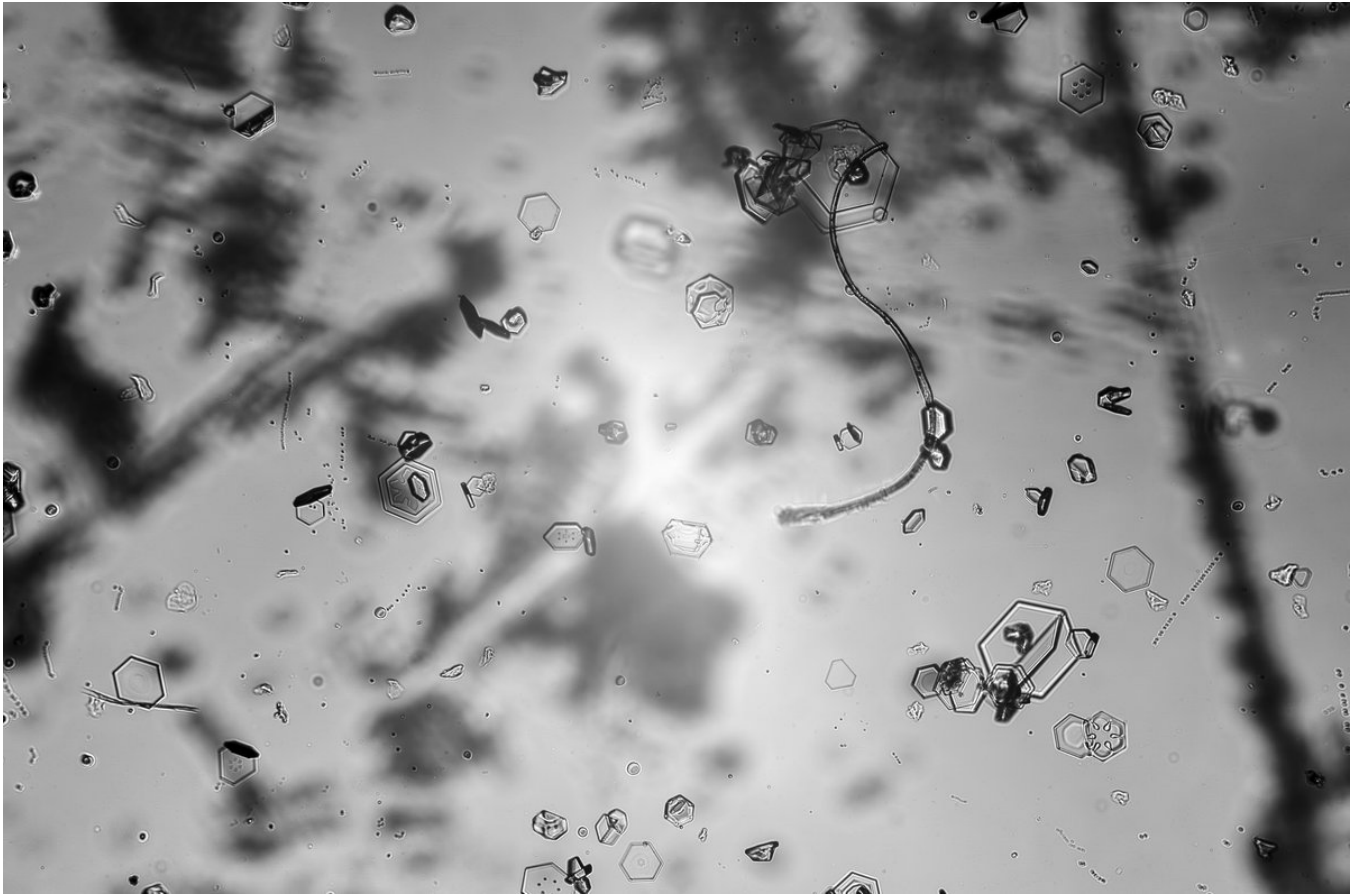


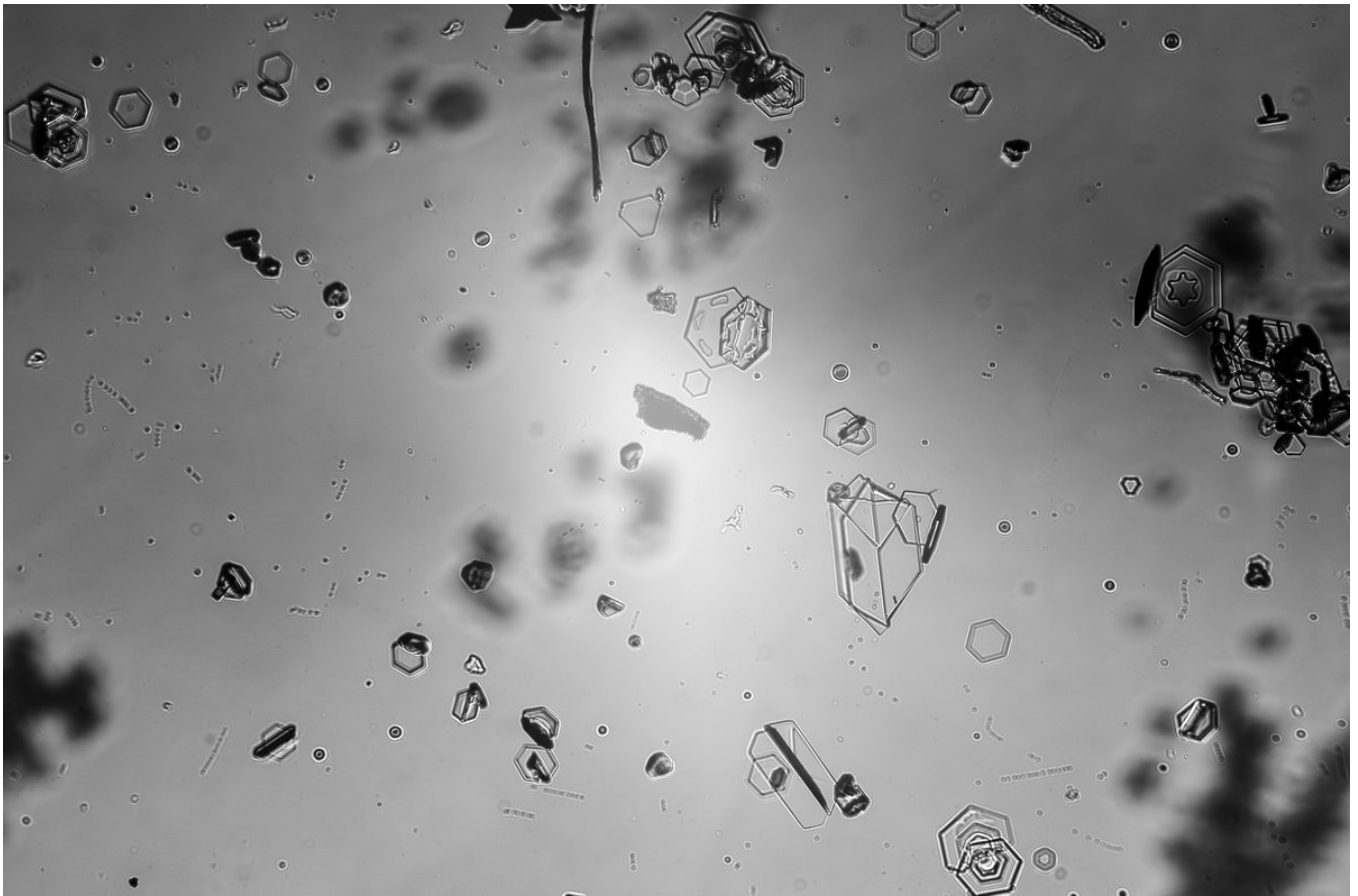


30/32 - 2026-03-01 11:59 UTC

22/23 Feb 2026, Rovaniemi. More 3rd stage crystals.

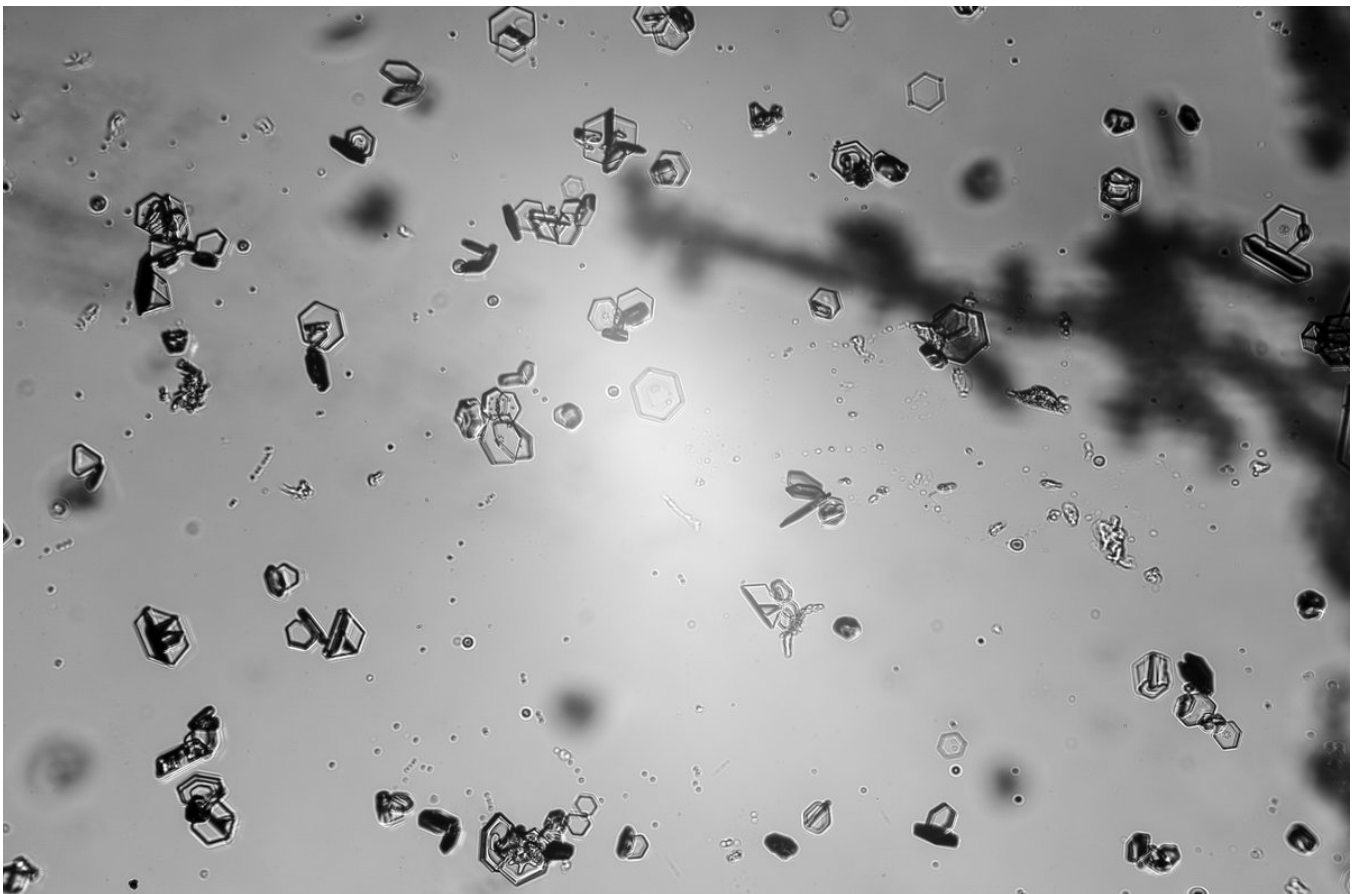


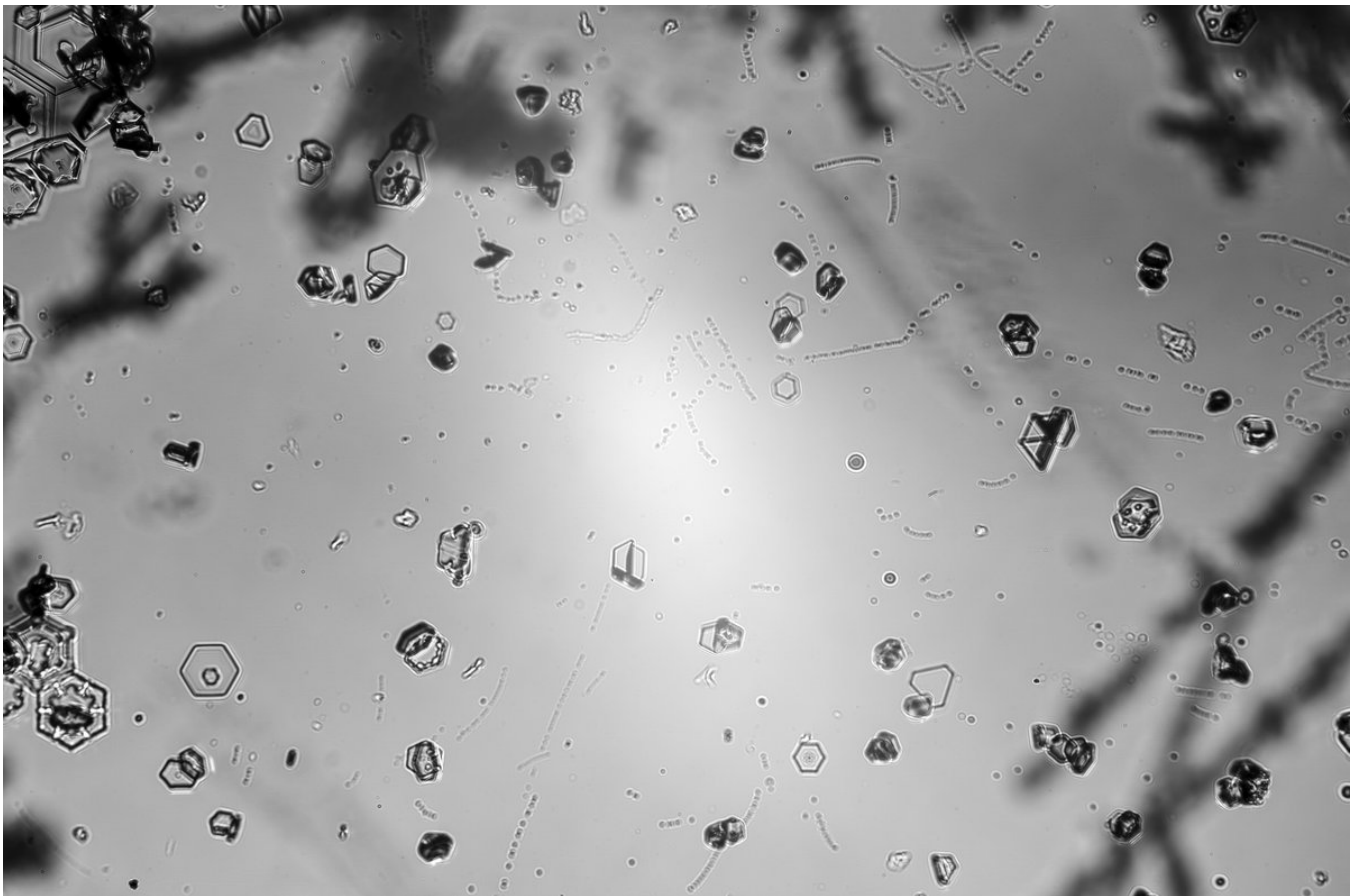
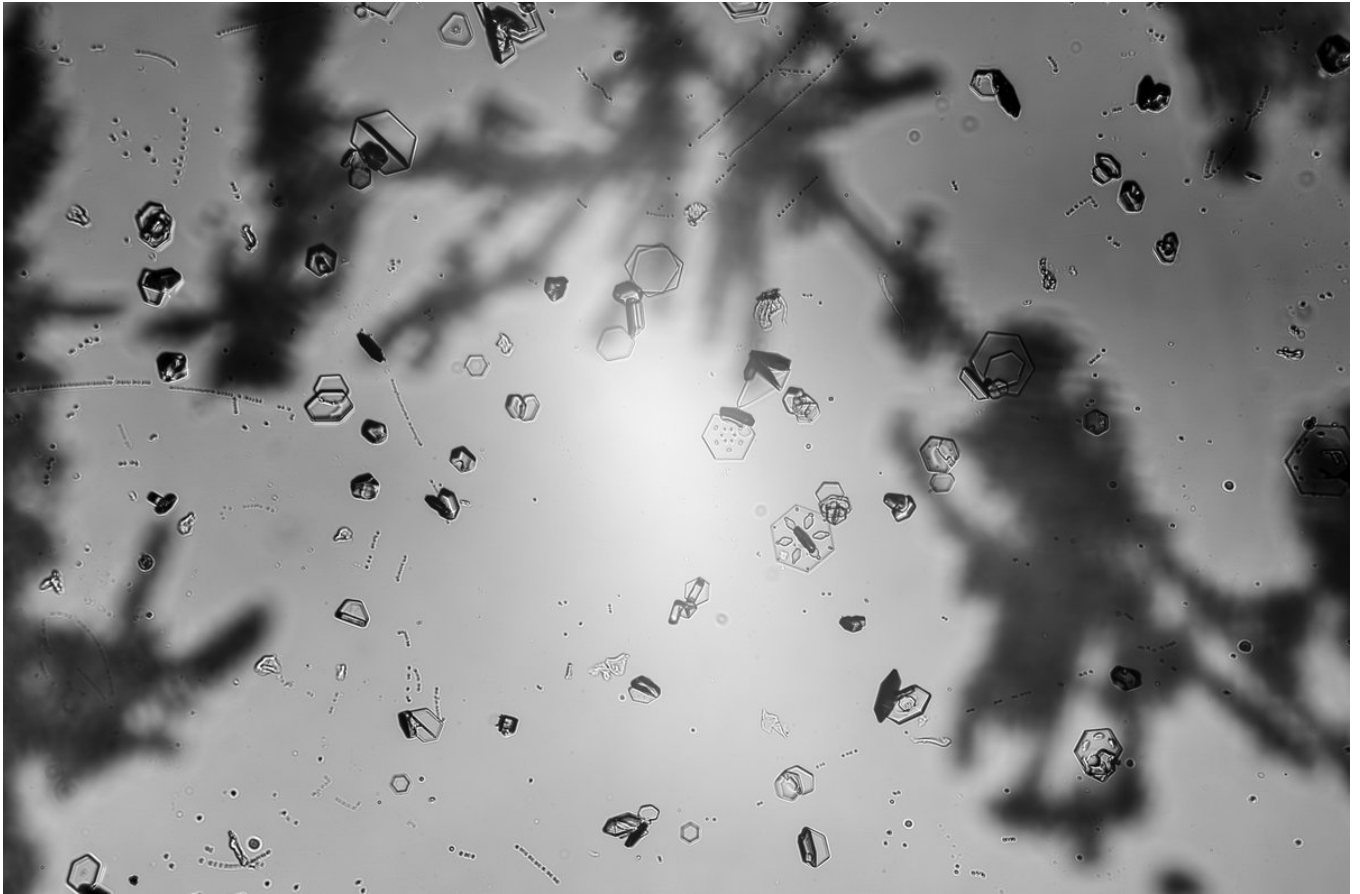


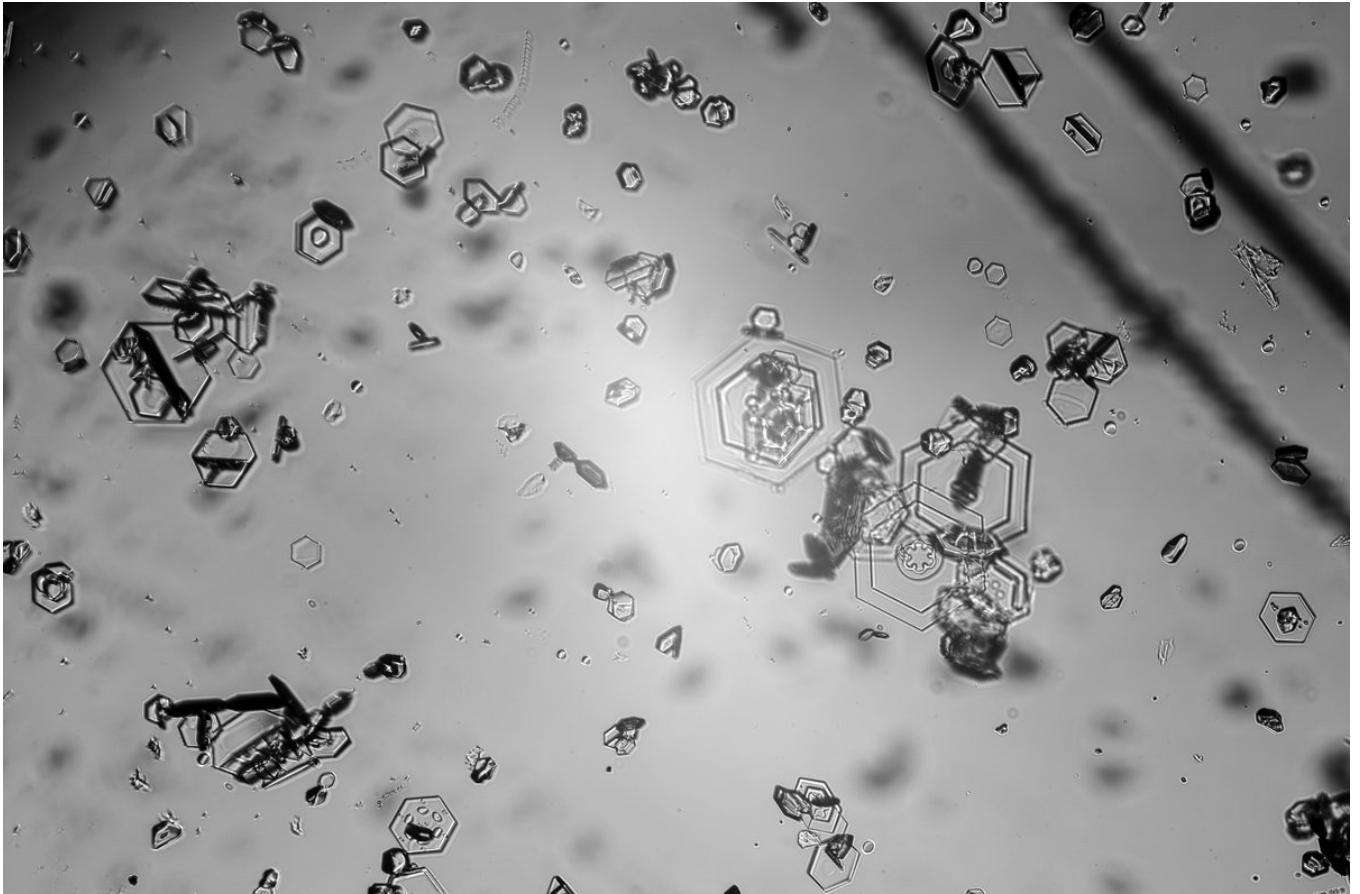


31/32 - 2026-03-01 11:59 UTC

22/23 Feb 2026, Rovaniemi. Another set of 3rd stage crystals.







32/32 - 2026-03-01 12:00 UTC

22/23 Feb 2026, Rovaniemi. Last set of 3rd stage crystals.

